# Supplemental Material to "The two families of 84 and the number of generations"

Draft, 3 November 2026

This is the supplement of the paper *The two families of 84 and the number of generations*. Its parts keep the numbers of the full Supplemental Material, so that a citation of a part in the paper refers to the part of the same name here.

**Parts contained.** Lemma parts: 18, 21. Eleven-dimensional files: README, 03, 04, 05, 09. Other parts: R4, R7, R21 (the letters H, L and R mark the history of the field, the literature check and the referee reports).

The full Supplemental Material, with all parts of the project, exists as `supplemental.pdf` and `supplemental_mobile.pdf` in the same folder as this file. A reference inside a part to a part not listed here points to the full version.

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# Part 18

## T5b: vector masses in the spectral action at fixed cutoff

Prompt: prompts/18 (STRATEGY T5b). Attempted by gpt-6-astra (codex 0.164, xhigh, 3 Nov 2026), working alone. Checked by Claude Opus 5.5 with Python (numpy, scipy), by computations independent of the kernel derived here: a direct heat trace with constant a reproduces equation (2) for five choices of (m_1, m_2), to about 0.03%; a direct heat trace on Fourier modes with a_2(x_1) = a cos(q x_1), including the F_{12} and [T, V] vertices, reproduces the kernel (37) at three values of (m, q), to 0.1-0.3%, the size of the discretization error. The closed forms (16)-(19), the bound (28), the limits (27), the neutral matrix (30) recomputed from Standard Model charges, and the identities behind (31) also check. The largest value of the Taylor mass is M_W^2 = 0.275 Lambda^2, at m_t^2 = 2.02 Lambda^2. Refereed by Terra 6.1 (gpt-6.1-terra, effort max, 3 Nov 2026; report): all ten numbered results are correct, with small amendments incorporated below. The referee added Propositions 11 to 14 and Remark 15, which widen the scope in both directions; the ratio in Proposition 14 was added by Claude.

**Scope.** Within the three-generation, zero-Majorana, top-only family at fixed cutoff, these results refute an infinitely heavy W, Z, t limit for the stated quadratic

mass definitions. They do not rule out other finite data (Proposition 14), a different action or constraint, a joint limit with Lambda -> infinity, or a quantum effective theory. They do not show unbroken SU(2) x U(1) on the light space (Remark 15).

For the heat cutoff, on $\mathbb{R}^4$, we obtain the exact quadratic vector form without expanding in $m/\Lambda$. In the three-generation Standard Model finite representation with only the top Dirac mass nonzero, both the mass defined by the $q^0/q^2$ Taylor ratio and the unique positive time-like pole mass satisfy $M_W \sim \Lambda^2/m_t$ as $m_t/\Lambda \to \infty$ at fixed $\Lambda$. The broken vectors consequently do not decouple by becoming heavy, although the heavy fermion contribution disappears from the gauge coefficients. These statements concern the classical bosonic quadratic action about a prescribed constant background; moreover, the positive Laplace class considered below does not itself select a finite nonzero stationary background when radial variations of that background are allowed.

**1. Theorem (exact quadratic kernel).**

Let

$$D_A = -i\gamma^\mu(\partial_\mu + iA_\mu) + \gamma_5 V, \qquad V = V_0 = V_0^\dagger,$$

on $\mathbb{R}^4$, with Hermitian Euclidean gamma matrices, $\{\gamma^\mu, \gamma^\nu\} = 2\delta^{\mu\nu}$, $\gamma_5^2 = 1$, and four spin components. Let the finite trace be the literal trace on $H_F$, including every stated multiplicity, without a subsequent projection; this agrees with the bosonic trace of Chamseddine-Connes-Marcolli, which counts particles, antiparticles and four spin components. Take $A$ smooth and compactly supported.

We use the vacuum-subtracted action, or equivalently its quadratic variation, since the constant-background trace on $\mathbb{R}^4$ has infinite volume. Write

$$A_\mu(q) = a_\mu(q)T, \qquad T = T^\dagger, \qquad q^\mu a_\mu(q) = 0,$$
$$\int_q = \int \frac{d^4q}{(2\pi)^4}.$$

Diagonalize $V_0$:

$$V_0 e_i = m_i e_i, \qquad m_i \in \mathbb{R}.$$

The signed eigenvalues are needed; their squares alone do not determine the answer.

For $S_t = \operatorname{Tr} e^{-tD_A^2}$, define

$$b(u) = u(1-u), \quad d_{ij} = m_i - m_j,$$

$$I_t(a,b;q^2) = \int_0^1 e^{-t[(1-u)a+ub+u(1-u)q^2]}\, du.$$

Then

$$S_t^{(2)} = \int_q |a_T(q)|^2\, C_{t,T}(q^2),$$

where

$$\boxed{C_{t,T}(q^2) = \frac{1}{8\pi^2}\sum_{ij}|T_{ij}|^2 \left[\left(\frac{2}{t} + d_{ij}^2 + q^2\right) I_t(m_i^2, m_j^2; q^2) - \frac{e^{-tm_i^2} + e^{-tm_j^2}}{t}\right].} \tag{1}$$

For several generators, replace $|T_{ij}|^2$ by $\operatorname{Re}((T_a)_{ij}(T_b)_{ji})$.

In particular, writing $C(q^2) = C_0 + q^2 C_1 + O(q^4)$, we obtain

$$\boxed{C_{0,t} = \frac{1}{8\pi^2}\sum_{ij}|T_{ij}|^2 \left[\left(\frac{2}{t} + d_{ij}^2\right) I_t(m_i^2, m_j^2; 0) - \frac{e^{-tm_i^2} + e^{-tm_j^2}}{t}\right],} \tag{2}$$

and

$$\boxed{C_{1,t} = \frac{1}{8\pi^2}\sum_{ij}|T_{ij}|^2 \int_0^1 e^{-t[(1-u)m_i^2+um_j^2]}\big[1 - 2b(u) - td_{ij}^2 b(u)\big]\, du.} \tag{3}$$

These formulas also apply, after integration over $t = s/\Lambda^2$, to

$$f(z) = \int_{(0,\infty)} e^{-sz}\, d\nu(s), \quad 0 < f(0) = \nu((0,\infty)) < \infty, \quad \int (s^{-2} + s^2)\, d\nu(s) < \infty, \tag{4}$$

where $\nu$ is positive. These hypotheses suffice for the Euclidean action density and the Taylor coefficients used below. They do not, by themselves, guarantee analytic continuation to every negative $q^2$.

*Proof.* With $\nabla_\mu = \partial_\mu + iA_\mu$,

$$D_A^2 = -\nabla^2 + V_0^2 - \frac{i}{2}\gamma^{\mu\nu}F_{\mu\nu} + \gamma^\mu\gamma_5[A_\mu, V_0].$$

Let $P_0 = -\partial^2 + V_0^2$ and write $D_A^2 = P_0 + P_1 + P_2$, according to degree in $A$. Duhamel expansion and cyclicity give

$$S_t^{(2)} = -t\,\mathrm{Tr}(e^{-tP_0}P_2) + \frac{t^2}{2}\int_0^1 \mathrm{Tr}\,\left(e^{-(1-u)tP_0}P_1 e^{-utP_0}P_1\right) du. \tag{5}$$

The linear vertex from $j, p$ to $i, p+q$ is

$$T_{ij}\left[(2p+q)\cdot a + \gamma^{\mu\nu}q_\mu a_\nu + (m_j - m_i)\gamma^\mu\gamma_5 a_\mu\right].$$

Completing the momentum square produces

$$(4\pi t)^{-2}e^{-t[(1-u)m_i^2 + um_j^2 + b(u)q^2]}.$$

For transverse $a$, the averaged spin trace of the vertex product is

$$4\left(\frac{2}{t} + d_{ij}^2 + q^2\right)|a|^2.$$

The $P_2$ term contributes

$$-\frac{1}{4\pi^2 t}\sum_{ij}|T_{ij}|^2 e^{-tm_i^2}|a|^2.$$

Symmetrizing in $i, j$ proves (1). Differentiation proves (2)–(3).

The divided-difference structure is explicit:

$$I_t(a,b;0) = \frac{e^{-ta} - e^{-tb}}{t(b-a)}, \qquad I_t(a,a;0) = e^{-ta}, \tag{6}$$

and

$$\int_0^1 u(1-u)e^{-t[(1-u)a+ub]}\,du = \frac{1}{t^2}\partial_a\partial_b I_t(a,b;0). \tag{7}$$

Thus (2) uses a first divided difference, while (3) also uses its mixed derivative. The full momentum dependence follows from the convergent series

$$I_t(a,b;q^2)=\sum_{n=0}^{\infty}\frac{(-q^2/t)^n}{n!}\partial_a^n\partial_b^n I_t(a,b;0). \tag{8}$$

Repeated-node values are obtained continuously.

For completeness, the mass coefficient for (4) can be written entirely in dimensionless variables. Put

$$r_i=m_i/\Lambda,\quad x_i=r_i^2,\quad z(u)=(1-u)x_i+ux_j,$$
$$\Phi_1(z)=\int s^{-1}e^{-sz}\,d\nu(s).$$

The bracket replacing that in (2) is

$$\Lambda^2\left[2\int_0^1\Phi_1(z(u))\,du+(r_i-r_j)^2\int_0^1 f(z(u))\,du-\Phi_1(x_i)-\Phi_1(x_j)\right]. \tag{9}$$

The corresponding bracket in (3) is

$$\int_0^1\left[(1-2b)f(z)+(r_i-r_j)^2bf'(z)\right]du. \tag{10}$$

Each unweighted integral is a divided difference of an antiderivative. The moment assumptions justify the integrations and differentiations. □

On a finite flat torus, (5) remains exact, but the Gaussian momentum integral must be replaced by the spin-structure-dependent lattice sum. Explicitly, for a transverse polarization $a$, its two-vertex contribution per unit volume is

$$\frac{2t^2}{\mathrm{Vol}}\sum_{ij,p}|T_{ij}|^2\int_0^1 e^{-t[(1-u)(p^2+m_i^2)+u((p+q)^2+m_j^2)]}$$
$$\left(|(2p+q)\cdot a|^2+(d_{ij}^2+q^2)|a|^2\right)du,$$

and the quadratic-vertex contribution is

$$-\frac{4t}{\mathrm{Vol}}\sum_{ij,p}|T_{ij}|^2e^{-t(p^2+m_i^2)}|a|^2.$$

Consequently, the continuum formulas below are statements on $\mathbb{R}^4$, or in the infinite-volume limit. A finite

torus can have additional holonomy dependence, including for constant commuting connections.

**2. Lemma (gauge invariance and the Goldstone direction).**

Let scalar fluctuations include the gauge orbit of $V_0$. Then the nonzero vector coefficient $C_0$ in (2) is compatible with gauge invariance.

For one generator, parameterize the orbit by

$$V(x) = e^{i\pi(x)T} V_0 e^{-i\pi(x)T}.$$

With our conventions,

$$a_\mu \longmapsto a_\mu - \partial_\mu \alpha, \qquad \pi \longmapsto \pi + \alpha.$$

The invariant combination is

$$B_\mu = a_\mu + \partial_\mu \pi. \tag{11}$$

*Proof.* Gauge covariance gives

$$D_{A^U, V^U} = U D_{A,V} U^{-1}.$$

Applying $U = e^{-i\pi T}$ transforms $V$ to $V_0$ and $a$ to $B$. Hence

$$S[a, e^{i\pi T} V_0 e^{-i\pi T}] = S[B, V_0]. \tag{12}$$

This establishes the Goldstone completion directly, without a derivative expansion.

For comparison, the longitudinal vector kernel is obtained from (1) by replacing the $q^2$ in its prefactor by $(1-2u)^2 q^2$ inside the integral. If $[T, V_0] = 0$, its integral vanishes by

$$\int_0^1 [2 + tq^2(1-2u)^2] e^{-tq^2 u(1-u)}\, du = 2.$$

For a broken generator the longitudinal kernel need not vanish; it belongs to the invariant combination (11).

If $V_0$ is not stationary, a Cartesian scalar Hessian alone need not annihilate the linear gauge-orbit tangent. The quadratic term in $e^{i\pi T} V_0 e^{-i\pi T}$, contracted with the scalar tadpole, must also be retained. Equation (12) includes

that term. Transverse vectors do not mix quadratically with scalar fluctuations on this translation-invariant background. □

**3. Corollary (the asymptotic normalization).**

Let all finite eigenvalues be small relative to $\Lambda$, let $f$ satisfy (4), and let $Q$ below be positive definite on the chosen generator space (otherwise pass to the quotient by its null space). Define

$$Q_{ab} = \mathrm{Re}\,\mathrm{Tr}(T_a T_b), \qquad K_{ab} = \mathrm{Re}\,\mathrm{Tr}([T_a, V_0]^\dagger [T_b, V_0]).$$

At leading order,

$$C_0 = \frac{f(0)}{8\pi^2} K, \qquad C_1 = \frac{f(0)}{12\pi^2} Q,$$

and therefore

$$\boxed{\mathcal{M}^2_{\mathrm{Taylor}} = \frac{3}{2} Q^{-1/2} K Q^{-1/2}.} \tag{13}$$

This reproduces Proposition 13.5.

For the full three-generation Standard Model representation,

$$Q_{11} = 12, \qquad K_{11} = \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2),$$

so

$$\boxed{8M_W^2 = \sum_{\text{generations}} (m_\nu^2 + m_e^2 + 3m_u^2 + 3m_d^2).} \tag{14}$$

*Proof.* In (2), the orbital expression

$$\frac{2}{t} I_t(a, b; 0) - \frac{e^{-ta} + e^{-tb}}{t}$$

starts at order $t(a-b)^2$. The leading term is consequently $d_{ij}^2$. In (3), the leading integral is $\int_0^1 (1 - 2u(1-u))du = 2/3$, proving (13).

For $T_1 = \sigma_1/2$ on each left doublet, one generation of particles contributes 2 to $Q_{11}$. Charge conjugates and three generations give 12. A doublet with Dirac masses $m_u, m_d$ contributes $\frac{1}{2}(m_u^2 + m_d^2)$ to the particle commutator trace, before color multiplicity. This proves (14).

Equivalently, the gauge coefficient is

$$\frac{f(0)g_2^2}{2\pi^2} = \frac{1}{4}.$$

For comparison with CCM's $\operatorname{Tr} F(D/\Lambda)$, take $F(v) = f(v^2)$; then $F(0) = f(0)$. The normalization and mass constraint are precisely equations (4.10) and (4.16), with the latter stated in Lemma 4.6 of the checked preprint. CCM describes the relation as "valid at unification scale." Chamseddine–Connes–Marcolli, pp. 37–39. □

The leading bound $M^2 \le 6\|V_0\|^2$ in Lemma 14 follows from (13) and $K \le 4\|V_0\|^2 Q$. Formula (13), rather than an exact finite-cutoff identity, is its applicable approximation.

**4. Proposition (the exact charged-vector Taylor mass).**

Let the finite data be the full three-generation Standard Model representation, including charge conjugates. Let the Majorana matrix and every Dirac mass except one up-type color triplet be zero. Write

$$m = \mu\Lambda, \qquad x = \mu^2, \qquad r = e^{-x}.$$

Let $f(z) = e^{-z}$. Define

$$H(y) = \int_0^1 e^{-yu(1-u)}\,du,$$

$$g(y) = (2+y)H(y) - 2,$$

$$h_x(y) = (2+x+y)\int_0^1 e^{-xu-yu(1-u)}\,du - 1 - e^{-x}. \quad (15)$$

Then the full charged-vector kernel is

$$\boxed{C_W(q^2) = \frac{\Lambda^2}{4\pi^2}\left[\frac{9}{2}g(y) + \frac{3}{2}h_x(y)\right], \qquad y = q^2/\Lambda^2.} \quad (16)$$

Define

$$A(x) = h_x(0) = \frac{2[1-(1+x)e^{-x}]}{x},$$

$$w(x) = 3\int_0^1 u^2 e^{-xu}\,du = \frac{3[2-(x^2+2x+2)e^{-x}]}{x^3}. \quad (17)$$

Continuous values are understood at $x = 0$. The particle-trace kinetic coefficient is

$$K_W(x) = \frac{9}{2} + \frac{3}{2}w(x) > 0.$$

The Taylor mass, defined by $M^2_{W,\mathrm{Taylor}} = C_W(0)/C'_W(0)$, is

$$\boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{9A(x)}{4K_W(x)} = \frac{3[1-(1+x)e^{-x}]}{x[3+w(x)]}.} \tag{18}$$

Its limiting behavior is

$$\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{3}{8}x - \frac{23}{128}x^2 + O(x^3), \qquad \frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{1}{x} \quad (x \to \infty). \tag{19}$$

Consequently,

$$\frac{m}{M_{W,\mathrm{Taylor}}} \longrightarrow \sqrt{\frac{8}{3}} \quad (\mu \to 0), \qquad \frac{m}{M_{W,\mathrm{Taylor}}} \sim \mu^2 \quad (\mu \to \infty). \tag{20}$$

*Proof.* On the particle space, $Q_{11} = 6$. The selected quark doublet contributes $3/2$, while the remaining doublets contribute $9/2$. Diagonalizing the massive left-right block produces eigenvalues $+m, -m$. The charged generator connects these to the massless down states; every such pair has $d^2_{ij} = m^2$. Formula (1) therefore gives $h_x$ for the selected doublet and $g$ for the remaining doublets. Charge conjugation supplies the factor two in (16).

Integration by parts gives

$$h_x(y) = 2\int_0^1 u\,[x + 2y(1-u)]e^{-xu-yu(1-u)}\,du, \tag{21}$$

and

$$\boxed{h'_x(y) = 2\int_0^1 u^2[1-y(1-u)]e^{-xu-yu(1-u)}\,du.} \tag{22}$$

The boundary term proving (22) is $\partial_u[u(1-u)e^{-xu-yu(1-u)}]$. Thus

$$h'_x(0) = \frac{2}{3}w(x), \qquad g'(0) = \frac{2}{3}.$$

This proves (18). Expanding the elementary expressions at zero and taking their large-$x$ limits proves (19)–(20). □

The same Taylor result extends to every $f$ in (4). Put

$$\ell = f(0), \qquad w_f(x) = \int w(sx)\, d\nu(s),$$
$$A_f(x) = \int \frac{A(sx)}{s}\, d\nu(s).$$

Then

$$\boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{9A_f(x)}{4\left(\frac{9}{2}\ell + \frac{3}{2}w_f(x)\right)}.} \tag{23}$$

All denominators are positive, and dominated convergence gives

$$M^2_{W,\mathrm{Taylor}} \sim \frac{3}{8}m^2 \quad (x \to 0),$$
$$\boxed{\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} \sim \frac{\int s^{-2}\, d\nu(s)}{\ell x} \quad (x \to \infty).} \tag{24}$$

Indeed, $A(z)/z \to 1$, $w(z) \to 1$ at zero, while $zA(z) \to 2$, $w(z) \to 0$ at infinity.

If the other Dirac masses are merely small and nonzero, the answer depends on those additional parameters and is given by (1). In particular, the vanishing limit in (19) assumes they are zero: masses of surviving light doublets can leave a residual vector mass.

**5. Theorem (the charged-vector pole).**

Under the heat-cutoff hypotheses of Proposition 4, define a timelike pole by analytic continuation of the quadratic inverse propagator:

$$C_W(-M^2_{W,\mathrm{pole}}) = 0. \tag{25}$$

For every $x > 0$, this equation has exactly one solution with $M^2_{W,\mathrm{pole}} > 0$. It is simple and has positive residue in the convention that the Euclidean propagator near the pole is a positive multiple of $(q^2 + M^2_{W,\mathrm{pole}})^{-1}$.

Let $z(x) = M^2_{W,\mathrm{pole}}/\Lambda^2$. Its exact specification is the unique positive root

$$\boxed{\frac{9}{2}g(-z) + \frac{3}{2}h_x(-z) = 0.} \tag{26}$$

Furthermore,

$$z(x) \sim \frac{3}{8}x \quad (x \to 0), \qquad z(x) \sim \frac{1}{x} \quad (x \to \infty), \tag{27}$$

and both mass definitions obey

$$\boxed{0 < M^2_{W,\mathrm{Taylor}},\ M^2_{W,\mathrm{pole}} \leq \frac{\Lambda^2}{e}.} \tag{28}$$

*Proof.* Let

$$F_x(y) = \frac{9}{2}g(y) + \frac{3}{2}h_x(y).$$

Equation (22) implies $h'_x(y) > 0$ for $y \leq 0$, and also $g'(y) > 0$, since $g = h_0$. Hence $F_x$ is strictly increasing on that half-axis. We have

$$F_x(0) = \frac{3}{2}A(x) > 0.$$

As $y \to -\infty$, the prefactors in (15) become negative and both integrals grow exponentially, so $F_x(y) \to -\infty$. This proves existence, uniqueness, simplicity, and the residue sign.

For $y \leq 0$, equation (22) gives $g'(y) \geq 2/3$, and therefore

$$F'_x(y) \geq 3.$$

The mean-value theorem yields

$$z(x) \leq \frac{1}{2}A(x).$$

The Taylor denominator gives the same bound for $M^2_{W,\mathrm{Taylor}}/\Lambda^2$. Finally,

$$A(x) = 2\int_0^1 xu\, e^{-xu}\, du \leq \frac{2}{e},$$

proving (28).

As $x \to 0$, the bound implies $z(x) \to 0$, and $F'_x(y) \to 4$ near the root. As $x \to \infty$, $z(x) \to 0$, $h'_x(y) \to 0$ near the root, and $F'_x(y) \to 3$. Together with $F_x(0) \sim 3x/2$ and $F_x(0) \sim 3/x$, respectively, these prove (27). □

The pole mass governs the linearized dispersion relation of this analytically continued quadratic action. The Taylor mass governs its two-derivative truncation; they need

not coincide near the cutoff. We assert no closed-form inversion of (26), no classification of complex poles, and no Lorentzian unitarity theorem.

The pole proof also extends to positive measures supported in a compact subset of $(0,\infty)$, using

$$F_{f,x}(y) = \int \frac{1}{s}\left[\frac{9}{2}g(sy) + \frac{3}{2}h_{sx}(sy)\right] d\nu(s).$$

It gives the analogues of (24) for the pole mass. For the larger class (4), only the Euclidean and Taylor conclusions have been established here; its moment assumptions alone do not justify that continuation.

**6. Proposition (unbroken coefficients, neutral vectors, and weak angles).**

Retain the heat-cutoff Standard Model family of Proposition 4. Normalize color generators by $\mathrm{tr}_{\mathbf{3}}\, t_a t_b = \frac{1}{2}\delta_{ab}$, and use $Q = T_3 + Y$. Define the full-trace coefficients by

$$S^{(2)}_{\text{unbroken}} = \int \left(\kappa_{\text{em}} F_{\text{em}}^2 + \kappa_s \sum_a G_a^2\right).$$

Then

$$\boxed{\begin{aligned} &\kappa_{\text{em}} = \frac{1}{24\pi^2}\frac{16}{3}(5+r), \qquad \kappa_s = \frac{1}{24\pi^2}2(5+r), \\ &\frac{\kappa_{\text{em}}}{\kappa_s} = \frac{8}{3}. \end{aligned}} \tag{29}$$

Both coefficients approach 5/6 of their all-massless values.

Let $K$ denote the particle-trace neutral $q^2$ coefficient in coordinates $(A_3, A_Y)$. Then

$$\boxed{\begin{aligned} K_{33} &= \frac{21}{4} + \left(\frac{3}{4} - \frac{3x}{8}\right) r, \\ K_{YY} &= \frac{103}{12} + \left(\frac{17}{12} - \frac{3x}{8}\right) r, \\ K_{3Y} &= -\frac{1}{4} + \left(\frac{1}{4} + \frac{3x}{8}\right) r. \end{aligned}} \tag{30}$$

The matrix $K$, the charged coefficient $K_W$, and the unbroken coefficients are positive for every $x \geq 0$.

Define the neutral-current angle from the current of the massive direction after diagonalizing the two-derivative neutral kinetic form:

$$J_Z \;\propto\; T_3 - \sin^2\theta_{\mathrm{NC}}\, Q.$$

Then

$$\boxed{\cos^2\theta_{\mathrm{NC}} = \frac{K_{YY} + K_{3Y}}{K_{33} + 2K_{3Y} + K_{YY}} = \frac{5}{8}.} \tag{31}$$

A different continuation, namely

$$\cos^2\theta_{\mathrm{match}} := \frac{K_{YY}}{K_W + K_{YY}},$$

satisfies

$$\boxed{\cos^2\theta_{\mathrm{match}}(0) = \frac{5}{8}, \qquad \cos^2\theta_{\mathrm{match}}(\infty) = \frac{103}{157}.} \tag{32}$$

Thus these definitions cease to describe one common electroweak angle.

*Proof.* Photon and color generators commute with $V_0$. Their coefficients therefore follow from (3) with $d_{ij} = 0$ on each invariant mass block. The charge traces give (29).

A Dirac species of mass $m$, with neutral charges $q_L^a, q_R^a$, contributes to $K$

$$e^{-x}\left[q_L^a q_L^b + q_R^a q_R^b - \frac{x}{2}(q_L^a - q_R^a)(q_L^b - q_R^b)\right]. \tag{33}$$

Applying (33) to the three top colors, and retaining all other states, proves (30).

Write

$$v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad B_\infty = \begin{pmatrix} 21/4 & -1/4 \\ -1/4 & 103/12 \end{pmatrix},$$
$$U = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 17/12 \end{pmatrix}.$$

Then

$$K = B_\infty + rU - \frac{3xr}{8} vv^T.$$

We have $B_\infty \geq 5I$, $U \geq 0$, and $xr \leq 1/e$. Hence

$$K \geq \left(5 - \frac{3}{4e}\right) I > 0.$$

Positivity of $K_W$ follows from (17).

The neutral mass direction is $v$, while the photon direction is $(1,1)$. The kinetic-orthogonal massive field is proportional to $K^{-1}v$, giving the first expression in (31). Directly,

$$K_{33} + K_{3Y} = 5 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(5 + r),$$

which proves (31). Taking limits proves (32). □

The neutral mass also tends to zero. More precisely,

$$\boxed{\frac{M^2_{Z,\mathrm{Taylor}}}{\Lambda^2} = \frac{9}{4} x r \, v^T K^{-1} v \sim \frac{2}{3} x e^{-x} \quad (x \to \infty).} \tag{34}$$

For $x > 0$ there is likewise one positive massive neutral pole, besides the photon pole. To prove this, put $B = B_\infty + rU$. The exact neutral kernel, apart from the common factor $\Lambda^2/(4\pi^2)$, is

$$B\, g(y) + \frac{3}{2} x r \, H(y) v v^T. \tag{35}$$

After congruence by $B^{-1/2}$, its massive eigenvalue is

$$g(y) + \beta H(y), \qquad \beta = \frac{3}{2} x r \, v^T B^{-1} v > 0.$$

Its timelike root obeys

$$z - 2 + \frac{2}{H(-z)} = \beta. \tag{36}$$

The left side increases from zero to infinity, because

$$\frac{d}{dz}\left( z - 2 + \frac{2}{H(-z)} \right) = 1 - \frac{2\,\mathbb{E}_z[b]}{H(-z)} \geq \frac{1}{2},$$

where $\mathbb{E}_z$ uses density proportional to $e^{zb(u)}$. The pole is therefore unique and has positive residue. Since $v^T B_\infty^{-1} v = 8/27$,

$$\frac{M^2_{Z,\mathrm{pole}}}{\Lambda^2} \sim \frac{2}{3} x e^{-x}.$$

In particular, $M_W^2/M_Z^2$ does not continue to equal $5/8$.

For every $f$ in (4), equations (29)–(33) extend by replacing 1 by $\ell = f(0)$, $r$ by $f(x)$, and $xr$ by $-xf'(x)$.

Positivity follows by averaging the positive heat-cutoff matrices. The ratio 8/3, the neutral-current value 5/8, and the matching limit 103/157 persist. At the endpoint both neutral masses vanish, so the distinguished massive neutral direction—and hence its angle—requires continuation from finite $x$.

**7. Counterexample (positive cutoff does not ensure positive broken kinetic coefficients).**

Let $H_F = \mathbb{C}^2$,

$$V_0 = m\sigma_3, \qquad T = \sigma_1, \qquad x = m^2/\Lambda^2,$$

with the same spacetime and trace conventions. For the heat cutoff,

$$C_T(q^2) = \frac{\mathrm{Tr}(T^2)\Lambda^2}{8\pi^2} e^{-x} \left[(2 + 4x + y)H(y) - 2\right]. \quad (37)$$

Its $q^2$ Taylor coefficient is proportional to

$$e^{-x}(1 - x).$$

It vanishes at $x = 1$ and becomes negative for $x > 1$. Thus the two-derivative truncation has a wrong-sign vector kinetic term in this example.

Nevertheless, the full quadratic form has a unique positive real timelike pole with positive residue, satisfying

$$\boxed{4x < \frac{M^2_{\mathrm{pole}}}{\Lambda^2} < 4x + 2.} \quad (38)$$

Consequently $M_{\mathrm{pole}} \sim 2m$ diverges at fixed cutoff.

*Proof.* The generator connects the eigenvalues $m, -m$, so $d^2_{ij} = 4m^2$ while both squared eigenvalues equal $m^2$. Substitution in (1) proves (37). Its Taylor ratio is

$$M^2_{\mathrm{Taylor}} = \frac{6m^2}{1 - x},$$

where this is a positive mass only for $x < 1$.

The pole equation is (36) with $\beta = 4x$. The monotonicity proof there gives uniqueness and the residue sign. Evaluating (37) at $y = -4x$ gives $2H(-4x) - 2 > 0$, whereas at $y = -4x - 2$ it gives $-2$; this proves (38).

For general $f$ satisfying (4), the Taylor kinetic coefficient in this example is proportional to

$$f(x) + xf'(x) = \frac{d}{dx}[xf(x)]. \tag{39}$$

It must be negative somewhere: $xf(x) > 0$ for $x > 0$, but $xf(x) \to 0$ at infinity under (4). A nonnegative derivative everywhere would contradict these facts. □

This counterexample concerns arbitrary finite data, not the top-only Standard Model family. It also distinguishes a wrong-sign **Taylor truncation** from a negative-residue pole of the **full** form factor. We have not proved absence of complex-pole or other nonlocal problems in either model.

**8. Counterexample (heavy fermions do not dynamically select the centralizer).**

Let the data be those of Proposition 4, and let

$$\mathfrak{g}_{V_0} = \{X \in \mathfrak{g}_{\mathrm{SM}} : [X, V_0] = 0\}.$$

For nonzero top mass this is the represented unbroken gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{u}(1)_{\mathrm{em}}$. The limit $x \to \infty$ at fixed $\Lambda$ does **not** produce a theory containing only these vector directions by quadratic mass decoupling.

*Proof.* For fixed $y$, equation (21) implies

$$h_x(y) \longrightarrow 0.$$

Consequently

$$\boxed{C_W(q^2) \longrightarrow \frac{\Lambda^2}{4\pi^2}\frac{9}{2}g(q^2/\Lambda^2).} \tag{40}$$

For positive $y$,

$$g(y) = 4y \int_0^1 u(1-u)e^{-yu(1-u)}\,du > 0.$$

Thus a charged generator outside $\mathfrak{g}_{V_0}$ retains a finite nonzero quadratic kinetic form and a nonvanishing propagator at every fixed positive momentum. Its mass approaches zero. Equation (35) similarly leaves the positive matrix $B_\infty g(y)$, including the additional neutral direction. These facts contradict decoupling of the broken

vectors through diverging masses or vanishing propagators. □

A narrower decoupling statement is true. If we **restrict the connection from the outset** to the centralizer, the finite mass blocks are invariant and the heat trace factorizes. On the heavy block its contribution carries $e^{-sm^2/\Lambda^2}$, which tends to zero. The restricted action therefore converges to the light-block spectral action. This proves decoupling of a finite fermion block under that restriction; it does not prove dynamical elimination of the other gauge fields.

The amendment to Lemma 03 Part C is therefore:

- Its transverse kinetic weights, including $w_L(x,0) = w(x)$, and its two weak-angle formulas are correct under their stated conventions.
- They must be accompanied by the mass coefficient (2), including the quadratic-vertex subtraction.
- A centralizer-only endpoint does not follow from those weights. In the present exact quadratic family, the broken-vector masses vanish while their remaining kinetic coefficients are positive.

**9. No-go theorem (finite radial vacuum for the positive Laplace class).**

Let $f$ satisfy (4), and suppose the allowed scalar configurations include

$$V = cV_*, \qquad c > 0, \qquad V_* \neq 0,$$

with $A = 0$. Let the action be the literal spectral action, without an added scalar potential or a constraint fixing $c$. Then no finite $c > 0$ is stationary along this radial direction.

*Proof.* On $\mathbb{R}^4$, the constant-background action density is

$$\mathcal{U}(c) = \frac{\Lambda^4}{4\pi^2} \int s^{-2} \operatorname{Tr}_{H_F} e^{-sc^2 V_*^2/\Lambda^2}\, d\nu(s).$$

Differentiating gives

$$\boxed{\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \int s^{-1} \operatorname{Tr}_{H_F} \left( V_*^2 e^{-sc^2 V_*^2/\Lambda^2} \right) d\nu(s) < 0.} \tag{41}$$

The inequality is strict because $V_* \neq 0$ and $\nu$ is nonzero and positive. The same argument applies on a finite flat torus before performing its positive momentum sum. □

Thus the masses above are conditional quadratic-background quantities. In the top-only family, the ordinary Higgs radial variation supplies the direction used in this theorem. A finite symmetry-breaking minimum of the polynomial asymptotic scalar potential cannot automatically be regarded as a minimum of this full positive-Laplace spectral action.

**10. Corollary (comparison of heavy limits).**

Let the background family and mass definitions be those above.

At fixed $\Lambda$, with $\mu \to \infty$:

| quantity, heat cutoff | behavior |
|---|---|
| $m_t$ | $\mu\Lambda \to \infty$ |
| $M_{W,\mathrm{Taylor}},\ M_{W,\mathrm{pole}}$ | $\sim \Lambda/\mu \to 0$ |
| $M_{Z,\mathrm{Taylor}},\ M_{Z,\mathrm{pole}}$ | $\sim \sqrt{2/3}\,\Lambda\mu e^{-\mu^2/2} \to 0$ |
| $m_t/M_W$ | $\sim \mu^2 \to \infty$ |

(42)

At fixed $0 < \mu < \infty$, with $\Lambda \to \infty$, all four vector masses in (42) instead grow linearly with $\Lambda$, as does $m_t$. A controlled asymptotic family can also satisfy

$$\Lambda \to \infty, \qquad \mu \to 0, \qquad \mu\Lambda \to \infty;$$

then $M_W, m_t \to \infty$ while

$$m_t/M_W \to \sqrt{8/3}.$$

*Proof.* The first assertions follow from (19), (27), and (34)–(36). At fixed positive $\mu$, their dimensionless mass functions are fixed and positive, so multiplication by $\Lambda$ proves the joint-limit assertions. The final assertion follows from the small-$\mu$ limits. □

Accordingly, the obstruction has three distinct scopes. The CCM identity cannot be extrapolated to $\mu \to \infty$ at fixed cutoff because it belongs to the asymptotic truncation. The exact top-only quadratic action obstructs a fixed-cutoff heavy-$W$ limit for the heat cutoff under

both mass definitions, and for class (4) under the Taylor definition. A family of actual nonzero stationary vacua of the unmodified positive-Laplace action encounters the separate obstruction (41). Counterexample 7 excludes extending the first of these exact mass conclusions to arbitrary finite representations.

**11. Proposition (the radial derivative for general $f$; referee).** On flat $\mathbb{R}^4$, for a sufficiently integrable real cutoff $f$, with $\lambda_j^2$ the eigenvalues of $V_*^2$,

$$\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \sum_j \lambda_j^2 \int_{c^2\lambda_j^2/\Lambda^2}^{\infty} f(z)\, dz. \tag{43}$$

For nonnegative $f$, a finite $c > 0$ is stationary exactly when every tail associated with a nonzero $\lambda_j$ vanishes. So if $f$ has nonzero support arbitrarily far out, there is no finite stationary point on this ray; if $f$ has compact support ending at $R$, there is a stationary plateau once every nonzero $c^2\lambda_j^2/\Lambda^2 \geq R$. Sign-changing cutoffs can have cancellations. A nonconstant cutoff flat near zero lies outside (4), since a nonzero positive measure gives $f'(0) < 0$; (43) covers the nonnegative ones. A radial path with an additional fixed noncommuting mass matrix need not have the form $cV_*$. □

**12. Proposition (the top-only Taylor mass for general $f$; referee).** For nonnegative smooth compactly supported $f$ with $f(0) > 0$, the formulas of Proposition 4 hold with

$$A_f(x) = \frac{2}{x} \int_0^x z f(z)\, dz, \qquad w_f(x) = \frac{3}{x^3} \int_0^x z^2 f(z)\, dz,$$

by approximating $f$ with its derivatives by real combinations of $e^{-nz}$. Hence

$$\frac{M_{W,\mathrm{Taylor}}^2}{\Lambda^2} \sim \frac{\int_0^\infty z f(z)\, dz}{f(0)\, x}.$$

This includes cutoffs flat near zero. For a nonanalytic cutoff there need be no analytic continuation defining the pole. □

**13. Proposition (several heavy doublets; referee).** For the heat cutoff, let $P$ be the total particle-trace weight of complete massless doublets, and let heavy doublet $i$, of weight $d_i > 0$, have one Dirac member of mass

$m_i$ and one massless member. Then

$$C_W(y) = \frac{\Lambda^2}{4\pi^2}\Big(Pg(y) + \sum_i d_i h_{x_i}(y)\Big),$$
$$\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{3}{2}\,\frac{\sum_i d_i A(x_i)}{P + \sum_i d_i w(x_i)}.$$

For finitely many doublets, if $P > 0$ and all $x_i \to \infty$, the Taylor mass tends to zero. If $P = 0$ and all $x_i = x$, then $M^2_{W,\mathrm{Taylor}} \sim m^4/(2\Lambda^2)$, the form factor has a unique positive pole with $M^2_{W,\mathrm{pole}} \sim m^2/2$, and the whole unnormalized kernel tends to zero at every fixed Euclidean momentum. The divergent Taylor and pole scales differ, and neither alone establishes a healthy limiting gauge theory. □

**14. Proposition (the fully massive family; referee; a complete proof by Astra 6 is in the validation, item 11).** Give both members of every Standard Model doublet the same Dirac mass $m$, with no Majorana masses, and let $z_W, z_Z$ be the charged and massive neutral poles in units of $\Lambda^2$. The charged form factor is proportional to $e^{-x}[g(y) + 2xH(y)]$, and

$$2x < z_W < 2x + 2, \qquad \tfrac{16}{5}x < z_Z < \tfrac{16}{5}x + 2.$$

So both pole masses diverge with $m$ at fixed cutoff, and

$$\frac{M^2_{W,\mathrm{pole}}}{M^2_{Z,\mathrm{pole}}} \longrightarrow \frac{5}{8} \qquad (x \to \infty),$$

as also for $x \to 0$, where $z_W \sim 3x$ and $z_Z \sim \frac{24}{5}x$. In between the ratio differs, reaching about 0.66 near $x = 3$ (Python, solving the pole equations). The kernel tends to zero at every fixed Euclidean momentum, and the charged Taylor kinetic coefficient is proportional to $e^{-x}(1 - x/2)$, negative for $x > 2$.

*Proof.* The species with equal mass have eigenvalues $\pm m$. The charged generator connects them with $d^2 = 0$ and $d^2 = 4m^2$ in equal weights, which gives the charged form. For the neutral generators the diagonal charge has $d = 0$ and the off-diagonal charge $(q_L - q_R)/2 = T_3 v/2$ has $d^2 = 4m^2$; on the particle trace the kernel is $e^{-x}[\mathrm{diag}(6, 10)g + 12xHvv^T]$, so $\beta = 12x\, v^T \mathrm{diag}(6, 10)^{-1} v = \frac{16}{5}x$, and for the charged direction $\beta = 2x$. Equation (36) gives the bounds, since $H(-z) > 1$. □

This is a counterexample to any general no-go for diverging pole locations at fixed cutoff. It has no light fermions, and the physical quality of that limit is open.

**15. Remark (the light space does not carry unbroken $SU(2) \times U(1)$; referee).** In the top-only limit the broken masses vanish, but this does not restore the electroweak group on the whole light space. The light space retains $b_L$ after removing $t_L$. If $P_{\text{light}}$ projects off the top mass block, then on $b_L$ we have $P_{\text{light}} T_1 P_{\text{light}} = P_{\text{light}} T_2 P_{\text{light}} = 0$ and $P_{\text{light}} T_3 P_{\text{light}} = -\frac{1}{2}$, which do not obey the $SU(2)$ relations. Correspondingly $K_W = 9/2$, $K_{33} = 21/4$, $K_{3Y} = -1/4$ in the limit, while a linearly realized unbroken action would have equal weak coefficients and no mixing. The remaining complete massless doublets carry the ordinary representation, and their contribution is gauge invariant. If both members of every removed doublet are heavy, the retained light space can be invariant under the full group; a full limiting theory still needs control of the remaining fields, normalizations, interactions and vacuum. "The heavy fermion leaves the loop" is shorthand for the suppression in this classical trace.

The light-space statement should be read together with the separate role of the top (lemma 21): among single charged species only an up-type quark can be removed while keeping cos^2 theta_NC = 5/8 and the ratio 8/3, and removing it leaves the five other quarks and the leptons, on which SU(3) x U(1)_em acts with b_L paired with b_R.

**16. Remark (the CCM relation as the small-mass form of a weighted sum rule; Claude).** For the heat cutoff and doublets with one massive member, Proposition 13 reads

$$8\kappa M^2_{W,\text{Taylor}} = \sum_f N_f m_f^2 \, \rho(m_f^2/\Lambda^2),$$
$$\rho(x) = \frac{A(x)}{x} = \frac{2[1-(1+x)e^{-x}]}{x^2},$$

with $N_f = 2d_f$ and $\kappa = (P + \sum_i d_i w(x_i))/6$. As $x \to 0$, $\rho, \kappa \to 1$, which is the CCM relation (14). The relation is therefore the leading term of an expansion, and its range is $m \ll \Lambda$. For the top alone, $\kappa = (3+w)/4$; $m/M_W$ departs from $\sqrt{8/3}$ by 5% at $m \approx 0.45\Lambda$, and $M_W$ peaks

at about $0.52\Lambda$ at $m \approx 1.42\Lambda$ (Python). The maximum is unique, at $x = 2.024$ with $M_W/\Lambda = 0.5242$ (referee, by a sign argument and Python). A massive diagonal contribution to the trace is exponentially suppressed above the cutoff, while the mixed loop with a massless partner has the algebraic $\rho \sim 2/x^2$ at large $x$; a vacuum with no other mass leaves the W mass with no source. Whether the classical action is meaningful for $m > \Lambda$ is a separate question, on which these formulas do not decide. In the top-only heat-cutoff family, the ratio differs from $\sqrt{8/3}$ for every $x > 0$, since $3 + w - 4\rho$ vanishes at zero and is increasing, so the limit with $M_W$ and $m_t$ diverging together and $m_t/M_W \to \sqrt{8/3}$ needs $\Lambda \to \infty$ faster than $m_t$ (Corollary 10). This necessity is for that family: with one complete massless lepton doublet and one member of each other doublet of common mass $m$, the weighted sum equals the unweighted CCM sum at some $x_* > 0$ (referee).

As algebraic checks, we used **Python 3 with SymPy** to verify the integration-by-parts identities, the displayed small-$x$ Taylor series, and $v^T B_\infty^{-1} v = 8/27$. The arguments above provide the proofs; these computations introduce no fitted constants.

**Consequence for T5b.** At fixed cutoff, the specified top-only Standard Model spectral action does not realize an infinitely heavy $W, Z, t$ family: the top grows while both vector masses tend to zero, for the heat cutoff and the stated quadratic mass definitions. A joint limit with $\Lambda \to \infty$ permits all three background mass scales to diverge, but neither centralizer-only decoupling nor a stationary Higgs vacuum follows from that scaling. These conclusions concern the classical bosonic action and contain no quantum corrections or claims about the physical Standard Model at those scales. Other finite data behave differently: with every doublet heavy and no light fermions, the pole masses diverge at fixed cutoff with M_W^2/M_Z^2 -> 5/8 (Proposition 14). With a massless complete doublet retained, the Taylor mass tends to zero (Proposition 13). So in the families treated here, a heavy W and Z at fixed cutoff is found only without a massless complete doublet. Light nonzero masses are not treated: they leave a residual vector mass that depends on them.

# Part 21

## The separate role of the top: removal identities and counting

Prompted by the observation that the top has a separate role, which matters more than the loss of an electroweak partner for b_L. Proposition 21.1 and the counts below are exact calculations by Claude Opus 5.5, in Python with exact fractions. The readings in the last section are proposals, and nothing there is derived. Refereed by Terra 6.1 (effort max, 3 Nov 2026; report): one error, in Proposition 21.2, and several amendments, incorporated below.

**Outcome.** The spectral action detects the top: in the exact quadratic action of lemma 18, among single charged species only an up-type quark can be removed while keeping both the neutral-current value cos^2 theta_NC = 5/8 and the ratio 8/3 of the photon and gluon coefficients. Removing it leaves 84 of the 96 states of the finite Hilbert space. Those 84 are the seed of an SO(30), inside SO(32), whose adjoint contains the charged scalar counts of three generations, and 84 is also the number of on-shell components of the three-form of eleven-dimensional supergravity.

### Removal identities

For a Dirac species with charges q_L, q_R in the coordinates (T_3, Y), equation (33) of lemma 18 gives a contribution to the neutral kinetic matrix K with weight e^{-x}. The off-diagonal charge is (q_L - q_R)/2 = (T_3/2)(1, -1), which is orthogonal to the photon direction (1, 1). So the combinations K_33 + K_3Y and K_YY + K_3Y depend on x only through the factor e^{-x} of each massive species, and cos^2 theta_NC = (K_YY + K_3Y)/(K_33 + 2 K_3Y + K_YY) is determined by the sums of T_3 Q and Y Q.

**Proposition 21.1.** Per species, with colour multiplicity, on the particle trace for the first two columns and with charge conjugates and both chiralities for the last two:

| species | sum T_3 Q | sum Y Q | Q^2 coefficient | colour coefficient |
|---|---|---|---|---|
| up-type quark | 1 | 5/3 | 16/3 | 2 |
| down-type quark | 1/2 | 1/6 | 4/3 | 2 |
| charged lepton | 1/2 | 3/2 | 4 | 0 |
| neutrino | 0 | 0 | 0 | 0 |
| three generations | 6 | 10 | 32 | 12 |

Hence the ratios of the whole, 5/3 and 8/3, are those of an up-type quark alone. As one species is removed, that is, as its mass tends to infinity at fixed cutoff in the setting of lemma 18:

| removed | cos^2 theta_NC | photon/gluon coefficient |
|---|---|---|
| up-type quark | 5/8 | 8/3 |
| down-type quark | 59/92 | 46/15 |
| charged lepton | 17/28 | 7/3 |

So among single charged species only an up-type quark preserves both values. For an up-type quark they hold at every x, as in Proposition 6 of lemma 18. Within one generation, the down-type quark and charged lepton together contribute exactly what the up-type quark contributes: (1, 5/3; 16/3, 2).

*Proof.* Arithmetic with the Standard Model charges, Y = Q - T_3. The last statement is direct addition. ■

The scope is that of lemma 18: classical bosonic trace, with these definitions of the coefficients and of the angle, three generations, and the other species with the mass assignments of that calculation. Other definitions of the weak angle do not stay at 5/8 (Proposition 6 there). At the endpoint where the neutral mass vanishes, the massive neutral direction is no longer distinguished, so its angle is read as the limit from finite mass, as lemma 18 states.

**The state count in the action**

**Proposition 21.2.** With three generations and nu_R, dim H_F = 96 (Chamseddine-Connes-Marcolli). With only an up-type Dirac mass m and no Majorana mass, the constant-background density of lemma 18 has finite trace Tr exp(-V_0^2/Lambda^2) = 84 + 12 e^{-x} for the heat cutoff. So the coefficient of the vacuum density moves from 96 to 84 along the radial direction, with no finite nonzero stationary point on this free radial ray (No-go theorem 9 and Proposition 11 of lemma 18). This concerns the literal flat-space action with no added potential or constraint. Compact support can give a plateau; sign-changing cutoffs, additional terms, constraints and paths with an additional fixed noncommuting mass lie outside it. At m = 0 the derivative is zero.

*Proof.* The top block has 3 colours, 2 chiralities and charge conjugates, so 12 states, each with V_0^2 = m^2. ■

This also gives the top-only family a place in the radial problem, as the endpoint of a descending direction. A radial direction giving other species masses descends further, so this is the endpoint for that direction only.

**SO(30)**

Let the five quarks other than the top, in three colours, with their conjugates, form the vector 30 = (5, 3) + (5-bar, 3-bar) of SU(5) x SU(3) inside SO(30). The adjoint, of dimension 435, is

```
Lambda^2(30) = (24, 8) + (24, 1) + (1, 8) + (1, 1) +
↪  (10, 6) + (15, 3-bar) + conjugates of the last
↪  two.
```

With Q = 2/3 for u, c and Q = -1/3 for d, s, b, that is, Q = diag(3/5, 3/5, -2/5, -2/5, -2/5) + (1/15) I_5, which includes the U(1) commuting with SU(5), the charges are:

| representation | charge: number |
|---|---|
| (15, 3-bar) | -2/3: 6; +1/3: 6; +4/3: 3 |
| (10, 6) | -2/3: 3; +1/3: 6; +4/3: 1 |
| (24, 1) + (1, 1) | +1: 6; -1: 6; 0: 13 |

Three generations of scalar partners in a supersymmetric Standard Model with nu_R need, for antisquarks in the 3-bar, 6 of charge -2/3 and 6 of charge +1/3, and for sleptons 6 of charge +1, 6 of charge -1 and 12 neutral real states. So (15, 3-bar) and (24, 1) + (1, 1) contain exactly the charged counts. What is left over in them is three triplets of charge +4/3 and one neutral state. The counting of 15 diquarks and 25 mesons, with the charge +4/3 states, and the remark that the top is unable to form them, go back to an early preprint (hep-ph/0512065), which finds neither SO(30) nor a role for SO(32). What SO(30) adds is one group whose adjoint contains both families. It also sits in SO(32), with 496 -> (435, 1) + (1, 1) + (30, 2) under SO(30) x SO(2), which hints at a role for SO(32); we have not explored this. These are representation and charge counts: identifying the entries with physical scalar partners needs a field construction, spin and chirality assignments, interactions and dynamics.

### The two families of 84

**Proposition 21.3.** Count four states per Dirac fermion and colour, and take Dirac neutrinos. Then

```
96 = 84 + 12:   five quarks and all leptons (60 + 12
 ↪  + 12), and the top (12);
96 = 84 + 12:   all electrically charged fermions
 ↪  (72 + 12), and the neutrinos (12).
```

The two families of 84 overlap in 72 states, and each is the other with the top and the neutrinos exchanged. Without nu_R, the first family has 78 states and the second stays at 84. In eleven dimensions, the three-form C_3 has C(9, 3) = 84 on-shell components, and its dual C_6 has C(9, 6) = 84. The 84 in H_F is a complex finite-Hilbert-space dimension, and the three-form count is a real bosonic polarization count, so an identification needs a map of representations and degrees of freedom.

### Readings, as proposals

These are proposals to test, with no derivation behind them.

1. **Seiberg duality.** For SU(3) with five flavours, N_f = 5 lies in the conformal window 3 N_c/2

< N_f < 3 N_c, and the magnetic dual is SU(N_f - N_c) = SU(2) with five flavours and a 5 x 5 gauge-singlet meson chiral superfield coupled by W = M q q-tilde (N. Seiberg, Nucl. Phys. B 435 (1995) 129, arXiv:hep-th/9411149, checked by the referee). The removal of the top is what gives N_f = 5, and SU(2) is the magnetic group there. To become a statement this needs an N = 1 supersymmetric theory with the required massless matter and dynamics, for example an emergent one, with an operator and anomaly map. Five-flavour ordinary QCD alone does not supply the duality, and the magnetic SU(2) needs a separate identification with any electroweak group.

2. **Branes.** The first family of 84 could relate to C_3 or to an M2 or M5 brane, and the second family to the dual object. The exchange of the top and the neutrinos would then be the content of the duality on the Standard Model side. The counts match, and equality of the number is insufficient: a representation and dynamics must relate the finite fermionic states to the bosonic form or brane degrees of freedom. The pairing of item 1 is one candidate. The first family also needs Dirac neutrinos for its 84, and the second does not.
3. **Relation to the odd labels.** The removal of the top leaves the five quarks and the leptons, on which SU(3) x U(1)_em acts with b_L paired with b_R. So the loss of the electroweak partner of b_L has no role at the label 9, where SU(2) is absent. A statement here needs the required involution on this finite sector, including its action on D, J, the algebra and its opposite; the vectorlike charges alone do not establish it, and the neutral states still need treatment.

**Numerical check**

Python with exact fractions, run on 3 Nov 2026 in a temporary directory, not kept: all tables above, and the dimension 435 = 224 + 1 + 105 + 105 with 105 = 60 + 45 and 224 = 192 + 24 + 8.

# Part 11D, README, referee pending

## Eleven dimensions from massless W and Z

Branch `11D`, opened 3 Nov 2026. This folder studies how the limit M_W, M_Z -> 0 of hypothesis H could give an eleven-dimensional theory, with proposed connections to Witten's seven-manifolds and the M2 and M5 readings of lemma 21 and STRATEGY T12. The branch also contains the independent referee report and related ledger and status updates.

Attempted by Claude (Anthropic) on 3 Nov 2026. The Astra 6 audit of the same day (reviews/2026-11-03_astra_audit_d11.md, of snapshot b2ddf98) supplies the joint-limit family and corrections to Proposition 1.4 and Proposal 1.3. GPT-6.1 Terra has addressed the requested items analytically in the referee report, with amendments and explicitly conditional source comparisons. The full EPJC paper and several cited primary sources were unavailable, and the original localized-family numerical table was not recomputed. The general third-charge extension of the referee has been checked by Claude Sonnet 5.5. The original scoped no-go results had two-model support after the Terra report; Claude Sonnet 5.5 has since checked the Terra sections on files 01 to 04 a second time (second check), so the no-go results of those files have three model identifiers, the ledger's general-consensus threshold, each in its stated scope. Files 05 to 09 are being checked in the same way. Opening statements in the numbered files that say "referee pending" are historical; their adjacent referee notes and the table below give the current status. prompts/terra_check_11d.md is the referee request. The conversation in which the folder was written is recorded at https://claude.ai/code/cshk_01BpscANdX3GG8D45SQ1BxsaHviSe7ufxxCLNiNaDv4fW

### The picture

H reads the two sheets of the finite geometry as the minimal truncation of an interval with opposite chiral walls, and v -> 0 as the interval growing. The labels 9, 10 and 11 of H are KO labels; the heat dimension at the

11 endpoint is 5. To make 11 a metric count, the finite KO-6 factor is resolved into a six-dimensional base, B_6 = CP^2 x S^2, which has isometry SU(3) x SU(2) and is the base of Witten's spaces:

```
11 = 4 (spacetime) + 1 (the interval of H) + 6
↪  (B_6).
```

The following table summarizes two proposed completions of 4 + 6 and their brane readings. The $M^{111}$ identification uses the recalled fibration, and the physical dictionaries and the single-family construction remain open:

| | electric (M2) completion | magnetic (M5) completion |
|---|---|---|
| eleven-dimensional vacuum | AdS4 x Y_{3,2}, Y_{3,2} = M^{111}, a circle bundle over B_6 | AdS5 x CP^2 x S^2 |
| the interval of H | the fibre circle, antiperiodic, two walls at the fixed points of a reflection | the radial direction of AdS5, infrared end at z_IR ~ 1/v |
| SU(3) x SU(2) | isometries | isometries |
| hypercharge | the vector of C_3 on the harmonic two-form (b_2 = 1) | one combination of the two C_3 vectors (b_2 = 2) |
| v -> 0 | the M2-brane moves to the apex; AdS4 x Y_{3,2} with the Standard Model group unbroken | the infrared end moves to infinity |

**Results**

| # | result | status | file |
|---|---|---|---|
| 1.1 | The chiral interval of length l is unitarily the scalar antiperiodic circle of radius R = l/pi; the grading is reflection, the walls its fixed points, m = 1/(2R) | Proposition independently checked, with counting clarification | 01; referee |
| audit §3, §4 | Joint family: D_F = epsilon D_*, l ~ epsilon^{-alpha}, alpha < 1, gives a finite window and an opening tower; a localized variant uses a bulk mass and logarithmic length factor | Window, Weyl constant and localized asymptotics independently checked analytically | 01; referee |
| 1.3 | The circle of H as the M-theory circle: v -> 0 along the joint family is strong coupling with the Wilson-line angles going to zero, the 9 endpoint has no bulk fermions as in type 0 strings | Proposal | 01 |
| 1.4 | The angle-dependent one-loop term of lemma 24 is -93 zeta(5) n_f/(512 pi^2 l^4) at its minimum, relative to theta = 0, where M_W = M_Z = 0 at fixed l | Proposition independently checked in the scope of lemma 24 | 01; referee |
| 1.5 | An isolated stationary angle of a scale-free potential with theta = l m gives l ~ 1/m; flat directions allow other stationary paths. The specified one-loop determinant has a small-angle obstruction when n_f != 54/5 | Corrected conditional result; universal claim has a counterexample | 01; referee |

| # | result | status | file |
|---|---|---|---|
| 2.1 | The hypercharge quotient of the 5-bar is the circle bundle of class ±(3,2), with effective group (U(3) x U(2))/U(1)_Y and a trivial-canonical cone; its identification with M^{111} uses the recalled fibration | Quotient checked; literature identification conditional, with metric and dual-action amendments | 02; referee |
| 2.2 | Only the 5 and 5-bar of SU(5) give a seven-dimensional free hypercharge quotient; among the specified circle quotients only (3, 2) has the required trivial canonical character | Proposition independently checked | 02; referee |
| 2.3 | Y_{a,b}: H^2 = Z (+ torsion), H^4 of order a^2 (Z/9 for (3,2)), spin iff a odd and b even; H^2(B_6) -> H^2(Y_{3,2}) is (x, y) -> 2x - 3y | Proposition independently checked; literature comparison pending | 02; referee |
| 2.4 | The Z_6 is the kernel of the displayed SU(3) x SU(2) action on Y_{3,2}; tensor Kaluza-Klein modes on an invariant background contain no triplet or doublet | Proposition independently checked in the tensor-mode scope | 02; referee |
| 2.5 | No generation from one abelian line bundle per field on CP^2 x CP^1 with fluxes along Y and B - L; the asserted equal images and unsigned linearity need common ten-dimensional chirality | Scoped no-go independently checked, with amendment | 02; referee |

| # | result | status | file |
|---|---|---|---|
| 2.5, third charge | The original bounded third-charge search is excluded analytically; the proof extends to any real third charge and independent source chiralities | Bounded claim independently checked; the extension to any real third charge checked by Claude Sonnet 5.5 (two model identifiers) | 02; referee |
| 2.6 | A line-bundle index gives one generation using a universal charge, chi and B - L; the colour-action convention changes the quark bidegrees, and compatible global flux quantization remains to be specified | Conditional index result independently checked, with amendment | 02; referee |
| 3.1 | Hypercharge is the Betti vector of Y_{3,2}; the N = 1 fivebrane baryons are d^c and L | Proposal | 03 |
| 3.2, 3.3 | One generation is Λ^odd(5-bar); its realization as a Fock space of wrapped fivebranes with 6Y as homology class is conjectural | Standard representation count independently checked; brane interpretation remains a conjecture | 03; referee |
| 3.4 | The Higgs as an M2 position is a proposal; the FI quotient has residual Z_3 or Z_2 stabilizers, and the fixed-N tower remains finite | Coordinate stabilizer independently checked; orbifold and physical-identification amendments | 03; referee |
| 3.5 | AdS5 x CP^2 x S^2 with the displayed magnetic G_4 solves the classical field equations for scaled Einstein factors iff abs(f_1) < 2 abs(f_2); quantization and stability remain open | Proposition independently checked | 03; referee |

| # | result | status | file |
|---|---|---|---|
| 3.6, 3.7 | Two completions of one base; the M2/M5 exchange as the electric-magnetic duality of hypercharge, which does not permute the 96 states | Proposals | 03 |
| 3.8, 3.9 | The 84 of C_3 is not colour-equivariantly either family of 84, through the specified little-group action or as tensor Kaluza-Klein modes on Y_{3,2} | Scoped no-go and corollary independently checked | 03; referee |
| 4.1 | The ratio-preserving 12-state blocks of whole species are one up-type quark, or the three neutrinos | Proposition independently checked | 04; referee |
| 4.2, 4.3 | The colour contents of F1 and F2 differ, and Lambda^3 C^9 is neither family for any linear SU(3) action; Lambda^3 C^9 has only 1, 3, 3bar exactly for C^9 = 9 (1), 3 + 6 (1) or its conjugate | Scoped no-go independently checked, with amendments | 04; referee |
| 4.A, 4.4 | Two commuting order-3 gradings of e8 place the left-handed 10 and H_u in the C_3 sector and (d^c, L), nu^c in the other sectors; top and antitop each have chiralities in both sectors | Branching and counts independently checked, with chirality amendment | 04; referee |
| 4.5 | Up-type grade selection is within the C_3 sector and the other Yukawa triples use one field per sector; the physical top mass needs family-breaking or compactification data | Grade selection independently checked, with family-tensor amendment | 04; referee |

| # | result | status | file |
|---|---|---|---|
| 4.6 | The Chevalley involution exchanges conjugate sectors and preserves both families; a grading-preserving E8 automorphism cannot exchange the specified top and neutrino blocks. An antiunitary J needs an extra conjugation; its physical duality identification is a proposal | Scoped no-go independently checked, with amendment | 04; referee |
| 4 §7, §8 | nu^c as the gravitational state selects t = +1/2; a discrete Z_3 from the M-grading | Proposal; discrete-grade selection independently checked | 04; referee |
| 5.1 | As configurations of aligned three-family E8 roots the two families of 84 have equal pair counts but are not isometric; no element of W(E8) exchanges them | Scoped no-go independently checked | 05; referee |
| 5.2 | With four families, M = -P in W(E8) transposes colour with family and maps the up-type quark block of family 0 onto the neutrino blocks of families 1, 2, 3 | Proposition independently checked, with setwise-invariance clarification | 05; referee |
| 5.3 | With a unitary colour-family identification, the flip commutes with gamma and J and conjugates the heavy-top and three-heavy-neutrino limits; the displayed algebras are not flip-invariant | Proposition independently checked, with unitarity amendment | 05; referee |

| # | result | status | file |
|---|---|---|---|
| 5.4 | No assignment of the four types to three of four family slots, aligned or misaligned, admits the exchange as an isometry of the specified E8 roots; the original universal neighbour counts need correction | Scoped no-go independently checked analytically, with amendment | 05; referee |
| 5.5 | Families as zero modes of an SU(4) bundle escape the single-root-slot assumptions; the index counts net chirality and a compatible Weyl action remains open | Proposal, independently reviewed with amendments | 05; referee |
| 6.1 | The SO(32) charged Wilson-line sector has 30 copies of the lemma-22 channel spectrum; its light mass is a/R for 0 <= a <= 1/2 | Spectrum independently checked, with multiplicity and range amendments | 06; referee |
| 6 §2, §3, §5 | The heterotic web and type-I' dictionary as a route to the target family | Proposals reviewed; Wilson-line map, winding duality and slope rule confirmed against the two sources in the second check | 06; referee |
| 7.1 | A channel spectrum has fundamental domain [0, pi/2], with doubly degenerate periodic and antiperiodic endpoint spectra | Proposition independently checked, with endpoint and scope clarifications | 07; referee |
| 7.2 | The SO(32) endpoint has centralizer S(O(30) x O(2)); the lemma-24 one-loop minimum is a = 1/2 under its stated hypotheses | Proposition independently checked, with disconnected-centralizer amendment; brane interpretation remains a proposal | 07; referee |

| # | result | status | file |
|---|---|---|---|
| 7.3 | A doubled flux torus gives KO-6 signs, a 2p-dimensional kernel and a generated M_p(C) corner; adding a scalar summand gives a unital finite datum | Error in the stated unital triple; signs and matrix corner independently checked | 07; referee |
| 7 §2, §4, §5, §6 | Readings of dualities for finite geometry; the torus flux and M2 wrapping need a constructed identification | Proposals reviewed; duality recalls conditional on unavailable primary sources | 07; referee |
| 8.1 | The descended antiunitary gives O(n) or USp(n) on each eigenspace of d^2; preservation of the rest of the data requires further conditions | Proposition independently checked, with symmetry-scope clarification | 08; referee |
| 8.2 | A signed channel eigenvalue has multiplicity k in the interior and 2k at the endpoints; the corresponding abs(D) counts differ | Proposition independently checked, with counting clarification | 08; referee |
| 8.3, 8 readings | Wall slopes and 2L/3 placement follow from the recalled rule; E-type enhancements require tuned coupling, and the electroweak identification remains a proposal | Arithmetic independently checked; slope rule and E-series confirmed against the sources in the second check, physical readings conditional | 08; referee |
| 9.1 | The integrated profile and far-wall condition follow from the recalled slope rule; its spectral trace uses sixteen selected channels | Integration independently checked; slope rule confirmed against Polchinski-Witten in the second check, channel dictionary linear only at constant dilaton | 09; referee |

| # | result | status | file |
|---|---|---|---|
| 9.2 | Antitriplet counts are (6, 6, 3) in the antisymmetric block and (3, 6, 1) in the symmetric block; the single-line antisymmetric component vanishes | Representation counts independently checked; orientifold projection and physical identification conditional | 09; referee |
| 9.3 | The stated KO signs force a particle-antiparticle block into S^2 V, excluding nonzero Lambda^2 V blocks; the full symmetric spectrum and interval scalar modes require further construction | Scoped no-go independently checked, with amendment | 09; referee |
| 9 §1 | SO(32) branching and the SU(16) alternative; the archived abstract also states an extra chiral +4/3 quark per generation | Algebra independently checked; detailed full-paper comparison pending | 09; referee |
| 9 §5 | E-type wall and del Pezzo correspondence; the two degree-eight surfaces have distinct root lattices and engineering assignments | Error in the E1 surface alternatives; dP_2 comparison conditional | 09; referee |
| 9 §5(2) | The E8 Weyl transposition conjugates a Wilson line and acts on the Narain lattice; a physical exchange must preserve the full background | Gauge action independently checked, with background-scope clarification | 09; referee |
| 5 Readings | Three families as the obstruction; M exchanges the two heavy directions of the algebra chain; 128 = 44 + 84 under Spin(9) | Proposals | 05 |

The author's numerical checks are in checks_11d.py (Python 3, exact fractions). The independent referee arguments are analytic.

**Added on master, 4 Nov 2026, by Claude Opus 5.5, with one independent reading by Claude Sonnet 5.5 (review):** 10 tests the dilaton as second scale (negative in scope; computes a_0; corrects the dictionary of 06 to 09: nonlinear angle map, T-dual lengths, Romans mass) and 11 computes the SO(30) fluctuations (the antisymmetric block needs M_15(C) at both walls, and the window sees its commutator with the masses).

**How the massless limit gives eleven dimensions here**

Vanishing W and Z masses open a dimension only along a specified path (audit §2.2): at fixed l, both theta -> 0 and theta -> pi/2 make them massless with the tower fixed. The criterion used here is the joint family of the audit, m_weak « E_epsilon « 1/R -> 0, with finite gauge couplings.

1. In the circle language of Proposition 1.1, the joint family is R -> infinity for an antiperiodic circle with two fixed points, with the Wilson-line angles of D_F going to zero. Read as the M-theory circle (Proposal 1.3), it is the strong-coupling limit, with the weak scale below the momentum tower of the eleventh dimension.
2. What selects the path: an isolated stationary angle of a potential with l and the angles as its only scales gives alpha = 1, the original l ~ 1/M_W of H, on which the finite window does not separate. A flat direction permits other paths without selecting one (referee of Proposition 1.5). A second scale could select a joint family; the localized family of the audit (bulk mass mu, mu l ~ (1 - alpha) log(1/epsilon)) is the first candidate, and the four-dimensional couplings must stay finite, which on the circle needs g_5^2 ~ R.
3. In the brane language of Proposal 3.4, v -> 0 moves the M2-brane to the apex of the cone over the Witten space with Killing spinors, and the near-horizon geometry is AdS4 x Y_{3,2}: 4 + 7 = 11 with SU(3) x SU(2) from isometries and U(1)_Y from C_3 (Proposal 3.1). This is the family of manifolds with the Standard Model group, singled out by the 5-bar (Theorem 2.1, Proposition 2.2). At fixed N it is the fixed-tower path; a joint family there needs

N -> infinity, in tension with the N = 1 matter content.

4. On the magnetic side, AdS5 x CP^2 x S^2 carries the same base, and v -> 0 sends the infrared end of AdS5 to infinity (Proposal 3.6), which is the interval of H growing.

The angle-dependent one-loop term (Proposition 1.4) is minimized at the massless point at fixed l; the direction in which l moves depends on the angle-independent Casimir coefficient, which is not yet computed.

**Next computations, ranked**

0a. **The main point** (06, PLAN task B): one family realizing the three regimes; first steps listed in 06 §6 (type I' interval with the D8 configuration 15 + 1 as the parent of the lemma-22 window; Chan-Paton algebras against C + H + M_3(C); inner fluctuations against Λ^2(30)). 0b. **A nine-dimensional spectral triple on the type I' interval** (07 §7): sixteen D8 positions as the eigenvalues of D_F, and its strong-coupling limit with E8 walls. File 09 gives a conditional dilaton profile and representation counts; the physical string dictionary and whether the coupling can select the path remain open. 0. **A potential that selects alpha < 1** (Proposition 1.5, its referee amendment and audit §7.1): the one-loop potential of the localized family, G(l m, l mu), from the exact secular equations of lemma 09, and the angle-independent Casimir coefficient a_0 of the flat family; then the vector kinetic matrix and mass Hessian along the selected path (audit eq. (17)), with g_4 finite. 1. **Remaining referee work** (report, second check): compare the EPJC paper and cited primary sources when available, check the original localized numerical table if needed, and compare the EPJC paper when available; the second check of files 01 to 09 and of the third-charge extension is complete. 2. **Spinors on Y_{3,2}** (prompts/astra_11d_01.md): the lift of (U(3) x U(2))/U(1)_Y to the spin bundle, whether fermionic Kaluza-Klein modes can carry triality or isospin 1/2, and which spin structures give the antiperiodic fibre momenta (k + 1/2)/R of Proposition 1.1. 3. **W and Z masses from the resolution**: specify the partial orbifold resolution, any further smoothing, and the brane background; compute the broken SU(2) x U(1)_Y vector masses as functions of zeta and N and

compare M_W/M_Z with the cos^2 theta = 5/8 of the spectral action. 4. **The 9D reduction of Proposition 2.6**: with SU(2)_R broken by wall data, derive the mass basis of u and d against the charge basis, and the vectorlike limit at the 9 label, where the left-handed interactions are off. 5. **Fermionic zero modes of wrapped fivebranes** (Conjecture 3.3) at N = 1, and three generations. 6. **The magnetic completion**: flux quantization, stability and the two C_3 vectors of AdS5 x CP^2 x S^2 against the fluxed pair (chi, B - L) of Proposition 2.6. 7. **A four-family completion with the fourth family decoupled** (PLAN, steps 4-5): a finite geometry or Hořava-Witten configuration in which the colour-family transposition M of Proposition 5.2 is a symmetry, so that the three-family statement of lemma 21 follows as a limit; Propositions 5.1 and 5.4 (one root per state) show that with three families the exchange is not an isometry.

**The Astra 6 audit of 3 Nov 2026**

The audit (reviews/2026-11-03_astra_audit_d11.md) reviews the master snapshot b2ddf98 and proposes a route from the massless electroweak limit to eleven dimensions with X = D(Λ^2_- T^*S^4) as principal candidate and M^{111} as comparison. Its use here:

| audit item | use in this folder |
|---|---|
| §2.2, the massless limit must specify a path | adopted as the criterion; Proposal 1.3 revised; Proposition 1.5 shows what a scale-free action selects |
| §3, joint family | restated in circle language in 01, checked against Theorems 22.1 and 22.4; the constant 2 r l E/pi of its Weyl count confirmed (Python, section H) |
| §3.1, vector poles of the lemma-23 toy | quoted in 01: M_W^2/ M_Z^2 -> 5/8 along the family |
| §4, localized family and tail identity | table of roots and the identity (8) reproduced (Python, section H, bisection), independently of the audit's script |

| audit item | use in this folder |
|---|---|
| §7.1, modulus target | In the stated flat one-loop family the electroweak term is m^2/l^2 (q = -2); a second scale could select the audit's q > 0 path. General scale-free potentials can have flat directions |
| §7.2, gauge normalization | test 2 of Proposal 1.3, and the large-N couplings in 03 |
| Appendix A, M^{111} | agrees with Theorem 2.1 and Proposition 2.3 for (3, 2), and with the placement of hypercharge in the charged module (03) |
| §2.1, §5, §6, X and lemma 28 | outside this folder; the 7-dimensional parent X is a third possible completion, M-theory on R^4 x X with the Bryant-Salamon comparison and the circle of solidification §7 as a candidate M-circle, listed as a direction only |
| §2.4, docs and distribution | outside this branch: refreshing the PDFs, adding lemmas 17 and 28 to paper/ build_supplement.sh and repairing the supplement layout belong to master's build, and LaTeX is not available in this session |

**References used here**

Read in this session (3 Nov 2026), from the arXiv abstract pages or the ar5iv text through a summarizing fetch: D. Fabbri, P. Fré, L. Gualtieri and P. Termonia, *M-theory on AdS4 x M^{111}: the complete Osp(2|4) x SU(3) x SU(2) spectrum from harmonic analysis*, Nucl. Phys. B 560 (1999) 617, hep-th/9903036 (coset, Killing spinors only for p = q, R-symmetry, Betti multiplet, Kaluza-Klein states neutral under it); D. Fabbri, P. Fré, L. Gualtieri, C. Reina, A. Tomasiello, A. Zaffaroni and A. Zampa, *3D superconformal theories from Sasakian seven-*

*manifolds*, hep-th/9907219 (D-term 2|U|^2 = 3|V|^2, U(1) fibration over P^1 x P^2, baryons det U and det V as wrapped M5-branes, E_0 = 4N/9 and N/3, b_2 = b_5 = 1). M. Kreck and S. Stolz, *A diffeomorphism classification of 7-dimensional homogeneous Einstein manifolds with SU(3) x SU(2) x U(1)-symmetry*, Ann. of Math. 127 (1988) 373 (title and scope only).

Recalled and not reread, to be checked before any of them carries an argument: E. Witten, *String theory dynamics in various dimensions*, hep-th/9503124 (D0 states as momentum modes); O. Bergman and M. Gaberdiel, *Dualities of type 0 strings*, hep-th/9906055; M. Fabinger and P. Hořava, *Casimir forces between end-of-the-world branes*, hep-th/0002073; E. Witten, *SL(2, Z) action on three-dimensional conformal field theories with Abelian symmetry*, hep-th/0307041; C. N. Pope and P. van Nieuwenhuizen, *Compactifications of d = 11 supergravity on Kähler manifolds*, Commun. Math. Phys. 122 (1989) 281; J. Gauntlett, D. Martelli, J. Sparks and D. Waldram, *Supersymmetric AdS5 solutions of M-theory*, hep-th/0402153.

Already in the corpus: Witten 1981 and Castellani-D'Auria-Fré (LITERATURE.md §F); Hořava-Witten, Atiyah-Witten and Acharya-Witten-type cones (solidification §6).

Prior work for the observation of Theorem 2.1, that the weights of M^{111} are the hypercharges of the 5-bar: a thin web search of 4 Nov 2026 found no statement of it, as recorded in LITERATURE.md; this does not establish priority.

**Files**

- 01_interval_and_eleventh_dimension.md: the "1".
- 02_witten_spaces_hypercharge_quotient.md: the "6 + 1" of Witten's spaces, the Z_6, chirality from fluxes.
- 03_branes_hypercharge_and_the_84.md: C_3, M2, M5 and the 84.
- 04_two_families_m2_m5.md: the two 12-state blocks, two gradings of E8, Yukawa grades, and a proposed M2/M5 reading. An antiunitary J requires more than the complex-linear Chevalley involution.

- 05_colour_family_exchange.md: the two families as E8 root configurations, the colour-family transposition with four families.
- 06_main_point_heterotic_route.md: the main point: target, the heterotic web 9/10/11, dictionary, Proposition 6.1.
- 07_dualities_9_11.md: recalled dualities, the proposed D8-position dictionary, the conditional one-loop endpoint, and the doubled flux-torus matrix corner with a unitality amendment.
- 08_orientifold_sign.md: antiunitary commutants, a proposed orientifold interpretation, and conditional wall-enhancement arithmetic.
- 09_dilaton_spectral_and_so32.md: a conditional type I' coupling profile, SO(32) branching and diquark counts, and proposed string and del Pezzo readings; the full EPJC comparison remains pending.
- PLAN.md: the handover plan of the line.
- checks_11d.py: exact checks.
- ledger_11d.pl: ledger facts, included by ../ledger.pl.
- prompts/: the referee request and the first Astra job.
- ../reviews/2026-11-03_astra_audit_d11.md: the Astra 6 audit used in the second version.

# Part 11D, 03 branes hypercharge and the 84, referee pending

## Hypercharge from the three-form, matter from branes, and the two families of 84

Part of the 11D line of research (README). Propositions 3.2 and 3.5 and Corollary 3.9 are proved here, and Proposition 3.8 is a negative result in a stated scope, refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5; the numbered proposals are conjectures to test. Attempted by Claude (Anthropic), 3 Nov 2026. Checks: checks_11d.py, sections B, E and G, Python with exact fractions. Referee pending.

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** Proposition 3.8 and Corollary 3.9 have been independently

checked by hand in their stated scopes; the tensor-mode argument needs the displayed colour action to preserve the background. See the referee report.

**Referee note on Proposition 3.5 (GPT-6.1 Terra).** The classical field equations and curvature condition are independently checked; flux quantization, supersymmetry and stability remain open. See the report.

**Referee note on Proposition 3.2 (GPT-6.1 Terra).** The exterior-algebra representation and charge counts are independently checked. Their realization by wrapped branes remains a proposal; see the report.

### Setting

02 realizes the Witten space with Killing spinors as Y_{3,2} = (5bar)//U(1)_Y. Its isometries give SU(3) x SU(2) x U(1)_R, and no tensor Kaluza-Klein mode on it, on an invariant background, carries triality, weak isospin 1/2 or fractional hypercharge (Proposition 2.4). The M-theory vacuum AdS4 x Y_{3,2} is the near-horizon geometry of N M2-branes at the apex of the cone C(Y_{3,2}), with N units of flux of *G_4 through Y_{3,2}. Lemma 21 proposed relating its two families of 84 states to the three-form C_3, to the M2 or M5 brane, and to their duality. This file connects the two.

### Proposal 3.1 (hypercharge is the Betti vector of Y_{3,2})

In AdS4 x M^{111} the massless vectors are those of the isometries, SU(3) x SU(2) x U(1)_R, and one more from C_3 on the harmonic two-form (b_2 = 1), the Betti multiplet; the Kaluza-Klein states are neutral under it, and only non-perturbative states carry its charge (Fabbri et al., hep-th/9903036 and hep-th/9907219). By Proposition 2.3 the Betti class pairs with the divisors {l(U) = 0} and {m(V) = 0} as 2 and -3, the values of 6Y on d^c and L.

**Proposal.** Identify the Betti U(1) of Y_{3,2} with hypercharge. The gauge group of the vacuum is then

```
SU(3) x SU(2) (isometries) x U(1)_Y (from C_3) x
 ↪  U(1)_R (isometry),
```

thirteen vectors, the Standard Model group and one R-symmetry. With massless W and Z this is the eleven-dimensional vacuum of the v -> 0 limit; the identification is consistent with three facts already established:

1. Tensor Kaluza-Klein modes on an invariant background have Y = 0, as they must by Proposition 2.4, which leaves them only PSU(3) x SO(3) representations; spinors are open.
2. The states with Y != 0 are non-perturbative: M5-branes wrapped on five-cycles and M2-branes wrapped on the two-cycle.
3. The M5 baryons of Fabbri et al. are det U on one five-cycle and det V on the other, with baryonic charges 2N and -3N and representations Sym^N(3bar) and Sym^N(2). For N = 1 they are U in (3bar, 1)_{1/3} and V in (1, 2)_{-1/2}: d^c and L, the 5-bar.

In this reading hypercharge is the homology class of a wrapped fivebrane, 6Y = [Sigma_5] in H_5(Y_{3,2}; Z) = Z, and Witten's 1981 chirality argument, which assumes gauge fields acting by coordinate transformations, no longer covers U(1)_Y. It still covers SU(3) x SU(2) for Kaluza-Klein fermions, so chiral doublets must also be brane states.

**Proposition 3.2 (one generation is the odd part of the exterior algebra of the 5-bar)**

**Statement.** As SU(3) x SU(2) x U(1)_Y representations,

```
Λ^1(5bar) + Λ^3(5bar) + Λ^5(5bar) = (d^c + L) + (Q +
↪  u^c + e^c) + nu^c = 16,
Λ^0 + Λ^2 + Λ^4 = the conjugates,
```

and the charge of SO(10)/SU(5) is X = 5 - 2k on Λ^k. The 32 states are the finite Hilbert space of one generation with nu^c (dim H_F = 96 for three).

**Proof.** Enumeration with U, V of charges 2, -3 (Python, section B); the identification of the 16 of SO(10) with the odd forms on C^5 is standard. ■

The new element is the reading of the generators: U and V are the coordinates whose zero loci are the baryonic five-cycles of Y_{3,2}, so a state of Λ^k carries the total

class of k such cycles, which is its 6Y. For example Q = U^1 U^2 V^1 has class 2 + 2 - 3 = 1.

**Conjecture 3.3 (a generation as a Fock space of wrapped fivebranes).** At N = 1, the states of one generation and its conjugate are the states of an exterior algebra generated by five fermionic zero modes, one for each of the cycles {U^i = 0} and {V^A = 0}, with hypercharge the total homology class and X the number of fivebranes. Needed: the fermionic statistics of the wrapped states (from fermionic zero modes of the M5 worldvolume theory on Sigma_5), the regime N = 1, where the supergravity description fails and the M2-brane theory is the abelian quotient of 02, and three generations.

**Proposal 3.4 (the Higgs is the position of the M2-brane, and the massless limit is the apex)**

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The stated coordinate stabilizer is independently checked under the proposed charge assignment. The FI quotient has residual $\mathbb{Z}_3$ or $\mathbb{Z}_2$ stabilizers on the respective exceptional loci, so it is a partial orbifold resolution. See the referee report.

Resolve the cone with a Fayet-Iliopoulos parameter: 2|U|^2 - 3|V|^2 = zeta. For zeta < 0 the locus U = 0 is the projective line of the doublet, P(V) = CP^1; for zeta > 0 the locus V = 0 is CP^2.

**Statement (proved part).** On the brane coordinates (U, V), on which the Betti U(1)_Y of Proposal 3.1 acts with charges (2, -3), the value V = (v, 0), U = 0 on the doublet line is fixed by SU(3) and by T_3 + Y, and moved by T_3 - Y. (In the quotient geometry, where U(1)_Y is divided out, the same point is fixed by T_3, which is T_3 + Y up to the quotient.) Q = T_3 + Y gives U the charge +1/3 of d^c and V the charges (0, -1) of (nu, e). (Python, section B2.)

**Proposal.** The scalar that breaks SU(2) x U(1)_Y to U(1)_em is the position of the M2-brane on the doublet line, with the quantum numbers of V, that is of H^* (or of a sneutrino). As zeta -> 0 and the brane moves to the apex, the full group is restored and the near-horizon geometry is AdS4 x Y_{3,2}. So the eleven-dimensional theory reached at v -> 0 is the Freund-Rubin vacuum

on the Witten space with Killing spinors, with SU(3) x SU(2) from its isometries and U(1)_Y from C_3. The relation between M_W and zeta, or the brane position, is the next computation (README, item 3).

**Which path this is.** In AdS4 x Y_{3,2} the Kaluza-Klein scale is 1/L with L ~ N^{1/6} l_p, independent of zeta. At fixed N, the limit zeta -> 0 therefore restores the group at a fixed tower: it is the path of the Astra audit (§2.2) on which masslessness opens no dimension, and the eleven dimensions are present above 1/L at every v. A joint family in the sense of 01 (weak masses vanishing faster than the tower) needs L/l_p -> infinity, so N -> infinity, with the isometry couplings, g^{-2} proportional to the Killing-vector norms times the volume of Y in eleven-dimensional units, going to zero unless compensated (audit §7.2). Large N conflicts with item 3 of Proposal 3.1, which needs N = 1 for the baryons to be d^c and L rather than Sym^N(3bar) and Sym^N(2). This tension, between the regime in which the matter content is right and the regime in which the dimension opens, is the main obstruction of the M2 side in its present form, stated for the Freund-Rubin vacuum with N units of flux; escapes are N = 1 matter on a separate stack, a varying flux along the family, or a different vacuum.

**Agreement with the audit (its Appendix A).** The audit states independently that the circle bundle of M^{111} has c_1 = 3h + 2u, that the total space is spin, that its geometric U(1) is an R-symmetry, and that hypercharge belongs to the charged module and the gauge lift, since a fixed circle-charge sector acts by a scalar phase. This agrees with Theorem 2.1, Proposition 2.3 for (3, 2), and the placement of hypercharge in Proposal 3.1. The audit also notes that Toeplitz compression on the first holomorphic sector H^0(O(3, 2)), the (10, 3) of Theorem 2.1, item 4, generates the full M_30(C), a noninformative algebra in the sense of STRATEGY T0.

### Proposition 3.5 (a magnetic completion: AdS5 x CP^2 x S^2)

**Statement.** On AdS5 x CP^2 x S^2 with product metric and four-form

```
G_4 = f_1 vol(CP^2) + f_2 J ^ vol(S^2),
```

with J the Kähler form of CP^2 and orthonormal-frame components f_1, f_2, the equations of eleven-dimensional supergravity hold with Ricci components

```
R_AdS = -(f_1^2 + 2 f_2^2)/6,   R_CP2 = f_1^2/3 +
 ↪  f_2^2/6,   R_S2 = 2 f_2^2/3 - f_1^2/6,
```

so a solution exists with round factors exactly when |f_1| $<$ 2|f_2|.

**Proof.** dG_4 = 0, and *G_4 = vol(AdS5) ∧ (f_1 vol(S^2) + f_2 J), which is closed. G_4 ∧ G_4 is an eight-form on the six-dimensional factor, so it vanishes and d*G_4 + G_4 ∧ G_4/2 = 0. The Einstein equation R_MN = (G_MPQR G_N^PQR - g_MN G^2/12)/12 gives the displayed components and zero off the diagonal (Python, section G, by summing all frame components); each factor is Einstein, and the radii follow from the three values. ■

Supersymmetry, stability and flux quantization are not checked. The product with flux on KE_4 x S^2 appears in Pope and van Nieuwenhuizen (1989), and supersymmetric AdS5 solutions fibre the S^2 over the Kähler-Einstein base (Gauntlett, Martelli, Sparks and Waldram, hep-th/0402153); both recalled, not reread here.

**Proposal 3.6 (two completions of one base)**

The base B_6 = CP^2 x S^2 has isometry SU(3) x SU(2), and its dimension is the KO-dimension of the finite factor. It has two eleven-dimensional completions, both of the form 4 + 1 + 6:

| | electric completion | magnetic completion |
|---|---|---|
| geometry | AdS4 x Y_{3,2}, Y a circle bundle over B_6 | AdS5 x B_6 |
| flux | *G_4 through Y: N M2-branes | G_4 through CP^2 and CP^1 x S^2: M5 charges (N_1, N_2) |
| the “1” | the fibre circle (Proposition 1.1 of 01) | the radial direction of AdS5 |

| | electric completion | magnetic completion |
|---|---|---|
| vectors from C_3 | b_2(Y) = 1: U(1)_Y | b_2(B_6) = 2: U(1)_{h_1} x U(1)_{h_2} |
| further vectors | SU(3) x SU(2) x U(1)_R | SU(3) x SU(2) |
| v -> 0 | the M2-brane at the apex | the infrared end of AdS5 moves to infinity |

Under reduction along the fibre of class (3, 2), the combination of U(1)_{h_1} x U(1)_{h_2} along (3, 2) is replaced by the fibre isometry, and the quotient Z^2/(3, 2) = Z, (x, y) -> 2x - 3y, is hypercharge. The magnetic side therefore carries two abelian factors, a candidate for the fluxed pair needed by Proposition 2.6 together with the universal twist. The magnetic completion realizes the counting of H directly: four-dimensional spacetime on the M5 worldvolume, the interval as the conformal radial coordinate z in (z_UV, z_IR) with M_KK ~ 1/z_IR, as in Higgsless models on an interval, and v -> 0 as z_IR -> infinity.

**Proposal 3.7 (the M2/M5 exchange acts on the Betti vector)**

In AdS4, M2-branes on the two-cycle and M5-branes on five-cycles are electric and magnetic for the Betti vector (C_3 = A ∧ omega_2 and C_6 = Ã ∧ omega_5), with Dirac pairing given by the intersection of H_2 and H_5. Which of the two is electric depends on the boundary condition of the bulk vector, and the S operation of Witten's SL(2, Z) action on three-dimensional theories with a U(1) symmetry exchanges them (hep-th/0307041, recalled). With Proposal 3.1 this exchange acts on hypercharge as electric-magnetic duality.

**Test against lemma 21.** S maps charges to monopole charges; it does not permute the 96 states of H_F. The exchange of the top and the neutrinos in lemma 21 is a permutation, so it does not come from this S. In the brane reading, the 84 states of the second family (Q_em != 0) are the states whose fivebrane class 6Y is not cancelled by

T_3, after the breaking of Proposal 3.4; the first family (all states except the top) has no reading here. This points away from the M2/M5 duality as the origin of the exchange, and toward a property of the top (lemma 21, Proposition 21.1) that the present construction does not see.

**Proposition 3.8 (no colour-equivariant identification of the 84 with the three-form)**

**Statement.** For every homomorphism SU(3) -> SO(9), Λ^3 R^9 ⊗ C is not isomorphic, as an SU(3) representation, to the colour content of either family of 84 in lemma 21: 10 (3 + 3bar) + 24 (1) for the five quarks and all leptons, 12 (3 + 3bar) + 12 (1) for the charged fermions. The same holds for Λ^6 R^9, the polarizations of C_6.

**Proof.** A real nine-dimensional representation has a self-conjugate complexification of dimension 9, which is 9 (1), 3 + 3bar + 3 (1) or 8 + 1 (the sextets and two triplet pairs already exceed 9). In the first case Λ^3 has only singlets. In the second, Λ^2(3) ⊗ 3bar = 3bar ⊗ 3bar contains the 6bar. In the third, Λ^3(8) = 1 + 8 + 10 + 10bar + 27. Both families contain only 1, 3 and 3bar. Λ^6 R^9 is isomorphic to Λ^3 R^9 under SO(9). (Python, section E, by weights.) ■

**Scope and escapes.** A linear action of colour on the nine transverse directions of the eleven-dimensional little group. Escapes: colour acting on the internal space instead of the little group, a nonlinear or brane realization, or a map that is not colour-equivariant, in which case the equality of the two 84s is an equality of numbers only.

**Corollary 3.9 (the coloured states of the 84 are not Kaluza-Klein modes of C_3 on Y_{3,2})** Under the quotient realization, on a background invariant under the displayed action, every tensor Kaluza-Klein mode of C_3 (or of the metric) on Y_{3,2} carries a representation of PSU(3) x SO(3) (Proposition 2.4), so none is a colour triplet; spinor modes are open. The 60 coloured states of the first family, and the 72 of the second, are therefore not modes of C_3 on Y_{3,2}. Together with Proposition 3.8 this excludes, in their stated scopes, the two direct routes, the little group and the tensor Kaluza-Klein tower, for a colour-equivariant iden-

tification. The route left open is the brane one: wrapped M5 and M2 states, which carry triality through their hypercharge class (Proposition 2.4 (a)).

**Summary of the direction**

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The physical identifications in this summary are proposals. The colour exclusions have the stated linear-representation and tensor-mode scopes; other internal spaces, nonlinear actions and brane realizations remain directions. See the wording audit.

The M2 and M5 details of lemma 21 enter in a specific place. Hypercharge is the C_3 vector of the Witten space with Killing spinors, the 5-bar consists of the N = 1 five-brane baryons, the Higgs is the M2 position, and the M2/M5 exchange is the electric-magnetic duality of hypercharge. The count 84 of C_3 does not survive as a colour-equivariant identification. The next computations are in the README.

# Part 11D, 04 two families m2 m5, referee pending

## The two families of 84 and the M2 and M5 sides

Part of the 11D line of research (README, PLAN). Attempted by Claude (Anthropic), 4 Nov 2026. Propositions 4.1 to 4.3 and Theorem 4.4 and Propositions 4.5, 4.6 are proved here by hand (branching rules and counting); the negative results 4.3 and 4.6 were refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5, with the amendments below; readings are labelled as proposals.

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** Proposition 4.1 and the colour obstructions in Propositions 4.2(a) and 4.3 are independently checked. The tensor-factor conclusion of 4.2(b) needs a factor-preserving map, and the "exactly" clause in 4.3 also needs nonzero triplets: the trivial nine-dimensional representation gives 84 singlets. See the referee report; the other checked results are noted below.

**Referee note on Proposition 4.6 (GPT-6.1 Terra).** The stated sector and block obstructions hold in the scopes in the referee report. The Chevalley involution is complex-linear; an antiunitary $J$ needs an additional conjugation. Its identification with physical M2/M5 duality remains a proposal.

### The question

Lemma 21, Proposition 21.3: with three generations and Dirac neutrinos, dim H_F = 96 = 84 + 12 in two ways,

```
F1 = all states except the top          (the top: 3
 ↪  colours x 4 = 12 states),
F2 = all states except the neutrinos    (the
 ↪  neutrinos: 3 generations x 4 = 12 states),
```

where 4 counts chirality and particle/antiparticle. The two families overlap in the 72 states of the charged fermions other than the top. Lemma 21 proposed relating one family to C_3 (or the M2-brane) and the other to C_6 (or the M5-brane), with the exchange of the top and the neutrinos as the content of the duality on the Standard Model side, and recorded that equal numbers are insufficient. The files 02 and 03 showed that colour-equivariant identifications through the little group or through Kaluza-Klein modes on the Witten space fail (Proposition 3.8, Corollary 3.9). This file asks the question in the Hořava-Witten wall group E8 and in the finite geometry.

### 1. What singles out the two blocks

**Proposition 4.1.** Consider blocks of 12 states made of whole Dirac species (all colours, both chiralities, particles and antiparticles; 12 states for a quark species, 4 for a lepton species). The blocks whose removal keeps both spectral-action values of lemma 21, cos^2 theta_NC = 5/8 and kappa_em/kappa_s = 8/3, are exactly: one up-type quark (u, c or t), and the three neutrinos together. So F1 (with the top, the heavy up-type quark) and F2 are the two kinds of solutions: a charged one and a neutral one.

**Proof.** A 12-state block of whole species is one quark species or three lepton species. By Proposition 21.1, with the totals (sum T_3 Q, sum Y Q; Q^2, colour) = (6, 10;

32, 12) and cos^2 theta_NC = (sum Y Q)/(sum T_3 Q + sum Y Q), removing a quark keeps both values only for an up-type quark, (1, 5/3; 16/3, 2). For three lepton species with k charged leptons, each contributing (1/2, 3/2; 4, 0), the angle becomes (10 - 3k/2)/(16 - 2k), which is 5/8 only for k = 0; neutrinos contribute zero to all four sums. ■

**Reading at the 9 label.** At the 9 label the light gauge algebra is colour plus electromagnetism (canonical §3, lemma 21). H_F then splits into three parts: the 72 charged light states (F1 ∩ F2), the 12 neutrino states, which carry no charge of that algebra and are free, and the 12 top states, which are charged and infinitely heavy. F1 = charged light + free light; F2 = charged light + charged heavy. The exchange of lemma 21 exchanges the block that decouples through zero charge with the block that decouples through infinite mass.

**2. Colour obstructions**

**Proposition 4.2.** (a) As representations of colour SU(3), F1 = 10 (3 + 3bar) + 24 (1) and F2 = 12 (3 + 3bar) + 12 (1); they are not isomorphic. Hence no unitary of H_F commuting with colour maps F1 onto F2, and no automorphism of the finite spectral triple of the Standard Model does. (b) Any tensor-factor-preserving bijection of the top block onto the neutrino block that is linear in each index exchanges the colour index of the top (r, g, b) with the generation index of the neutrinos (e, mu, tau).

**Proof.** (a) Counting: F1 has the five quarks other than the top (5 x 2 triplet pairs, counting chirality) and 24 leptonic states; F2 has the six quarks and 12 charged-lepton states. Colour acts through A_F. (b) The top block is C^3_colour ⊗ e_3 ⊗ C^4 (third generation, 4 for chirality and conjugation) and the neutrino block is e_lepton ⊗ C^3_generation ⊗ C^4 in the Pati-Salam form C^4_colour ⊗ C^3_generation of the up-type sector; a map of the first onto the second that respects the tensor factors sends C^3_colour to C^3_generation. ■

So an exchange needs a structure in which colour and generation can be exchanged; a structure in which colour acts and generations do not admits none.

**Proposition 4.3 (complex case of Proposition 3.8).** For every homomorphism SU(3) -> SU(9), the colour content of Λ^3 C^9 is not that of F1 or F2. Λ^3 C^9 contains only 1, 3 and 3bar exactly when C^9 is 9 (1), 3 + 6 (1) or 3bar + 6 (1); for 3 + 6 (1), Λ^3 C^9 = 15 (3) + 6 (3bar) + 21 (1), and for 9 (1) it is 84 (1).

**Proof.** C^9 is a sum of irreducibles of dimensions 1, 3, 6, 8 with total 9. With a sextet: 6 + 3 (1) has the 6 in Λ^1(6) ⊗ Λ^2(3 (1)); 6 + 3 has Λ^3(6) = 10 + 10bar. With an octet: 8 + 1 has Λ^3(8) = 1 + 8 + 10 + 10bar + 27. With two triplets T, T' (each 3 or 3bar) and three singlets: Λ^1(T) ⊗ Λ^1(T') ⊗ Λ^1(1) contains T ⊗ T', which is 6 + 3bar, 6bar + 3 or 8 + 1. With three triplets: T ⊗ T' ⊗ T'' contains a 10, a 10bar, a 15 or a 15bar. With nine singlets there is no triplet. With one triplet and six singlets, Λ^3(3 + 6 (1)) = Λ^3(3) + 6 Λ^2(3) + 15 (3) + 20 (1) = 15 (3) + 6 (3bar) + 21 (1). Neither family has unequal numbers of 3 and 3bar. ■

Together with Proposition 3.8 (real nine-dimensional representations) this excludes, in that scope, the colour-equivariant route through Λ^3 of a linear nine-dimensional space. The positive construction below places the two blocks instead of identifying the two families.

## 3. Two commuting gradings of E8

E8 is the gauge group of each wall in Hořava-Witten theory, and it also models the C-field topologically (Witten 1996, "On flux quantization in M-theory"; Diaconescu, Moore and Witten 2000; recalled). Over C, e8 has an inner automorphism of order three with fixed algebra sl(9):

```
e8 = sl(9) ⊕ Λ^3 C^9 ⊕ Λ^6 C^9          (80 + 84 +
 ↪  84),
```

with [Λ^3, Λ^3] ⊂ Λ^6 (wedge product), [Λ^3, Λ^6] ⊂ sl(9) (contraction) and [Λ^6, Λ^6] ⊂ Λ^3. In the U-duality algebra e8(8) of eleven-dimensional supergravity on T^8, the decomposition under gl(8) by level (West's level, recalled) places the C_3 shifts at level 1 (Λ^3 of 8, 56 states), the C_6 shifts at level 2 (Λ^6 of 8, 28), the dual graviton at level 3 (8) and gl(8) at level 0; then Λ^3 C^9 = 56 (level 1) + 28 (level -2) and the grading above is the level mod 3. So M-grade b = 1 is one unit of M2

charge (or minus an M5), b = 2 is one unit of M5 charge (or minus an M2), and b = 0 holds the metric and the dual graviton (levels 0 and ±3). The bracket Λ^3 × Λ^3 -> Λ^6 is the algebraic form of the term C_3 ∧ G_4 in G_7 = dC_6 + (1/2) C_3 ∧ G_4, equivalent by duality to the Chern-Simons coupling; Λ^3 × Λ^6 -> sl(9) adds levels 1 + 2 = 3, the dual graviton. Under SO(9) ⊂ SL(9), 248 = 36 + 44 + 84 + 84, the SO(9) content of so(9), the graviton, C_3 and C_6. We call the three pieces the gravity, C_3 and C_6 sectors.

A second element of order three, the centre of SU(3) in E8 ⊃ (SU(3) × E6)/Z_3, gives the family grading e8 = (su(3) ⊕ e6) ⊕ (3, 27) ⊕ (3bar, 27bar), with family grade a = 0, 1, 2. With this SU(3) as family group, (3, 27) holds three families of E6, each containing a generation 16 ⊂ 27, and (3bar, 27bar) holds their conjugates.

**Lemma 4.A (the two gradings commute).** Take SU(9) ⊃ SU(3) × SU(6) × U(1) with C^9 = (3, 1)_{-2} + (1, 6)_{1}, and let s_f be the centre of this SU(3). Then s_f commutes with the SU(9) grading, its centralizer is (SU(3) × E6)/Z_3, the U(1) is the Cartan of the SU(2) in E6 ⊃ SU(6) × SU(2) with charge q = 6 T_3, and the joint pieces g_{a,b} are

| | b = 0 (gravity) | b = 1 (C_3) | b = 2 (C_6) |
|---|---|---|---|
| a = 0 | su(3) + su(6) + u(1): 44 | (1,1)_{-6} + (1,20)_{3}: 21 | (1,1)_{6} + (1,20)_{-3}: 21 |
| a = 1 | (3, 6bar)_{-3}: 18 | (3, 15)_{0}: 45 | (3, 6bar)_{3}: 18 |
| a = 2 | (3bar, 6)_{3}: 18 | (3bar, 6)_{-3}: 18 | (3bar, 15bar)_{0}: 45 |

so that b ≡ a + q/3 = a + 2 T_3 (mod 3).

**Proof.** Λ^3 of (3,1)_{-2} + (1,6)_{1} is (1,1)_{-6} + (3bar,6)_{-3} + (3,15)_{0} + (1,20)_{3}; the adjoint is (8,1) + (1,35) + (1,1) + (3,6bar)_{-3} + (3bar,6)_{3}; Λ^6 is the conjugate of Λ^3. The column a = 0 sums to 86 = 8 + 78 and the row a = 1 to 81 = 3 x 27, which

matches E8 ⊃ SU(3) × E6 with E6 ⊃ SU(6) × SU(2): 27 = (15, 1) + (6bar, 2) and 78 = (35, 1) + (1, 3) + (20, 2), the SU(2) triplet being (1,1)_{-6}, the u(1) and (1,1)_{6}. ■

## 4. Placement of the generation

Let SU(5) ⊂ SU(6) ⊂ E6 be the Georgi-Glashow group, with 15 = 10 + 5 and 6bar = 5bar + 1. The commutant of SU(5) in E6 is SU(2) × U(1), and its SU(2) is the SU(2) of the lemma; we call it SU(2)_I. With SO(10) × U(1)_psi and SU(5) × U(1)_chi, 27 = 16_1 + 10_{-2} + 1_4, 16 = 10_{-1} + 5bar_3 + 1_{-5} and 10 = 5_2 + 5bar_{-2}. The Cartan T_3 of SU(2)_I vanishes on the singlets 10 and 5, so it is proportional to psi + chi: T_3 = t on 5bar_16 and on N = 1_{E6}, and T_3 = -t on 5bar_10 and on nu^c = 1_16, with t = ±1/2. The SU(2)_I doublets are (d^c, D^c), (L, H_d) and (N, nu^c).

**Theorem 4.4 (the generation in the three sectors).** For the generation 16 ⊂ 27 of family grade a = 1:

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The branching and counts are independently checked. The fields $t_L, t^c$ are in the left-handed basis; $t^c$ is an antitop field, so the top particle's two chiralities occupy different sectors. See the referee report.

1. the 10 = (Q, u^c, e^c) and the 5 = (D, H_u) lie in g_{1,1}, the C_3 sector;
2. (d^c, L) and N lie in g_{1,1+2t}, and (D^c, H_d) and nu^c lie in g_{1,1-2t}: one pair in the C_6 sector and the other in the gravity sector, according to the sign t; the two choices differ by the Weyl reflection of SU(2)_I;
3. the antiparticles lie in the conjugate pieces g_{2,-b}.

So every top state of a particle (t_L in Q, t^c in u^c), and the up-type Higgs, lie in the C_3 sector, and the antitop in the C_6 sector. The left-handed particle fields of the neutrinos (a = 1) lie outside the C_3 sector, nu_L and nu^c in different sectors (C_6 and gravity); their conjugates do not, since conj(nu_L) lies in C_3 for t = +1/2. With conjugates, the top block has (C_3, C_6, gravity) = (6, 6, 0) and the neutrino block (3, 3, 6), for

both signs of t; a conjugation-closed block always has equal C_3 and C_6 counts. With three families and antiparticles, the 96 states of H_F split as 45 (C_3) + 45 (C_6) + 6 (gravity: nu^c and its conjugate) for t = +1/2, and 33 + 33 + 30 for t = -1/2.

**Proof.** b ≡ a + 2 T_3, with T_3 = 0 on the 10 and the 5, and T_3 = ±t on the doublet members; for antiparticles a = 2 and T_3 changes sign. The counts follow per family: for t = +1/2, C_3 holds the 10 and the 5 of antiparticles (15), C_6 their conjugates, gravity nu^c and its conjugate; for t = -1/2, C_3 holds the 10 and the conjugate of nu^c (11), C_6 their conjugates, gravity the 5bar and the 5 (10). ■

## 5. Yukawa couplings as brackets

**Proposition 4.5.** Suppose the cubic couplings of the matter come from the structure constants of e8, as in heterotic and Hořava-Witten compactifications, and both gradings are symmetries, so that a coupling needs grades summing to (0, 0) mod 3. Take H_u in the 5 ⊂ g_{1,1} and H_d in 5bar_10 ⊂ g_{1,1-2t}. Then, for either sign t:

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The grade selection and bracket targets are independently checked. In the literal exact $SU(3)_f$ triplet realization, the family tensor is antisymmetric and does not by itself give an isolated top mass; the physical Yukawa matrix needs additional data. See the referee report.

1. the up-type Yukawa couplings Q u^c H_u have all three fields in the C_3 sector;
2. the down-type, charged-lepton and Dirac-neutrino couplings Q d^c H_d, L e^c H_d and L nu^c H_u have one field in each of the three sectors;
3. in Dirac form, the top mass is ad() : g_{1,1} -> g_{2,2}, the wedge Λ^3 -> Λ^6 with the Higgs trivector, while the Dirac-neutrino mass is a bracket that passes through the gravity sector: Λ^6 -> sl(9) (levels 1 + 2 = 3) for t = +1/2, sl(9) -> Λ^3 for t = -1/2.

**Proof.** Grades from Theorem 4.4: (1, 1) three times for the up type; for the others the b-grades are {1, 1 + 2t, 1 - 2t} = {1, 2, 0}. The Dirac masses pair the left-handed

field with the conjugate of the right-handed one, through the Killing form between g_{a,b} and g_{-a,-b}. ■

The family grading alone allows only epsilon_{ijk} in exact family SU(3); in compactifications the family label is a cohomology label and the couplings are intersection numbers. The M-grading survives when its element commutes with the structure group of the bundle: for the standard embedding, s_M = s_f exp(2 pi i (2/3) T_3), a product of the centre of the holonomy SU(3) and an element of SU(2)_I ⊂ E6, which does.

**Reading (proposal).** The up-type Yukawa couplings are the only Standard Model Yukawa couplings inside one sector, and that sector's bracket is the C_3 ∧ C_3 -> C_6 term, metric-independent; the other couplings use one field of each sector, including the gravity sector. The top, the only fermion with a Yukawa coupling of order one and the one singled out by the spectral action in lemma 21, takes its mass from the C_3 bracket.

## 6. The Chevalley involution and the real structure J

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** This heading describes a proposed physical identification. The Chevalley involution is complex-linear, and an antiunitary $J$ needs additional conjugation; constructing the finite triple remains open. See the wording audit.

**Proposition 4.6.**

1. No automorphism of e8 that preserves the family grading maps the C_3 sector of the matter, g_{1,1} (dimension 45), onto its C_6 sector, g_{1,2} (dimension 18).
2. The Chevalley involution, the element -1 of the Weyl group of E8, maps g_{a,b} to g_{-a,-b}. It normalizes SU(9), acting as complex conjugation (Λ^3 <-> Λ^6), and on the placed H_F it exchanges every state with its conjugate: it is the real structure J of the finite geometry. The family grade a is the four-dimensional chirality in the left-handed basis (a = 1 for the left-handed fields, a = 2 for their conjugates), and J inverts it, as KO-dimension six requires (J gamma = -gamma J).

3. J preserves F1 and F2, which are unions of species with their conjugates.
4. The Weyl reflection of SU(2)_I fixes s_f and replaces b by 2a - b: on the matter (a = 1) it fixes the C_3 sector and exchanges the C_6 and gravity sectors, together with (d^c, L) <-> (D^c, H_d) and nu^c <-> N.

**Proof.** (1) Dimensions. (2) -1 acts on a Cartan subalgebra by -1, so it inverts every element of the maximal torus containing s_f and s_M, and maps the joint eigenspace for (omega^a, omega^b) to that for (omega^{-a}, omega^{-b}); it maps the root system A_8 of SU(9) to itself. (3) Immediate. (4) The reflection inverts T_3 and commutes with SU(3) × SU(6). ■

**Negative result (scope: e8 with these two gradings, automorphisms preserving the pair up to inversion).** The Chevalley involution, read as the M2/M5 exchange (a proposal), maps each block to its conjugate and leaves the top and neutrino blocks in place. An exchange as in lemma 21 must therefore come from an automorphism that neither preserves nor inverts the family grading; for aligned families, Proposition 5.1 also excludes every element of W(E8) mapping F1 onto F2. The colour-family exchanges inside E8 ⊃ SU(3)^4, where the four SU(3) factors are permuted by the normalizer, are the remaining candidates (PLAN, step 4). What the gradings do give is a placement of the left-handed fields (a = 1): the top on the M2 side, the neutrinos on the M5 and gravity sides; with conjugates the counts are those after Theorem 4.4.

## 7. Agreement with files 02 and 03, and the finite geometry

- **nu^c as the gravitational state.** nu^c is the only Standard Model state with Y = 0, so on the Witten space it is the only one that can be a Kaluza-Klein mode (Proposition 2.4 with Proposal 3.1), while every hypercharged state is a brane state. In e8 with t = +1/2 it is the only Standard Model state in the gravity sector, and every hypercharged state lies in the C_3 or C_6 sector. The two placements agree for t = +1/2; for t = -1/2 the gravity sector holds (d^c, L) instead. We take the agree-

ment as selecting t = +1/2 (proposal).

- **The finite geometry.** On H_F, the family grading is the chirality, the Chevalley involution is J, and the Yukawa blocks of D_F are brackets with the Higgs: the top block maps the C_3 sector to the C_6 sector, the Majorana block acts inside the gravity sector (nu^c with nu^c, t = +1/2). A finite triple built on the graded e8, with D_F = ad(Higgs), is the construction to attempt next (PLAN, step 5).
- **The two families.** F1 removes a block of the C_3 sector (with its conjugate in C_6): the states joined by the cubic C_3 coupling. F2 removes the block that meets the gravity sector. In this sense, for the left-handed fields, the top belongs to the M2 side and the neutrinos to the M5 and gravity sides, which is the correspondence lemma 21 asked for, at the level of the removed blocks; with conjugates the blocks have counts (6, 6, 0) and (3, 3, 6). The two families of 84 are not themselves the 84 of C_3 and the 84 of C_6 (Propositions 3.8, 4.3, 4.6).

## 8. Remark: a discrete symmetry from the M-grading

For t = +1/2, s_M acts on the 27 by omega^b: omega on the 10 and on the 5 (H_u, D), omega^2 on (d^c, L) and N, and 1 on (D^c, H_d) and nu^c. It allows Q u^c H_u, Q d^c H_d, L e^c H_d, L nu^c H_u, N H_u H_d, the Majorana mass of nu^c and the Weinberg operator L L H_u H_u, and forbids the mu term H_u H_d, the dimension-five operators Q Q Q L and u^c u^c d^c e^c, and the trilinear couplings L L e^c, L Q d^c, u^c d^c d^c; the bilinear L H_u is allowed. This is the pattern of E6-inspired models in which the singlet N supplies the mu term. It is a symmetry of the low-energy theory only if the breaking of E6 commutes with s_M; for t = -1/2 the charges change. Stated as a remark, not used above.

*Scope of the remark (second check, 4 Nov 2026).* The selection rule is that of the single element s_M, and it is a symmetry only if the breaking of E6 commutes with s_M. The family grading would forbid (nu^c)^2, L L H_u H_u and L H_u, which s_M allows, since their family grades are 2, 1 and 2.

**9. What remains open**

1. Colour-family exchange in E8 ⊃ SU(3)^4 acting on the placed states: does an element map the top block onto the neutrino block and F1 onto F2? (PLAN, step 4.)
2. A finite real spectral triple on the graded e8 with D_F = ad(Higgs): order zero and order one, and comparison with C + H + M_3(C) (PLAN, step 5).
3. Hořava-Witten dynamics: whether an M5-brane moving into a wall (small-instanton transition) can remove one of the 12-state blocks of Proposition 4.1 (PLAN, step 6).
4. The E11 level decomposition beyond E8, where C_3, C_6 and the dual graviton sit at levels 1, 2, 3: test the assignment 10 -> level 1, (d^c, L) -> level 2, nu^c -> level 0 or 3 (PLAN, step 7).

**References for this file**

Recalled, not reread in this session: E. Witten, "On flux quantization in M-theory and the effective action", hep-th/9609122; D.-E. Diaconescu, G. Moore and E. Witten, "E8 gauge theory, and a derivation of K-theory from M-theory", hep-th/0005090; P. West, "E11 and M theory", hep-th/0104081, for the level decomposition and the commutator of the level-one generators; E. Cremmer and B. Julia (1979) for E8(8) in three dimensions; E. B. Vinberg and A. G. Elashvili (1978) for the Z_3-graded e8 and trivectors of C^9; the E6 subgroup chains are standard (R. Slansky, Phys. Rep. 79 (1981) 1).

# Part 11D, 05 colour family exchange, referee pending

## The exchange of the top and the neutrinos as a colour-family transposition in E8

Part of the 11D line of research (README, PLAN, step 4). Attempted by Claude (Anthropic), 4 Nov 2026. Proposition 5.1 (a negative result in its scope) and Proposition 5.2 are proved by hand, refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, re-

views/2026-11-04_sonnet_second_check_11d.md); an exploratory search in Python with exact fractions found the same results (no isometry for three families, signed coordinate permutations for four); the reruns of 4 Nov 2026 (Python, exact integers) are not kept, by the rule that checking scripts are not saved. Readings are labelled as proposals.

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** Proposition 5.1 has an independent analytic check of its neighbour and pair counts, and of its no-isometry conclusion in the aligned-root scope. See the referee report. Proposition 5.4 is addressed at its statement below.

### Setting

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The M2/M5 identification is a proposal, and the colour-family conclusion of Proposition 4.2 needs a factor-preserving map. See the wording audit.

Proposition 4.6 showed that the M2/M5 exchange of the graded e8 acts as J and cannot exchange the top with the neutrinos, and Proposition 4.2 that any such exchange must trade the colour index of the top for the generation index of the neutrinos. This file tests the colour-family exchanges inside E8.

Use E8 ⊃ SO(16) ⊃ SO(10) × SO(6), with the roots of E8 in the coordinates of R^8: ±e_i ± e_j, and (1/2)(±1, …, ±1) with an even number of minus signs. The 128 half-integer roots form the chiral spinor of SO(16), which under SO(10) × SU(4) is (16, 4) + (16bar, 4bar): four families of the Standard Model with ν^c, with their conjugates. Write a root as (χ | w | φ), with χ the signs of the three colour coordinates, w those of the two weak coordinates and φ those of the three family coordinates; a particle has an even number of minus signs among χ, w and an even number among φ. The four family patterns are $\mathbb{1} = (+++)$ (family 0) and the three patterns with two minus signs (families 1, 2, 3); the colour patterns with one minus sign are the quark colours, the pattern $-\mathbb{1}$ is the lepton. Then, for a colour pattern c with one minus sign and a family f,

| state | root (times 2) |
|---|---|
| u_L, d_L | (c \| −+ \| f), (c \| +− \| f) |
| u^c, d^c | (−c \| ++ \| f), (−c \| −− \| f) |
| ν_L, e_L | (−𝟙 \| −+ \| f), (−𝟙 \| +− \| f) |
| ν^c, e^c | (𝟙 \| ++ \| f), (𝟙 \| −− \| f) |

and antiparticles are the negatives. Three families give the 96 roots of H_F, closed under negation; the four family slots are permuted by the Weyl group of SU(4), so the choice of three is immaterial. The same 96 roots are (16, 3) + (16bar, 3bar) in E8 ⊃ E6 × SU(3) ⊃ SO(10) × U(1) × SU(3), the embedding of file 04.

**Proposition 5.1 (the two families of 84 are not isometric)**

**Statement.** Let H_F be the 96 roots of three families, the top block T the 12 roots of the up-type quark of one family f_t with its conjugate, and the neutrino block N the 12 roots of the neutrinos of the three families with their conjugates; F1 = H_F − T and F2 = H_F − N. Then F1 and F2 have the same numbers of pairs at each inner product (42 at −2, 816 at −1, 1812 at 0, 816 at +1), but no isometry of R^8 maps F1 onto F2. In particular no element of the Weyl group of E8 does, and no isometry of H_F maps T onto N.

**Proof.** For a root v of a set S let d_S(v) be the number of roots of S at inner product +1 with v; it is preserved by isometries. Two half-integer roots have inner product +1 exactly when their signs agree in six of the eight places.

1. d_{H_F} = 22 for every root. H_F is a single orbit of the group generated by the Weyl group of SO(10) (transitive on the 16, a minuscule representation), the permutations of the three families and the negation. For ν^c of family 0, (𝟙 | ++ | 𝟙), the roots of H_F with exactly two minus signs are: two in the first five places with family 𝟙 (10), two among the family places (2, the patterns of families 1 and 2 in H_F), and antiparticles with one minus sign in each block (5 x 2). The total is 22.
2. Removing N. For u_L = (c | −+ | f): the inner product with ν_L(g) is $(1 + f \cdot g)/4$ and with ν^c(g) is $(1 + f \cdot g)/4$, where $f \cdot g = 3$ for g = f and −1

otherwise; so u_L has exactly 2 roots of N at +1 (its own family's $\nu$_L and $\nu\hat{}$c), and the same holds for u$\hat{}$c and the conjugates. For d_L = (c | +− | f): the inner product with $\nu$_L(g) is $(-3 + f \cdot g)/4$, which is −1 for g ≠ f, so the two roots $-\nu$_L(g), g ≠ f, are at +1, and $\nu\hat{}$c(f) is at +1: 3 in all; the same count holds for d$\hat{}$c, e_L, e$\hat{}$c and the conjugates. So in F2, d = 20 on the 36 up-type quark roots and d = 19 on the 48 others.

3. Removing T. For a neutrino of the top's family, $\nu$_L(f_t) = (−𝟙 | −+ | f_t): its inner products with t_L(c) = (c | −+ | f_t) and with t$\hat{}$c = (−c | ++ | f_t) are (−1 + 2 + 3)/4 = 1 and (1 + 0 + 3)/4 = 1, six roots at +1; for g ≠ f_t, $\nu$_L(g) has inner product (−1 + 2 − 1)/4 = 0 with t_L and (1 + 0 − 1)/4 = 0 with t$\hat{}$c. The same holds for $\nu\hat{}$c. So in F1, d = 16 on the 4 roots of the neutrinos of the top's family and d = 22 on the 8 roots of the other neutrinos (the up-type quarks of the other families have 2 roots of T at +1 and the down-type quarks and charged leptons 3, by inner products of the same kind as in item 2, so d = 20 and d = 19).

F1 has roots with d = 16 and d = 22, and F2 has none. The sums of d agree, $4 \cdot 16 + 48 \cdot 19 + 24 \cdot 20 + 8 \cdot 22 = 48 \cdot 19 + 36 \cdot 20 = 1632 = 2$ x 816, which is the equality of the pair counts at +1. Both sets are closed under negation, so the pairs at −1 are as many as those at +1 and the pairs at −2 are the 42 antipodal pairs; the 1812 pairs at 0 are the remainder of the 3486 pairs. ■

**Scope.** The roots of H_F in E8, through the embedding above, which is unique up to W(E8) for the aligned (16, 3) + (16bar, 3bar); isometries of R$\hat{}$8 acting on the roots. Escapes: maps that are not linear on the roots (bijections of states with no geometric meaning), other realizations of H_F in a larger group, and a fourth family (Proposition 5.2). The tau neutrino carries the obstruction: in F1 the neutrino of the top's own family is the only state whose neighbourhood changes by six.

**Proposition 5.2 (four families: the transposition of colour and family)**

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The Weyl map and block exchange are independently checked. Family-zero lepton species blocks are fixed setwise; $\nu^c$ and $e^c$ individually go to their negatives. See the referee report.

**Statement.** Let P be the permutation of coordinates that exchanges the three colour coordinates with the three family coordinates and the two weak coordinates with each other, and M = −P. Then M is an involution in the Weyl group of E8 that preserves the 128 roots of the four families and SU(2)_L, and acts on the up-type and down-type states as the transposition of the four Pati-Salam colours (r, g, b, lepton) with the four families (1, 2, 3, 0), with the lepton paired with family 0 and the colour c paired with the family −c. In particular:

1. the up-type quark of family 0 (t_L and t^c, three colours, with conjugates) goes onto the neutrinos of families 1, 2, 3 (ν_L and the conjugate of ν^c), and back;
2. the down-type quark of family 0 goes onto the charged leptons of families 1, 2, 3;
3. nu_L and e_L of family 0 are fixed, nu^c and e^c of family 0 go to their negatives, u^c and d^c go to conjugate roots, and the quarks of families 1, 2, 3 go to quarks of families 1, 2, 3 with colour and family transposed.

So the four-family sets 128 − T(0) and 128 − N(1, 2, 3), each of 116 roots, are exchanged.

**Proof.** P is an even permutation of coordinates and −1 is in the Weyl group of E8, so M is in it; M^2 = P^2 = 1. Both preserve the set of half-integer vectors with an even number of minus signs. P maps (χ | w_1 w_2 | φ) to (φ | w_2 w_1 | χ), so M maps it to (−φ | −w_2 −w_1 | −χ). The weak part (−+) goes to (−+) and (++) to (−−), so T_3 of SU(2)_L, (w_2 − w_1)/2, is preserved and T_3 of SU(2)_R is reversed. Then t_L(c) = (c | −+ | 𝟙) goes to (−𝟙 | −+ | −c) = ν_L(−c); t^c = (−c | ++ | 𝟙) goes to (−𝟙 | −− | c) = −ν^c(−c); b_L(c) = (c | +− | 𝟙) goes to (−𝟙 | +− | −c) = e_L(−c); ν_L(0) = (−𝟙 | −+ | 𝟙) and e_L(0) are fixed; u_L(c, f) = (c | −+

| f) goes to u_L(−f, −c), with −f a quark colour and −c a family pattern. ■

The search found further signed permutations with the same effect on the two blocks; P itself maps T(0) onto N(1, 2, 3) up to conjugation, and M = J ∘ P, with J = −1 the real structure of Proposition 4.6.

**Proposition 5.3 (the flip in a four-family finite geometry)**

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** Choose the factor isomorphism $\iota$ unitary, or unitary up to a common scalar. An arbitrary isomorphism need not make $F$ unitary. With this amendment the flip and block calculation are independently checked; see the report.

**Statement.** Let H_f = (C^2_L ⊕ C^2_R) ⊗ C^4_c ⊗ C^4_f, H = H_f ⊕ conj(H_f), with J the antilinear exchange of the two summands and γ equal to +1 on C^2_L and −1 on C^2_R in H_f and to its negative on conj(H_f), the signs of KO-dimension six. Fix a unitary isomorphism ι: C^4_c -> C^4_f sending the lepton line $\ell$ to the family line e_0 and the colour space C^3 onto the span of e_1, e_2, e_3, and let F be the flip x ⊗ y -> ι^{-1}(y) ⊗ ι(x) on C^4_c ⊗ C^4_f, extended by the identity on C^2_L ⊕ C^2_R and to conj(H_f) by J F J^{-1}. Then:

1. F is unitary, F^2 = 1, and F commutes with γ and J.
2. For the up-type Dirac blocks T_top = Y ⊗ P_q ⊗ P_0 and T_ν = Y ⊗ P_$\ell$ ⊗ (1 − P_0), where Y maps the up state of C^2_L to the up state of C^2_R (and Y^* back), P_$\ell$ and P_q = 1 − P_$\ell$ are the lepton and quark projectors of C^4_c and P_0 the projector on e_0, one has F T_top F^{-1} = T_ν, and likewise for the down-type blocks of family 0 and the charged leptons of families 1, 2, 3.
3. So the heavy limits D + t T_top and D + t T_ν, for D commuting with F, are conjugate by F: their light spaces, of 116 states each, correspond. The light algebras A ∩ {T}' correspond when the algebra is invariant under F; the Pati-Salam and Standard Model algebras act on C^4_c alone and are not.

**Proof.** F acts on the factor C^4_c ⊗ C^4_f only, where it conjugates a ⊗ b to ι^{-1} b ι ⊗ ι a ι^{-1}, so P_q ⊗ P_0 goes to P_ℓ ⊗ (1 − P_0) by the choice of ι. γ and Y act on the other factor, and J F J^{-1} on conj(H_f) is the conjugate flip. ■

F equals M on the left-handed states and M followed by the particle-antiparticle exchange on the right-handed ones (second check): F exchanges T_top and T_nu inside H_f, while M sends right-handed states to antiparticle roots and so flips gamma on 64 states; F lies outside W(E8). It also makes reading 2 below precise: F exchanges the projector P_ℓ, which the Majorana direction selects in colour space, with P_0, which the top Yukawa coupling selects in family space; as operators, it maps the top Yukawa block to the Dirac blocks of the neutrinos of families 1, 2, 3, and it does not map the Majorana block to a Dirac block, since the Majorana block joins H_f to conj(H_f).

**Proposition 5.4 (no three-slot assignment admits the exchange)**

**Referee note (GPT-6.1 Terra).** The no-isometry conclusion is correct by an independent analytic argument, but the neighbour counts in the statement below are false for misaligned slots. The corrected formulas and a counterexample are in the referee report.

**Statement.** Place the top in slot 0, and let the up-type quarks, the down-type quarks, the charged leptons and the neutrinos each occupy three of the four family slots, independently (the up-type quarks in slots containing 0). This includes the aligned families of Proposition 5.1 and the configurations of the 9 label, where SU(2)_L is off and does not tie the slots of up-type and down-type quarks. For none of the 192 configurations S is there an isometry of R^8 mapping S − T onto S − N. S − N has d in {13, 17, 19, 20}, and {19, 20} only when the down-quark and charged-lepton slot sets coincide (48 of the 192); each neutrino root of S − T has d = 16 + 6 (1_A(f) − 1_0(f)), with A the slot set of the up-type quarks, so the values are 16 and 22 in the patterns (16, 22, 22) and (16, 16, 22).

**Proof.** Exhaustive computation of the neighbour counts d (Python, exact integers, rerun twice on 4 Nov 2026 and

not kept; also the analytic proof in the referee report); the counts of S − T and S − N differ for every configuration, so no isometry exists. ■ This is a computer-assisted statement; Proposition 5.1 is its hand-proved aligned case.

*Correction (4 Nov 2026).* The conclusion holds on rerun; the sentence on the values of d in the Statement is wrong for misaligned configurations. See 10, section on corrections.

**Scope.** It assumes that each family occupies one slot, so that every state is a single root. Four slots then cannot be reduced to three while keeping the exchange as an isometry of the roots.

**Proposal 5.5 (four slots, three zero modes)**

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** General bundle zero modes are outside the root-slot hypotheses. The index gives net chirality, and split bundles do not automatically reduce to three distinct root slots; the Weyl action on a specified bundle and its zero modes remains open. See the referee report.

In heterotic compactifications on a Calabi-Yau threefold with E8 ⊃ SO(10) × SU(4), 248 = (45, 1) + (1, 15) + (16, 4) + (16bar, 4bar) + (10, 6), the SU(4) is the structure group of a rank-four bundle V, and the families are the zero modes, the 16s coming from H^1(X, V); their net number is the index |c_3(V)|/2, which is 3 in the heterotic Standard Models built on SU(4) bundles (Braun, He, Ovrut and Pantev, hep-th/0501070; recalled, not reread). In that reading the four slots of this file are the fibre of V, and the three families are not three of the slots: a zero mode is a section of V and in general involves all four. Propositions 5.1 and 5.4 do not apply to it, since they assume one slot per family.

The transposition M of Proposition 5.2 then exchanges the Pati-Salam SU(4), a gauge group, with the structure group SU(4) of V: colour and holonomy trade places, and SO(10) goes to the commutant SO(10)' of the other SU(4). The top-neutrino exchange becomes a statement about the fibre, which an index map must carry to the zero modes. The test is an explicit bundle. For a split bundle V = L_0 + L_1 + L_2 + L_3 (line-bundle models), slot i carries n_i = −χ(L_i) families with n_0 + …

+ n_3 = 3; such models fall under Proposition 5.1 only for n = (1, 1, 1, 0) up to order; for every n_i >= 0 with sum 3 the degree multisets still differ (second check, 40 cases), so the obstruction persists for aligned split bundles. A non-split V, whose zero modes are spread over the slots, is the case to compute.

**Readings (proposals)**

1. **Three families are the obstruction.** The exchange of lemma 21 is a symmetry of the E8 geometry exactly in the four-family completion, where the colour-family square is 4 x 4 and the top belongs to the family paired with the lepton colour. With three families in single slots the square is 4 x 3 and the exchange is not an isometry (Propositions 5.1 and 5.4, in their scope of one root per state). Lemma 21's 96 = 84 + 12 in two ways is the three-family shadow of 128 = 116 + 12 in two ways, exchanged by M.
2. **The two heavy directions of the algebra chain.** Canonical §2.2 and §3 obtain the chain Pati-Salam -> Standard Model -> colour plus electromagnetism from two heavy operators: the Majorana direction, which selects the lepton among the four Pati-Salam colours and $\nu$_R among the SU(2)_R states, and the top Yukawa coupling, which selects the heavy family and the up-type state. M exchanges the lepton colour with family 0 and keeps the up-type direction (F, its finite-geometry counterpart, differs from it on right-handed states): it exchanges the direction selected by the first heavy operator with the direction selected by the second. If the 11 label is the grand algebra M_2(H) + M_4(C) (lemma 28) and the 9 label colour plus electromagnetism, M exchanges the two steps of 11 -> 10 -> 9. This needs a finite geometry in which the generation space is C^4 with an algebra exchangeable with M_4(C), and a Dirac operator compatible with the exchange.
3. **Four families and eleven dimensions.** The 128 of the four families is the chiral spinor of SO(16) $\subset$ E8; under Spin(9) $\subset$ SO(16), embedded through its spinor, it decomposes as 44 + 84 in the class with 128 -> 44 + 84 (the other chirality of the spinor em-

bedding gives an irreducible 128), the graviton and the three-form of eleven-dimensional supergravity (Pengpan and Ramond, hep-th/9808190; Kostant's cubic Dirac operator on F4/Spin(9); recalled). The two families of 116 are not Spin(9) subrepresentations; whether the exchange M has a Spin(9) meaning is open.

3b. **The arithmetic 44 = 32 + 12 and the two placements of the top (observation of the user, 4 Nov 2026).** Observation with an open question; nothing beyond (a) to (d) is claimed.

(a) With four families, $128 = 44 + 84$ under the Spin(9) of Reading 3, and $128 - 84 = 44 = 32 + 12$: removing F1 (three families minus the top, 84 states) leaves the fourth family plus the top block, and removing F2 (three families minus the neutrinos) leaves it plus the neutrino block. In this count the top goes with the fourth family into the graviton number 44, and dually the neutrinos do: the four-family completion of the $96 = 84 + 12$ of lemma 21 in two ways. M of Proposition 5.2 exchanges the two blocks of 12 that fill the 44, T(0) and N(1, 2, 3); it sends each family to the states of one colour (family 0 to the leptons), so it does not carry one set of 44 onto the other.

(b) The fit is numerical in the present scope. Proposition 3.8 excludes a colour-equivariant identification of Λ^3 R^9 with either family of 84 for any linear action of colour on the 9. The colour SU(3) lies in no such Spin(9) (Claude Opus 5.5, 4 Nov 2026; checked by Claude Sonnet 5.5): the vector 16 of SO(16) is 3 + 3bar + 10 (1) under colour, while the spinor 16 of Spin(9) restricted to an SU(3) inside Spin(9) is 16 (1), 2 (3 + 3bar) + 4 (1) or 8 + 8, by the three real nine-dimensional representations of Proposition 3.8. The stabilizer in so(16) of the split 44 + 84 is spin(9) itself (the intermediate subalgebras would be submodules of 36 + 84, and so(16) is irreducible on the 128), so the split is not colour-covariant; and no set of 84 roots spans the 84, since a root partition has the rank-8 Cartan subalgebra in its stabilizer. Also, colour has 248 = 8 + 78 (1) + 27 (3 + 3bar), while the 44 of any so(9) of

e8 with 248 = 36 + 44 + 84 + 84 contains a 6 or a 27, so colour lies in no such so(9), including that of file 04. Scope: the regular colour SU(3) of the root table, in this class. Escapes: other colour embeddings; identifications that are not equivariant. A strengthening (Sonnet, exact integers, not kept): no isometry of R^8 maps the set (fourth family + top) onto (fourth family + neutrinos), since up to family permutations their degree multisets differ, {10^16, 12^12, 13^16} against {10^24, 13^16, 16^4}.

(c) The record holds two placements of the top, from different subgroups of E8, and they differ: under the Z_3 grading by SU(9) of file 04 (Lemma 4.A, Theorem 4.4) the left-handed top lies in the C_3 sector and ν^c in the gravity sector (t = +1/2, proposed in section 7 there); under the Spin(9) count above the top goes with the fourth family into the 44. SU(9) ⊂ E8 and Spin(9) ⊂ SO(16) ⊂ E8 are different subgroups and the sectors are different objects, so there is no contradiction, and no identification between the sectors is claimed. The two groups themselves are related: the Chevalley involution restricted to sl(9) is an outer involution with fixed algebra the so(9) of file 04, which is conjugate in E8 to the Spin(9) of this reading, with k = so(9) + 84 and p = 44 + 84, the 44 inside the gravity sector (Sonnet, 4 Nov 2026; a single-model statement, referee pending). How the 96 states of the three-family frame sit against the conjugated Spin(9) has not been worked out.

(d) Recalled analogy: Gell-Mann's colour-flavour SU(8) of N = 8 supergravity put the spin-1/2 fermions in Λ^3(8) = 56 with 8 = 3 + 5 (Gell-Mann, "From renormalizability to calculability?", 1983, and the SU(8) of Cremmer and Julia; recalled, not reread); here it is Λ^3 of 9 = 84 with the C-field, and Proposition 4.3 plays the role of the known failure of that content.

4. **NCG.** In the finite geometry with three generations the exchange is not an automorphism (Proposition 4.2). A Pati-Salam finite geometry with C^4_colour ⊗ C^4_family, where the flip of the

two factors exchanges the σ field with the top Yukawa direction, is the candidate in which it is (PLAN, step 5).

**What remains open**

- A finite spectral triple, or a Hořava-Witten configuration, with four families in which M is a symmetry and the fourth family is decoupled in a controlled way, so that the three-family statement of lemma 21 follows as a limit.
- Whether the fourth family can be made heavy by the same mechanism as the top (lemma 21 counts only one heavy up-type quark).

# Part 11D, 09 dilaton spectral and so32, referee pending

## The type I' dilaton as a spectral function, the SO(32) of the EPJC paper, and two more dualities

Part of the 11D line of research (README, PLAN, task C). Attempted by Claude (Anthropic), 4 Nov 2026, after the steer of the same day (the coincident-brane route is overused; the type I' dilaton and the colour-flavour idea are worth pursuing; compare with the SO(32) of the EPJC paper). Proposition 9.1 is proved from the recalled type I' slope rule; Proposition 9.2 is proved from representation theory and the recalled orientifold projections; Proposition 9.3 is proved (linear algebra and Connes's sign table); the comparison of section 1 is read from the arXiv text of the paper through a summarizing fetch; the rest are proposals. Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5 (reviews/2026-11-04_terra_check_11d.md, reviews/2026-11-04_sonnet_second_check_11d.md).

Correction (same day): the first version stated the slope rule for e^{−phi}. The piecewise-linear function is z, with e^{phi} = z^{−5/6}; Proposition 9.1 holds as stated with z in place of e^{−phi}, and infinite coupling is z = 0.

**1. The SO(32) of A. Rivero, Eur. Phys. J. C 84 (2024) 1058 (arXiv:2407.05397)**

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The branching algebra is independently checked. The archived original abstract also states an additional chiral charge-+4/3 quark per generation. The full paper was unavailable for direct comparison, so its detailed attribution remains pending. See the referee report.

That paper uses the adjoint of SO(32) to classify the scalars of a supersymmetric Standard Model with three generations, as composites: pairs of quarks as colour triplets and quark-antiquark pairs as singlets and octets, with three extra scalars of charge +4/3. Its pattern is SO(2n) ⊃ SO(n_1) x SU(n_2) x SU(3) x U(1) with 2n = n_1 + 6 n_2, which for SO(32) and n_2 = 5 gives SO(2) x SU(5) x SU(3) x U(1): the 30 = 6 n_2 are the five light quarks u, c, d, s, b ("turtles") in three colours with conjugates, and the top is the non-participating "elephant". Its explicit route also passes through SO(32) ⊃ SU(16) x U(1), with 496 = 1 + 120 + 120bar + 255 (quarks from the 120s, leptons from the 255), and it mentions open strings, orientifolds and an SU(9) for M2 and M5.

**Reading of the LaTeX source (Claude Sonnet 5.5, 4 Nov 2026; the earlier comparison used a summarizing fetch).** The paper uses colour singlets for mesons and colour antitriplets for diquarks, and keeps only (1, 24, 1), (1, 15, 3bar) and (1, 15bar, 3) of its decomposition; colour octets and sextets lie outside it. The top is the favoured non-binding quark ("we have not a concrete argument"), and the postulates force the non-binding quark to be up-type. The chain is SU(16) x U(1) to SU(15) to SU(5) x SU(3), with SO(30) and SO(32) reached by "adding singlets". The identification of the top with the SO(2) factor, the O8^- plane and the Chan-Paton reading belong to these files, and "orientifold" in the paper means planar orientifold QCD. The footnote on the 84 + 12 split and SU(9) for M2 and M5 anticipates Proposition 21.3 and 04. The abstract specifies a chiral +4/3 quark per generation; the body has three scalars of charge +4/3 and leaves their fermion partner open. The paper gives (15, 3bar) as the antisymmetric square of one fundamental without stating chirality, so

the question of Proposition 9.3 stays open; its remark on odd-parity diquarks is consistent with that proposition. The paper's selection of an up-type quark by its postulates and lemma 21's by the spectral-action values are independent arguments with the same outcome.

**Coincidence.** The chain SO(32) ⊃ SO(2) x SO(30) ⊃ SO(2) x SU(15) x U(1) ⊃ SO(2) x SU(5) x SU(3) x U(1), the 30 as five light quarks in three colours, and the top as the factor SO(2), are the same as lemma 21 and the dictionary of 06. Lemma 21 records the branching 496 -> (435, 1) + (1, 1) + (30, 2) and leaves SO(32) unexplored; the EPJC paper explores it through composites, and the files 06 to 09 through type I' branes. What the 11D files add: the type I' realization of the SO(2) as the top's D8-brane and of the 30 as the Chan-Paton labels of fifteen D8-branes on an O8^- wall (06); the top angle as the brane's position and the one-loop vacuum on the far wall (Propositions 7.1, 7.2); the orientifold type O^- matched by the seesaw sign (Proposition 8.1); the far wall's enhancement to SU(2) x U(1) at infinite coupling (Remark 8.3); the dilaton (Proposition 9.1); the diquark counts as strings between the stack and its image, which need the O^- type (Proposition 9.2); and the real-structure condition under which these counts arise, as connection components along the interval and not as finite inner fluctuations (Proposition 9.3). In type I' language the paper's other chain, SU(16) x U(1), is the configuration with all sixteen D8-branes coincident in the bulk, and the chain with SO(2) x SO(30) is the configuration with fifteen on a wall and one apart; the second is the one the corpus selects. The SU(9) of M2 and M5 in the paper is the grading of e8 used in 04.

**2. Proposition 9.1 (the type I' coupling function is a trace of a function of the finite Dirac operator)**

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The integration is independently checked conditional on the recalled slope rule. The trace is over sixteen selected channels, not automatically the full physical $D_F$ Hilbert space; direct comparison with the cited source was unavailable to the referee. (Second check, against Polchinski-Witten.) The formula z(L) = z(0) - c (8L - sum x_k) is exact in the positions; the dictionary x_k = (2L/pi) l m_k holds only for constant z, and in general theta_k = (pi/2)

W(x_k)/W(L) with W = int z^{-1/3} (10, Proposition 10.1). In the profile 15 + 1 the endpoint values 7cL and 8cL stand, and theta_t = pi/4 gives 7.358 cL in place of 7.5 cL. See the referee report.

**Setting.** Type I' on an interval [0, L], in the conformal coordinate x, with sixteen D8-branes at x_1, …, x_16 and O8^- planes at the ends. Recalled rule (Polchinski and Witten, hep-th/9510169): the dilaton is e^{phi} = z^{−5/6} for a function z that is continuous and piecewise linear in x, with slope c (8 − N(x)), where N(x) is the number of D8-branes to the left of x and c > 0; the coupling diverges where z vanishes. Dictionary (06, 07): x_k = (2L/pi) θ_k with θ_k = l m_k the angles of sixteen channels of the finite Dirac operator (fifteen light-quark channels with colour, and the top).

**Statement.**

1. z(x) = z(0) + c Σ_k min(x/2, x_k − x/2) for x in [0, L], so z is z(0) plus c Tr g_x(l |D_F|) over the sixteen channels, with g_x(θ) = min(x/2, (2L/pi) θ − x/2): a trace of a function of the finite Dirac operator, as the spectral action is.
2. At the far wall, z(L) = z(0) − c (8L − Σ_k x_k).
3. Infinite coupling at the far wall (z(L) = 0) holds exactly when z(0) = (2cL/pi)(4 pi − l Σ_k m_k): the coupling at the light-quark wall is fixed by the sum of the sixteen channel masses, the trace of |D_F| on these channels.

**Proof.** *∫0^x N(y) dy = Σ_k (x − x_k)+*, so z(x) = z(0) + c (8x − Σ_k (x − x_k)+) = *z(0) + c Σ_k (x/2 − (x − x_k)+)*, and x/2 − (x − x_k)_+ = min(x/2, x_k − x/2). Item 2 is x = L with Σ_k (L − x_k) = 16L − Σ x_k. Item 3 sets z(L) = 0 and substitutes x_k = (2L/pi) l m_k. ■

*Correction (4 Nov 2026).* The dictionary x_k = (2L/pi) theta_k is exact only for constant z; the map from angles to positions is nonlinear. See 10, Proposition 10.1.

**Reading (proposal).** With the light quarks near θ = 0, Σ_k m_k ≈ m_t and the condition is z(0) ≈ (2cL/pi)(4 pi − θ_t): 7cL at label 9 (θ_t = pi/2) and 8cL at label 11 (θ_t -> 0). The coupling therefore varies along the interval on a scale set by the spectrum of D_F, which is the kind of second scale that Proposition 1.5 requires to

select a joint family; whether the type I' coupling can play that role in the corpus's flat family is the next question. Since the slope decreases with x, z is concave, so z > 0 on [0, L) once z(0) > 0 and z(L) = 0; positivity of z(0) gives l Σ_k m_k < 4 pi, a bound on the trace of |D_F| in units of the tower scale, met when only the top is heavy (θ_t <= pi/2).

**3. Proposition 9.2 (diquarks as strings between the stack and its image)**

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The representation and charge counts are independently checked, conditional on the recalled projections. The shifts describe the specified gauge-field channels; $\Lambda^2$ of a line excludes that component, not all open-string states or strings to other branes. See the referee report.

**Setting.** V = C^5 ⊗ C^3, five light flavours in three colours, with Q = 2/3 on u, c and Q = −1/3 on d, s, b; the 30 of lemma 21 is V ⊕ Vbar. Fifteen D8-branes on an O8 plane carry the Chan-Paton space 30 (branes and images), with group SO(30) on an O8^- and USp(30) on an O8^+, and U(15) when the stack is moved off the plane (recalled).

**Statement.**

1. Under U(15), the adjoint of SO(30) is $\Lambda^2$V ⊕ $\Lambda^2$Vbar ⊕ (V ⊗ Vbar) and the adjoint of USp(30) is $S^2$V ⊕ $S^2$Vbar ⊕ (V ⊗ Vbar). The summands $\Lambda^2$V or $S^2$V and their conjugates are the strings between the stack and its image; V ⊗ Vbar are the strings with both ends on the stack.
2. $\Lambda^2$V = ($S^2$5, 3bar) ⊕ ($\Lambda^2$5, 6) and $S^2$V = ($S^2$5, 6) ⊕ ($\Lambda^2$5, 3bar).
3. In the colour antitriplet, $\Lambda^2$V has the charge counts (−2/3: 6, +1/3: 6, +4/3: 3) of lemma 21, and $S^2$V has (−2/3: 3, +1/3: 6, +4/3: 1). So the antisquark counts of three generations recorded in lemma 21 (6 of charge −2/3 and 6 of charge +1/3 in the 3bar) occur among the stack-image strings of an O8^- plane and not among those of an O8^+ plane.
4. With the angles θ_k = l m_k of 06, a string from channel i to the image of channel j has the spectrum

(n pi ± (θ_i + θ_j)))/l, and a string from channel i to channel j has (n pi ± (θ_i − θ_j))/l, n in Z.

5. A single brane has no string to its own image on an O8^- plane ($\Lambda^2$ of a line vanishes) and one on an O8^+ plane.

**Proof.** 1: 30 = V ⊕ Vbar as a U(15) module, $\Lambda^2$(V ⊕ Vbar) = $\Lambda^2$V ⊕ $\Lambda^2$Vbar ⊕ (V ⊗ Vbar) and likewise for $S^2$; the orientifold projection keeps the antisymmetric part for SO and the symmetric part for USp, and the mixed summand V ⊗ Vbar is the U(15) adjoint. 2: $\Lambda^2$(A ⊗ B) = ($S^2$A ⊗ $\Lambda^2$B) ⊕ ($\Lambda^2$A ⊗ $S^2$B) and $S^2$(A ⊗ B) = ($S^2$A ⊗ $S^2$B) ⊕ ($\Lambda^2$A ⊗ $\Lambda^2$B), with $\Lambda^2 3$ = 3bar. 3: split 5 = 2 ⊕ 3 by charge; $S^2 5$ = $S^2 2$ ⊕ (2 ⊗ 3) ⊕ $S^2 3$ has dimensions 3, 6, 6 with charges 4/3, 1/3, −2/3, and $\Lambda^2 5$ = $\Lambda^2 2$ ⊕ (2 ⊗ 3) ⊕ $\Lambda^2 3$ has dimensions 1, 6, 3 with the same charges. 4: Proposition 6.1 with charges (+1, +1) and (+1, −1) under U(1)_i x U(1)_j, whose holonomy is e^{2 pi i (a_i ± a_j)}, and θ = pi a. 5: dimensions. ■

**Reading (proposal).** (11, Theorem 11.3 and Proposition 11.4, qualify the selection below: the window sees the counts (3, 6, 1) also at s = +1, through the commutator with the masses, and parent scalars with the counts (6, 6, 3) need M_15(C) at both walls.) The diquark counts of lemma 21 and of the EPJC paper are the stack-image strings of an O8^- plane, the type that Proposition 8.1 associates with the factorization sign s = +1 selected by the seesaw sector (T2c); the O8^+ alternative gives the wrong number of up-type antisquarks. So the seesaw sign and the scalar counts select the same orientifold type. Two features follow from the projection. Diquark masses go with the sum θ_i + θ_j and meson masses with the difference θ_i − θ_j, so when the stack leaves the wall the diquarks become massive and the mesons stay massless: the diquarks are tied to the orientifold. And the top's single brane has no string to its own image, while its strings to the stack, the (30, 2), have masses near m_t (Proposition 6.1); in the EPJC paper the top's exclusion is an input, and here the O8^- projection with one brane for the top removes the top-top channel only (item 5); the top-to-stack strings (30, 2) remain, with masses near m_t, so the rest of the exclusion is a mass statement, as in the paper.

**4. Proposition 9.3 (the real structure decides between the two diquark counts)**

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The stated sign constraint excludes a nonzero antisymmetric block in the three KO labels. The signs permit all of S^2 V (second check: B ranges over exactly S^2 V for eps eps' = +1, with the diagonal block free); the algebra, grading and order conditions of a triple then select a subspace. Distinct-multiplet blocks remain related to their transposes. See the referee report.

**Setting.** E = V ⊕ Vbar with V as in Proposition 9.2, an antiunitary J on E that maps V onto Vbar and Vbar onto V (as particles and antiparticles are exchanged), $J^2$ = ε, and a Hermitian X on E with JX = ε' XJ, the condition that inner fluctuations obey in a KO-dimension with signs (ε, ε'). Let B be the block of X from Vbar to V, an element of V ⊗ V.

**Statement.**

1. B is symmetric (B in $S^2$V) if εε' = +1 and antisymmetric (B in $\Lambda^2$V) if εε' = −1.
2. In the KO-dimensions 5, 6, 7 of the finite part for the labels 9, 10, 11, (ε, ε') = (−1, −1), (+1, +1), (+1, +1), so εε' = +1: the diquark block of every J-compatible fluctuation lies in $S^2$V, whose colour antitriplet ($\Lambda^2$5, 3bar) has the counts (−2/3: 3, +1/3: 6, +4/3: 1).
3. For a connection on a circle whose real structure is complex conjugation (KO-dimension 1, (ε, ε') = (+1, −1)), the block lies in $\Lambda^2$V, whose colour antitriplet ($S^2$5, 3bar) has the counts (−2/3: 6, +1/3: 6, +4/3: 3) of lemma 21. The SO(32) Wilson line of Proposition 6.1 is of this type.

**Proof.** Choose a basis of V and let K: V -> Vbar be coordinatewise conjugation; J(v, w) = (ε K^{-1} w, K v) has $J^2$ = ε. Write X = [[A, B], [B*, D]]. The component of JX = ε' XJ that maps V to V gives ε K^{-1} B* = ε' B K, so B = εε' K^{-1} B* K^{-1} = εε' B^T in coordinates. Under the U(15)-equivariant identification of Hom(Vbar, V) with V ⊗ V, symmetric matrices form $S^2$V and antisymmetric ones $\Lambda^2$V. Items 2 and 3 use Connes's sign table, ε = (+, +, −, −, −, −, +, +) and ε' = (+, −, +, +, +, −, +, +) for KO-dimensions 0 to 7, and the

counts of Proposition 9.2. On the circle, D = −i d/dθ and J = conjugation give JD = −DJ, and a Hermitian connection A with JA = −AJ is i times a real antisymmetric matrix, an element of so(n), where the generator of R(2 pi a) lies. ■

Item 2 is a negative statement with the following scope: the blocks of J-compatible Hermitian operators between particles and antiparticles, for one multiplet V, in the three labels. It is the finite-geometry form of the symmetry of Majorana mass matrices, and of the Fermi-statistics constraint on a Lorentz-scalar bilinear ψ_i ψ_j of one Weyl multiplet without derivatives (recalled), under which the (15, 3bar) vanishes. Known escapes: item 3 (connection components along a circle or an interval), blocks between two distinct multiplets, which J does not constrain, and derivative couplings.

**Reading (proposal).** The counts of lemma 21 and of the EPJC paper, (6, 6, 3), belong to the antisymmetric block. In the corpus's setting this block arises as the component of an SO(30) connection along the circle or interval: the Wilson-line moduli of the type I' stack, whose stack-image part is Proposition 9.2. The finite triple alone, in any of the three labels, gives the symmetric block and the counts (3, 6, 1). So the composite scalars with the counts of lemma 21 point to the interval parent of lemma 22 and its gauge field, and the selection of O8^- in Proposition 9.2 concerns these connection components. This narrows the spin gap of Propositions 6.1 and 9.2: the $\Lambda^2 V$ states are scalars as circle components of a nine-dimensional gauge field, and they come with that field's vector and gaugino partners.

## 5. Two more dualities between 9 and 11 (recalled; proposals)

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The $E_1$ alternatives need correction: the canonical-orthogonal root lattices distinguish $\mathbb{P}^1 \times \mathbb{P}^1$ from $dP_1 = \mathbb{F}_1$, and the usual local engineering assigns the latter to $\widetilde{E}_1$. The $dP_2$ and brane duality claims remain conditional on the unavailable primary sources. The Weyl action is a gauge relabelling of a holonomy, subject to the full background data. See the report on the del Pezzo comparison and Weyl action.

1. **E-type walls and del Pezzo surfaces.** The E_{n+1} symmetry of an O8^- plane with n D8-branes at infinite coupling is the symmetry of the five-dimensional theory on a D4-brane probe, which M-theory realizes on a Calabi-Yau three-fold with a shrinking del Pezzo surface dP_{n+1} (Seiberg, hep-th/9608111; Morrison and Seiberg, hep-th/9609070; Intriligator, Morrison and Seiberg, hep-th/9702198). So the far wall with the top's brane (n = 1, E_2 = SU(2) x U(1)) has an eleven-dimensional description by a local dP_2, and the empty far wall (E_1 = SU(2)) by dP_1 or P^1 x P^1; the light-quark wall (n = 15) has none. The Witten base CP^2 x CP^1 of 02 is built from the same pieces (CP^2 = dP_0, and P^1 factors).
2. **Colour-flavour exchange as a T-duality.** In the heterotic string on S^1, the Weyl group of the gauge group acts on the Wilson lines as part of the T-duality group O(17, 1; Z). The colour-family transposition M of 05 is an element of the Weyl group of E8, so it is a T-duality of the nine-dimensional E8 x E8 string, and through Hořava-Witten a symmetry of the E8 holonomy on a wall in eleven dimensions. As a Weyl element it acts as a global gauge transformation, so by itself it relabels gauge-equivalent vacua; it exchanges colour and family physically only relative to a structure that tells them apart, such as the placement of H_F in 05. The four-family requirement of Propositions 5.1 and 5.4 remains.

## 6. Open

**Referee note (GPT-6.1 Terra, 4 Nov 2026).** The finite KO signs restrict the symmetric block without producing the full representation. Light interval scalar modes, their boundary conditions and the physical string dictionary remain to be constructed; the detailed EPJC comparison is pending.

- Whether the type I' coupling, as a spectral function of D_F (Proposition 9.1), supplies the second scale of Proposition 1.5 in the flat family of lemmas 23 and 24.
- The local dP_2 of the top's wall and its relation to the Witten base CP^2 x CP^1.

- Spin: in type I' the strings of Proposition 9.2 belong to nine-dimensional vector multiplets, with one real scalar each (the relative position), the same gap as in Proposition 6.1. By Proposition 9.3 a finite triple in the labels 9 to 11 gives $S^2V$, so the $\Lambda^2V$ scalars must come from the connection along the interval; the next computation is the inner fluctuations of the parent of lemma 22 with an SO(30) algebra, and whether its interval component carries $\Lambda^2V$ with the counts of lemma 21.
- Whether the (15, 3bar) of the EPJC paper could be read as bilinears of two distinct multiplets (the paper itself gives one fundamental with itself) (for example left-handed quarks and conjugates of right-handed ones arranged into one 30), which Proposition 9.3 does not constrain.

# Part R4 (referee report, lemma 18, two sheets, framework note)

## Referee report on lemma 18, the two-sheet section of lemma 16, and the background note

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_18_16b.md. Reproduced as delivered, with links relativized. The corrections are incorporated in lemma 18, lemma 16, the note and the paper. Claude checked with Python the equations (R1) and (R3), the fully massive kernel and its poles, and read Remark 2.9 of arXiv:0706.3688.

We checked the numbered statements in lemma 18, the two-sheet section of lemma 16, and the claims and table in current_framework.md. We read the relevant passages of the cited papers, identified below. We did not run a computation.

The principal conclusions are:

- The fixed-cutoff quadratic calculation in lemma 18 is correct in its stated top-only family. Its conclusion extends to a broad class of nonnegative smooth cutoffs, including compactly supported cutoffs flat near zero.

- The limiting top-only quadratic action does **not** by itself have unbroken $SU(2) \times U(1)$ acting on the entire light space. It contains an incomplete top-bottom doublet.
- The two-sheet construction reproduces the classified bimodule and real structure as abstract finite data. Example 16.9 does not derive its nontrivial grading as an even spectral truncation of the parent it specifies.
- Corollary 16.8 incorrectly says that its Dirac operator commutes with the whole centre in every case. It always commutes with the two *sheet projections*.

**1. Lemma 18, Theorem 1**

**Correct with amendment.** State suitable regularity and decay assumptions on $A$, for example smooth compactly supported fields, and interpret the trace on $\mathbb{R}^4$ as the stated relative trace or quadratic variation. For several generators, contract the coefficient matrix with the corresponding polarizations.

The Duhamel expansion has the two terms displayed in (5). For transverse $a$, shifting the loop momentum gives the orbital contribution $2/t$. At the two opposite vertices, the spin traces of the curvature and commutator terms give $q^2$ and $(m_i - m_j)^2$, respectively; their mixed spin traces vanish. The $A^2$ vertex contributes

$$-\frac{1}{4\pi^2 t} \sum_{ij} |T_{ij}|^2 e^{-t m_i^2} |a|^2.$$

Symmetrizing this term gives the subtraction in (1), including its factor. Differentiation gives (2)–(3). The divided-difference identities (6)–(8) also follow as written.

The moment assumptions (4) are sufficient for the displayed Euclidean coefficients: $\int s^{-2}\, d\nu$ controls the action density, while the positive moments provide the required differentiability and domination. They do not supply arbitrary timelike analytic continuation.

The trace convention agrees with CCM's bosonic spectral action. Their calculation explicitly counts both particle and antiparticle sectors and four spin components; see

CCM, Lemma 3.14 and §3.7.6. The fermionic chirality projection must not be inserted into that bosonic trace.

### 2. Lemma 18, Lemma 2

**Correct.** With the stated convention, $A_\mu \mapsto A_\mu - \partial_\mu \alpha$ and $\pi \mapsto \pi + \alpha$, so $A_\mu + \partial_\mu \pi$ is invariant. Conjugating by $e^{-i\pi T}$ proves (12), including the scalar contribution that is missed by using only the linear orbit tangent at a nonstationary background.

The stated longitudinal replacement follows from retaining the longitudinal part of the shifted orbital vertex. For a commuting generator, its integral vanishes by the displayed integration-by-parts identity. Translation invariance and rotation symmetry exclude quadratic mixing of a transverse vector with a scalar.

### 3. Lemma 18, Corollary 3

**Correct with amendment.** Formula (13) requires $Q$ to be positive definite on the chosen generator space. Otherwise it applies after quotienting by its null space.

The orbital part of (2) starts at second order in the squared-mass difference, whereas the commutator term starts with $d_{ij}^2$. Also $\int_0^1 (1 - 2u(1-u))\, du = 2/3$. This gives the stated $C_0$, $C_1$, and factor $3/2$.

For the full Standard Model finite trace, one generation contributes 2 to the particle trace of $T_1^2$; three generations and charge conjugation give $Q_{11} = 12$. The commutator trace gives (14). This agrees with CCM, equations (4.10) and (4.16), with $F(v) = f(v^2)$. It is a leading expansion, not an identity at fixed $m/\Lambda$.

### 4. Lemma 18, Proposition 4

**Correct.** In one color of the massive block, let

$$|+\rangle = (t_L + t_R)/\sqrt{2}, \qquad |-\rangle = (t_L - t_R)/\sqrt{2}.$$

Each has squared charged-generator matrix element $1/8$ to $b_L$, and the reverse ordered entries have the same value. Their total is $1/2$ per color. Thus the massive color triplet contributes $3/2$ to the particle trace, and

the other complete doublets contribute 9/2. The charge-conjugate block supplies the factor two used in (16).

Integration by parts gives $h'_x(0) = 2w(x)/3$ and $g'(0) = 2/3$. Consequently (18) follows. Expanding $A$ and $w$ gives the small-$x$ coefficient $-23/128$, and $A(x) \sim 2/x$, $w(x) \sim 6/x^3$ give the large-$x$ limit. The positive-Laplace extension (23)–(24) follows by dominated convergence under (4).

### 5. Lemma 18, Theorem 5

**Correct.** Equation (22) makes $h'_x$ and $g'$ strictly positive for $y \le 0$. Since $F_x(0) > 0$ and $F_x(y) \to -\infty$ as $y \to -\infty$, there is exactly one positive timelike root, and its derivative has the sign giving the stated Euclidean residue.

For $y \le 0$, $g'(y) \ge 2/3$, hence $F'_x(y) \ge 3$. The mean-value theorem gives $z \le A(x)/2$; the Taylor ratio has the same bound. Since $xue^{-xu} \le e^{-1}$, $A(x) \le 2/e$, proving (28). The limiting derivatives are 4 and 3 at small and large $x$, respectively, yielding (27). The result says nothing about complex poles or Lorentzian unitarity.

### 6. Lemma 18, Proposition 6

**Correct with amendment.** The statement about one *positive massive* neutral pole requires $x > 0$. At $x = 0$ there is only the massless neutral system.

For one Dirac species, diagonalizing the left-right mass block gives diagonal neutral charge $(q_L + q_R)/2$ and off-diagonal charge $(q_L - q_R)/2$. In (3), the latter connects eigenvalues $+m$ and $-m$, giving exactly (33). Adding three top colors gives (30). The commuting photon and color traces give (29), including charge conjugates.

The angle in (31) is a sensible **neutral-current definition**: the kinetic-orthogonal massive direction is $K^{-1}(1,-1)^T$, and its current can be written $T_3 - \sin^2\theta_{\rm NC} Q$. Here

$$K_{33} + K_{3Y} = 5 + r, \qquad K_{YY} + K_{3Y} = \frac{5}{3}(5 + r),$$

so the value $5/8$ is constant. This is a charge-trace identity for this family; it does not make all finite-cutoff definitions of the weak angle coincide.

The neutral mass form is rank one in $v = (1, -1)^T$. Its generalized eigenvalue gives (34), using $v^T B_\infty^{-1} v = 8/27$. For the pole, the left side of (36) has derivative at least $1/2$, so the positive massive root is unique and has positive residue. The substitutions stated for (29)–(33) are valid for (4).

### 7. Lemma 18, Counterexample 7

**Correct.** The only nonzero entries of $T$ connect $m$ to $-m$, so (37) follows from (1). Its $q^2$ coefficient is proportional to $e^{-x}(1-x)$, while its Taylor ratio is $6m^2/(1-x)$.

The pole equation is (36) with $\beta = 4x$. Since $H(-z) > 1$ for $z > 0$, its solution lies strictly between $4x$ and $4x + 2$, and the derivative is positive.

For (39), positive Laplace mixtures give $f(x) + xf'(x)$. The assumptions imply $xf(x) \to 0$, while $xf(x) > 0$ for $x > 0$; therefore its derivative is negative somewhere. A negative Taylor kinetic coefficient does not itself establish a negative-residue pole of the full form factor.

### 8. Lemma 18, Counterexample 8

**Correct.** For fixed $y$, (21) gives $h_x(y) \to 0$, leaving (40), which is positive at fixed positive $y$. The neutral matrix likewise leaves nonzero kinetic terms. The broken vector directions therefore do not disappear by their masses diverging or their fixed-momentum propagators vanishing.

The narrower decoupling statement is also correct: after restricting the connection to the centralizer, the mass blocks are invariant and the heavy block carries its heat suppression. "Decoupling" here is a statement about the classical bosonic quadratic action, not a quantum effective-action theorem. The symmetry of the unrestricted limiting light space is addressed below.

### 9. Lemma 18, No-go theorem 9

**Correct.** Differentiation of the heat-trace density gives (41), which is strictly negative for $c > 0$, nonzero $V_*$, and

the positive measure in (4). The theorem is fairly limited by the phrase "positive Laplace class."

The conclusion can nevertheless be sharpened on flat $\mathbb{R}^4$. For a sufficiently integrable real cutoff $f$, let $\lambda_j^2$ be the eigenvalues of $V_*^2$. Direct integration of the momentum variable gives

$$\mathcal{U}'(c) = -\frac{c\Lambda^2}{2\pi^2} \sum_j \lambda_j^2 \int_{c^2\lambda_j^2/\Lambda^2}^{\infty} f(z)\, dz. \tag{R1}$$

For **nonnegative** $f$, a finite $c > 0$ is stationary exactly when every tail in (R1) associated with a nonzero $\lambda_j$ vanishes. Thus:

- if $f$ has nonzero support arbitrarily far out, there is no finite stationary point on this ray;
- if $f$ has compact support ending at $R$, there is a stationary plateau once every nonzero $c^2\lambda_j^2/\Lambda^2 \geq R$.

This argument uses the continuum momentum integral; it is not the proof given for a finite torus. Sign-changing cutoffs can have cancellations in (R1), and a radial path with an additional fixed, noncommuting mass matrix need not have the form $cV_*$.

A nonconstant cutoff flat near zero cannot satisfy (4): a nonzero positive measure on $(0, \infty)$ gives $f'(0) = -\int s\, d\nu(s) < 0$. CCM use positive test functions and their asymptotic moments; profiles flat near zero are allowed and are outside the theorem's stated class. For nonnegative profiles, however, (R1) supplies the separate conclusion above. See CCM, Introduction and §6.2.

### 10. Lemma 18, Corollary 10

**Correct.** Equations (19), (27), and (34)–(36) give every entry of (42). At fixed positive $\mu$, all dimensionless pole and Taylor masses are fixed and positive, so their dimensionful values grow with $\Lambda$. In the last joint limit, the small-$\mu$ expansion gives $M_W \sim \sqrt{3/8}\, m_t$; the neutral masses also diverge, with $M_Z^2 \sim (3/5) m_t^2$.

These are scaling statements for prescribed backgrounds. They do not establish stationary vacua.

**11. Lemma 16, reducibility of Corollary 16.4's pair**

**Correct.** The projection

$$P = 1 \otimes \frac{1+\sigma_1}{2}$$

commutes with the represented algebra and with $J_F \otimes \sigma_1 K$, and is neither zero nor one. It proves reducibility of the **pair** $(A, J)$. The projection need not commute with the grading; irreducibility of the pair is the issue asserted in the text.

**12. Lemma 16, Theorem 16.7**

**Correct.** The graded flip is an antiunitary involution. The two direct-summand Dirac operators square to the corresponding sums of squares, and the graded-flip signs give $J\mathcal{D} = \mathcal{D}J$. The minus sign on the second grading gives $J\Gamma = -\Gamma J$.

For even $a, b$, the opposite action is on the other tensor factor, as displayed. Both order conditions follow from the tensor-factor separation. The product operators have compact resolvent under the spectral-triple hypotheses.

When the two finite spaces are ungraded and have equal dimension, the identification $\xi \otimes \bar{\eta} \leftrightarrow (v \mapsto \xi\langle \eta, v\rangle)$ gives left composition and $J(x, y) = (y^*, x^*)$, exactly as in Chamseddine–Connes, Proposition 2.8 and Remark 2.9.

**13. Lemma 16, Corollary 16.8**

**Error.** The KO signs, dimension, order conditions, real orientability formula, and the full-matrix special case are correct. The assertion that $D_F$ commutes with the **whole centre** in every case is false. It always commutes with the two projections distinguishing the sheets.

For a counterexample, take

$$V_X = \mathbb{C}^2, \quad \gamma_X = \sigma_3, \quad d_X = \sigma_1, \quad A_X = \mathbb{C} \oplus \mathbb{C}$$

acting diagonally, and take $V_Y = \mathbb{C}$, $d_Y = 0$. The element $(\mathrm{diag}(1, 0), 0)$ is central in $A_X \oplus A_Y$, but its commutator with $D_F$ is nonzero.

The orientability formula is the usual **real** zero-dimensional relation $\Gamma = \sum_i \pi(a_i) J \pi(b_i) J^{-1}$, with opposite-algebra coefficients. It is not the stronger assertion $\Gamma \in \pi(A)$, which can fail. In the full-matrix, equal-dimension special case, an invertible matrix in each summand is a separating vector; the pair is irreducible because a projection commuting with both left and right full matrix actions is scalar on each summand, and $J$ equates the two scalars.

### 14. Lemma 16, Example 16.9

**Error.** The two algebra-generation calculations and the abstract classified bimodule are correct. The claimed application of Corollary 16.8 with the subsequently chosen nontrivial grading is not justified by the specified even parent, and the compressed potential does not repair it.

For the generation calculations:

- The matrices $1, \Gamma_a, \Gamma_a \Gamma_b$ for $a < b$ are 16 real-linearly independent elements commuting with $j$. Since $\dim_{\mathbb{R}} \mathrm{End}_{\mathbb{H}}(W) = 16$, the real algebra generated by the five $\Gamma_a$ is $M_2(\mathbb{H})$.
- For Hermitian $B$, the fourth moment gives $Q(w^* B w / w^* w) = (B + \mathrm{tr}(B)1)/5$. This ranges over all Hermitian matrices. Their products generate $M_4(\mathbb{C})$ as a real algebra.

The use of two copies of the four-space is mathematically legitimate. It supplies independent direct summands and the 32-dimensional bimodule. For this particular grading of $W$ and trivial grading of $V$, the graded flip is the ordinary adjoint flip, and

$$\Gamma_F(\xi, \eta) = (\Gamma_5 \xi, -\eta \Gamma_5),$$

exactly the graded representation in Chamseddine–Connes, Proposition 3.2. The full algebra is a **graded prealgebra**; its even part is

$$\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C}).$$

The obstruction is that an even parent has $[P \gamma_X P, PaP] = 0$ for every compressed parent function. But the proposed grading $\Gamma_5 = 5Q(x_5)$ anticommutes

with $Q(x_1) = \Gamma_1/5$. Thus the full compressed $M_2(\mathbb{H})$ and that grading cannot both arise from the stated even parent. They can be chosen **after** truncation as abstract graded-algebra data, and the even subalgebra then gives the real zero-cycle orientability relation.

Similarly, $d = \mu\Gamma_1$ is the compression of a bounded real multiplication operator, but adding that operator to the original parent Dirac operator does not retain the original kernel as a spectral projection: multiplication by the nonconstant real coordinate does not preserve the holomorphic kernel. It also does not preserve the original parent grading as an odd perturbation.

Finally, $d$ commutes with the centre of the full prealgebra, namely $\mathbb{R} \oplus \mathbb{C}$. It does **not** commute with the whole centre after passing to the even algebra: for $e_+ = (1 + \Gamma_5)/2$, $[\Gamma_1, e_+] \neq 0$. It still commutes with the two sheet projections.

### 15. Framework note: real structure and KO-dimension

**Correct with amendment.** The KO-degree and antiparticle grading statements are right. The Euclidean Pfaffian construction uses the fermionic action on the appropriate chiral subspace. Barrett's Lorentzian treatment removes the quadrupling through compatible reality and chirality conditions; it should not be presented as the same Euclidean Pfaffian argument. See Barrett, §§1–3 and CCM, §4.4.

### 16. Framework note: classification of $(A, J)$

**Correct with amendment.** The classification is under order zero, a separating vector, irreducibility, and $J^2 = 1$. "Complexified algebra" here means the represented complex span used in the paper. The exclusion of the one-centre case also uses compatibility of the grading with the algebra—normalization in the graded-prealgebra formulation, or commutation in an even triple. Without compatibility, an arbitrary anticommuting grading is not excluded. See Chamseddine–Connes, §§2–3.

**17. Framework note: real form and grading**

**Correct with amendment.** The quaternionic grading and smallest choice $\dim_{\mathbb{H}} W = 2$ are hypotheses and choices in the source. The displayed full algebra is graded; $\mathbb{H} \oplus \mathbb{H} \oplus M_4(\mathbb{C})$ is its even part. On $\mathrm{Hom}(W, V)$, the grading acts by **right** composition, with the minus sign.

**18. Framework note: order one**

**Correct.** Under the hypotheses of Chamseddine–Connes, Theorem 4.1, the stated maximal even subalgebra is selected by the existence of a centre-connecting Dirac operator satisfying order one. The subsequent gauge group uses the represented unimodular lift; the statement is an algebraic selection, not a dynamical derivation of a vacuum.

**19. Framework note: unbroken symmetry**

**Correct with amendment.** CCM explicitly impose commutation with the nonscalar $\mathbb{C}_F = \{(\lambda, \lambda, 0)\}$ and identify its physical purpose with the massless photon; see CCM, equation (2.30) and Remark 2.19. The subalgebra is not unital in the full algebra, and the physical electromagnetic lift also uses the unimodularity condition. The nonscalar element in the kernel of $a \mapsto [D_F, a]$ explains the failure of the metric condition cited in the note; a treatment formulated only for complex algebras must specify its complexification.

**20. Framework note: beyond first order**

**Correct with amendment.** The quadratic fluctuation, gauge covariance, transitivity, and semigroup statements are supported by Chamseddine–Connes–van Suijlekom, §§I and III. The Pati–Salam construction is in the separate arXiv:1304.8050, §§III and VIII. Its discussion of a vacuum selecting the Standard Model subalgebra is model- and vacuum-dependent: the paper explicitly says that determining all vacua is difficult and checks a restricted Hessian. It should not be stated as a general theorem that every vacuum restores order one.

### 21. Framework note: quanta of geometry

**Correct.** In the conventions of Chamseddine–Connes–Mukhanov, equations (2)–(5), the two Clifford algebras in dimension four are $M_2(\mathbb{H})$ and $M_4(\mathbb{C})$, and the two-sided relation gives the sum of the pulled-back volume densities. This is a quantization condition with two maps, not a truncation result.

### 22. Framework note: twisted triples and grand symmetry

**Correct.** The larger algebra mixing spin and internal degrees of freedom and the extra scalar are described in Devastato–Lizzi–Martinetti, §§3–5. A twisted commutator replaces the ordinary bounded-commutator requirement in the later Devastato–Martinetti construction, §3. The note properly says that these constructions have not been studied here.

### 23. Table: noncommutative finite algebra

**Correct.** The interval and projective-space examples give the stated generated algebras, with algebra generation an additional operation. The distinction between $PAP$ as an operator system and its generated $C^*$-algebra is also emphasized in Connes–van Suijlekom, §1.

### 24. Table: inherited pointwise reality

**Correct.** Under Proposition 16.1's pointwise-reality and invariant-window hypotheses, order zero for the generated algebra holds exactly when that algebra is commutative. This is scoped to that inherited correspondence.

### 25. Table: order zero, order one, centre $\mathbb{C}$

**Error.** Corollary 16.3 gives the order conditions and dimension $k^2$, but it does not generally give centre $\mathbb{C}$. For its interval example $A_F = \mathbb{C} \oplus \mathbb{C}$, the centre is $\mathbb{C} \oplus \mathbb{C}$. The stated centre requires an additional condition, such as a simple generated algebra.

### 26. Table: KO-6 signs

**Correct.** Both constructions give the signs. The first pair is reducible and fails the displayed real orientability

relation; the two-sheet input is explicit.

**27. Table: centre $\mathbb{C} \oplus \mathbb{C}$, dimension $2k^2$**

**Correct with amendment.** Corollary 16.8 reaches this classified pair when the two spaces have equal dimension and the represented complex spans are full matrix algebras. Arbitrary generated algebras in that corollary can have larger centres and need not give an irreducible pair.

**28. Table: orientability**

**Correct.** The displayed zero-cycle works under the stated condition that both finite gradings lie in the even represented algebras. It is real zero-dimensional orientability, in the sense specified in section 13 of this report.

**29. Table: $M_2(\mathbb{H}) \oplus M_4(\mathbb{C})$**

**Correct with amendment.** The real generated algebras and the 32-dimensional abstract bimodule are reached. The nontrivial grading is added after the stated truncation, and the full algebra with that grading is a graded prealgebra. It is not an even spectral-triple algebra, nor is the grading derived from the specified even parent. The construction requires a **real** generated algebra on the $S^4$ side; complex $C^*$-generation would also include $i1$.

**30. Table: selection of $A_F$**

**Correct with amendment.** An off-diagonal term is missing, but an arbitrary off-diagonal term alone is not enough. One must impose the required order-one condition and the maximal-subalgebra hypothesis of the classification.

**31. Table: Yukawa block**

**Correct with amendment.** Theorem 12.7 gives the texture in a towerless limit, up to grading-preserving unitary identifications. An algebra and its representation must also be specified before calling the result a finite spectral triple. It is not embedded in Example 16.9.

**32. Table:** $[D_F, \mathbb{C}_F] = 0$

**Correct with amendment.** Proposition 17.3 explains the consequence **when** a chosen subalgebra commutes with $D_F$; it does not make an arbitrary interval or Yukawa block commute with a Standard Model representation. The appropriate common action on the paired blocks is additional data.

**33. Table: bosonic action**

**Correct with amendment.** The leading-order results are correctly described, but "near-cutoff in progress" is stale: lemma 18 now gives the exact quadratic heat-cutoff calculation and the stated positive-Laplace extension. Its scope and limitations are those checked above.

**34. Table: beyond first order**

**Correct.** The inherited two-point data supply an order-zero, first-order-failing example to which the generalized-fluctuation question can be posed. The table correctly records that the calculation has not been done and that relaxing order one does not cure order-zero failure.

**35. Table: quanta of geometry**

**Correct with amendment.** The same five Clifford matrices occur, and the projector identity is valid. For the quaternionic rank-one projector $\Pi = P_w + P_{jw}$, expansion in the real basis $1, \Gamma_a$ gives $\Pi = (1 + \sum_a x_a \Gamma_a)/2$. This algebraic connection does not derive the two-sided quantization condition from truncation.

**36. Table: twisted triples and grand symmetry**

**Correct.** The table records no result and marks the subject untested.

**Part 2: scope of the heavy-limit conclusion**

Within the three-generation, zero-Majorana, top-only family of Proposition 4, at fixed cutoff and for the heat function, **both** the charged Taylor mass and its unique positive timelike pole tend to zero. The neutral Taylor and pole masses also tend to zero. The result refutes an infinitely heavy $W, Z, t$ limit **for these back-**

**ground data and these quadratic mass definitions**. It also rules out a stationary nonzero radial vacuum in the positive-Laplace class. It does not rule out other finite data, a different action or constraint, a joint $\Lambda \to \infty$ limit, or a quantum effective theory.

The top-only Taylor conclusion is broader than (4). For a nonnegative smooth compactly supported $f$ with $f(0) > 0$, define

$$M_1 = \int_0^\infty z f(z)\, dz, \qquad M_2 = \int_0^\infty z^2 f(z)\, dz.$$

The heat formulas extend by linearity and smooth functional calculus to

$$A_f(x) = \frac{2}{x} \int_0^x z f(z)\, dz, \qquad w_f(x) = \frac{3}{x^3} \int_0^x z^2 f(z)\, dz. \tag{R2}$$

One way to justify the extension is to approximate a smooth compactly supported $f(z)$, with the derivatives needed for the quadratic variation, by finite real linear combinations of $e^{-nz}$: write $f(z) = e^{-z} g(e^{-z})$ and approximate the smooth $g$ and its derivatives by polynomials. The divided-difference formula for the variation permits passage to the limit. Thus

$$\frac{M_{W,\mathrm{Taylor}}^2}{\Lambda^2} \sim \frac{M_1}{f(0)x}.$$

This includes a nonnegative cutoff flat near zero. For a nonanalytic cutoff there need not be a prescribed analytic continuation defining the pole used in Theorem 5.

Several heavy doublets change the answer. Suppose, for the heat cutoff, that $P$ is the total particle-trace weight of complete massless doublets, and heavy doublet $i$, of weight $d_i > 0$, has one Dirac member of mass $m_i$ and one massless member. Then

$$C_W(y) = \frac{\Lambda^2}{4\pi^2}\left(Pg(y) + \sum_i d_i h_{x_i}(y)\right),$$
$$\frac{M^2_{W,\mathrm{Taylor}}}{\Lambda^2} = \frac{3}{2}\frac{\sum_i d_i A(x_i)}{P + \sum_i d_i w(x_i)}. \tag{R3}$$

For any fixed finite number of doublets, if $P > 0$ and all $x_i \to \infty$, the Taylor mass tends to zero. If $P = 0$ and all $x_i = x$, (R3) instead gives

$$M^2_{W,\mathrm{Taylor}} \sim \frac{m^4}{2\Lambda^2}.$$

The full heat form factor also has a unique positive pole in this equal-mass case. To find its scale, put $z = cx$, $0 < c < 1$, in (21). Concentration near $u = 0$ gives

$$h_x(-cx) \sim \frac{2(1-2c)}{x(1-c)^2}.$$

Monotonicity then places the root at $z/x \to 1/2$, so $M^2_{W,\mathrm{pole}} \sim m^2/2$. At every fixed Euclidean momentum, however, the whole unnormalized quadratic kernel tends to zero. The divergent Taylor and pole scales differ, and neither alone establishes a healthy limiting gauge theory.

There is also a fully massive example. Give **both** members of every Standard Model doublet the same Dirac mass $m$, with no Majorana masses. The charged form factor is proportional to

$$e^{-x}[g(y) + 2xH(y)].$$

Its positive pole obeys $2x < z_W < 2x + 2$. The neutral massive pole obeys $(16/5)x < z_Z < (16/5)x + 2$. Both pole masses therefore diverge with $m$. Yet the fixed-momentum action tends to zero, and the charged Taylor kinetic coefficient is proportional to $e^{-x}(1 - x/2)$, which is negative for $x > 2$. This is a counterexample to a general fixed-cutoff no-go for diverging pole locations, while leaving the physical quality of that limit open.

The phrase "the heavy fermion leaves the loop" is only a shorthand for the suppression in this **classical spectral**

**trace**. The light complete doublets supply the surviving charged kinetic form, but the top-only light space also retains $b_L$ after removing $t_L$. If $P_{\text{light}}$ is the projection off the top mass block, then on $b_L$

$$P_{\text{light}} T_1 P_{\text{light}} = P_{\text{light}} T_2 P_{\text{light}} = 0, \qquad P_{\text{light}} T_3 P_{\text{light}} = -\tfrac{1}{2}.$$

These compressed operators do not obey the $SU(2)$ commutation relations. Correspondingly, the limiting particle-trace coefficients have $K_W = 9/2$, $K_{33} = 21/4$, and $K_{3Y} = -1/4$; a linearly realized unbroken electroweak gauge action would require equal weak coefficients and no $3Y$ mixing. Vanishing vector masses therefore do **not** prove unbroken $SU(2) \times U(1)$ on the entire light space.

The remaining **complete** massless doublets do carry the ordinary electroweak representation, and their separate contribution is gauge invariant. If finite data instead make both members of every removed doublet heavy, the retained light subspace can be invariant under the full group; a full limiting theory still requires control of the remaining fields, normalization, interactions, and vacuum.

**Part 4: what the two-sheet construction establishes**

Example 16.9 is substantive as a concrete compression calculation: real base functions generate a quaternionic matrix algebra, real total-space functions generate a complex matrix algebra, and two copies give the classified 32-dimensional bimodule. The calculation does not select those ingredients dynamically. The restriction to base functions on one sheet, the second independent sheet, real algebra generation, the four-dimensional window, and the grading axis are inputs.

At the finite algebraic level, the two-sheet bimodule is the standard construction from two representations:

$$(W \otimes \bar{V}) \oplus (V \otimes \bar{W}).$$

The graded flip supplies the real structure and the block sign supplies the KO-6 relation. The calculation's addi-

tional content is its realization from the particular compressed function systems and its explicit matching to the classified data. The abstract bimodule and its adjoint-flip real structure are already stated in Chamseddine–Connes, Remark 2.9; the report makes no broader novelty claim.

The note appropriately lists several choices and says that no ordinary Kaluza–Klein vacuum is derived. It overstates the reach where it treats the chosen nontrivial grading and nonzero compressed $d$ as consequences of the stated even-parent truncation. It also labels the centre of Corollary 16.3 too generally. Lemma 16's later caveat that its generated algebras are complex and a quaternionic summand still needs additional real data is stale after Example 16.9's real-algebra construction.

For canonical section 4, the example closes the **abstract finite-module** problem after adding a second sheet: it turns the nested actions on one four-space into independent summands acting on two Hom spaces. It does not turn four global sections into rank-four fibres over $S^4$, derive the two sheets from the fibration, or produce the centre-connecting Dirac operator required for the subsequent algebra selection. The fibre-rank problem remains open.

**Amendments and errors**

- Add field regularity to Theorem 1, $Q > 0$ or a quotient by its null space to Corollary 3, and $x > 0$ to the massive neutral-pole assertion.
- Keep the neutral-current angle distinct from other finite-cutoff angle definitions.
- Replace Corollary 16.8's claim about the whole centre with commutation with the two sheet projections.
- In Example 16.9, distinguish the correct abstract graded prealgebra from an even spectral truncation; the proposed grading and nonzero compressed perturbation are not derived by the stated parent. The nonzero $d$ also fails to commute with the whole centre of the even algebra.
- Correct the table's unrestricted centre-$\mathbb{C}$ claim for Corollary 16.3, and add the full-matrix and equal-dimension hypotheses to its classified two-sheet row.

- Update the table's near-cutoff entry and qualify the claims about the Yukawa representation, off-diagonal Dirac operator, real-algebra generation, and the vacuum discussion beyond first order.
- Distinguish the Euclidean Pfaffian and Lorentzian fermion reductions, and retain the grading-compatibility hypothesis in the classification.
- On flat $\mathbb{R}^4$, equation (R1) extends the radial analysis to nonnegative non-Laplace cutoffs; compact support permits a stationary plateau.
- The top-only limiting quadratic action does not establish an unbroken electroweak representation on the full light space.

Overall: 36 statements checked: 18 correct, 15 correct with amendment, 0 gaps, and 3 errors.

# Part R7 (referee report, lemmas 20 and 21, Proposition I)

## Referee report on lemmas 20 and 21, the comparison in the steer note, and Proposition I

By Terra 6.1 (gpt-6.1-terra, effort max), 3 Nov 2026; prompt in ../lemmas/prompts/terra_check_20_21.md. Reproduced as delivered, with links made relative. The corrections are incorporated in lemmas 20 and 21, the steer note and the paper.

The review finds errors in the marked-point reading, the radial statement, parts of the scale comparison, Proposition I, and the no-go table. The connecting-operator calculations and the three consensus checks otherwise hold in their stated scopes.

The checks below are analytic; no numerical code was run. Source references identify the arXiv versions read.

**Part 1. Connecting operators 1. Correct.** In Result 1, block multiplication gives $T = cT^*c$ and $hT + Tg = 0$. The lower off-diagonal block of order one is (2); the other follows using adjoints and the involutive algebra. In $c$-paired bases, the first condition is complex symmetry.

The averaging proof gives $T = R + \tau(R)$, with $R$ intertwining the left actions and $\tau(R)$ the opposite actions. Conversely, such a decomposition gives order one on $\mathcal{A}(R)$, establishing (4). The final example has one arrow preserving the right character and the other preserving the left character; it satisfies order one although $B$ is not contained in $\mathcal{A}(T)$. No correction is needed.

**2. Correct.** For the displayed rank-one $T_\kappa$, with $\kappa \neq 0$, the two selected lines must reduce their respective actions and carry the same scalar. This gives (6), of real dimension $2 + 4 + 18 = 24$. The self-adjoint off-diagonal operator has complex rank 2.

Chamseddine–Connes, §4, equations (4.6)–(4.7) use an auxiliary intertwiner, also called $T$ there. Its compression $K : \mathbb{C}^2 \to \mathbb{C}^4$ has rank one or two. Rank two gives dimension at most $4 + 4 + 8 = 16$; rank one gives dimension 24, uniquely up to automorphism. For the displayed block, $T_\kappa x = \kappa K x K$. For a general admissible connector, the paper's $K$ is obtained from an auxiliary intertwiner and need not equal the Dirac block. No correction is needed.

**3. Correct.** Result 3 and Counterexample 4 have the stated domain-preservation equations. In the scalar example, the parent has signs $(+, +, -)$ and satisfies both order conditions. The two displayed chiral vectors form the window for

$$\theta/l < \Lambda_0 < (\pi - \theta)/l.$$

Their compressed algebra is the full diagonal $\mathbb{C}^2$, and for $p = \mathrm{diag}(1, 0)$,

$$[[m\sigma_1, p], p^\circ] = -m\sigma_1.$$

The negative conclusion concerns compression of the full smooth scalar gluing algebra in this window. Restricting the parent algebra, or changing the window or representation, changes the problem; Result 5 gives a positive constant-algebra construction.

**4. Correct.** In Result 5, (18) and (19) preserve the boundary domain. The equality $cX = X^* c^{-1}$ is the required compatibility for $J$. The window contains $2n - 2$ uncoupled zero modes and the pair $\pm m$, giving (20). Constant $B = \mathcal{A}(S)$ multipliers commute with the parent Dirac operator, so the stated order-one claim follows.

With the displayed conventions, $U(x) = e^{mxF}$ carries the uncoupled domain to the coupled one, and differentiation gives (21), including its sign. Its compression gives $m(S+S^*)$. The construction assumes the Hom fibres, grading, constant multipliers and coupled channel.

**5. Correct with amendment.** The rank bound in Result 6 follows from the factorization $T = \mathrm{ev}_2^* X \,\mathrm{ev}_1$. A point with $N$ fibre channels can give rank $N$. An exact change of boundary domain does not itself establish that factorization.

The separation argument in Result 7 is valid for unchanged finite windows when independent test functions can be supported away from the gluing sets. It applies to the stated point and fibre examples, including $S^4$ pullbacks after fibre integration. For a general subset upstairs in the twistor space, add the hypothesis that its image in $S^4$ has empty interior, so that the required independent base test functions exist. A new coupled window, boundary degrees of freedom, or a changed observable representation lies outside the result. The direction is to construct and analyze such a coupled parent.

**6. Correct.** Hořava–Witten, §2, states $\Gamma_{11}\chi = \chi$ for the intrinsic boundary gaugino. With a common ten-dimensional chirality convention and the standard common orbifold lift, the same-chirality comparison uses two type-(a), or two type-(b), fields. Opposite localization masses can place them near opposite ends.

For unbroken $E_8 \times E_8$, the displayed Hom space vanishes, and an operator odd for the common chirality has no matrix elements between the two retained same-chirality modes. Compactification, symmetry breaking, or additional bulk fields may give different effective couplings; those require a separate construction.

**7. Correct.** The three checks in Results 9–11 hold:

- The assumed weighted mass relation gives $m_{\max}/(2\sqrt{2}) \le M_W \le \sqrt{N_g}\, m_{\max}$. It forces the maximum to grow, with a fixed species necessarily unbounded only along some subsequence. The relation belongs to the specified asymptotic matching action; a full finite-cutoff calculation is a direction beyond it.

- With pointwise reality and a $J_0$-invariant window, inherited order zero holds exactly when the algebra generated by the compressed functions is commutative. A separate opposite action, as in the coarse correspondence, changes that hypothesis.
- If $P\gamma P$ is an actual grading, its square forces $[P,\gamma]=0$. Thus it commutes with compressed even observables, whereas $\Gamma_5$ fails to commute with $Q(x_1)$. An added finite grading and restriction to its even algebra evade that particular inheritance claim.

The cited repository reports record earlier checks by another model. This review is a further solo check of the displayed arguments.

**Part 2. Readings 8. Correct with amendment.** In the CCM preprint, §2.5–2.6, especially (2.34), the right-handed up lepton is $\nu_R$, and the off-diagonal block maps it to its conjugate through a symmetric Majorana matrix. After the stated identification, and an overall grading convention if needed, the one-generation connector has that form.

The interval construction gives one mass $m = \arctan(t)/l < \pi/(2l)$, with next positive absolute level $(\pi - \arctan t)/l$. Three identical copies give three equal masses. A general three-generation CCM Majorana matrix requires additional flavour data; it is not supplied by the single rank-one construction.

**9. Error.** The projector identity and

$$2\Pi_p - 1 = p_a\Gamma_a = 5Q(p\cdot x)$$

are correct. The pair of projective points determines the two lines, but it does **not** determine the connector's magnitude. Rephasing $w$ and $v$ also changes $KxK$ by a phase. At fixed nonzero $|\kappa|$, the line choices are conjugate under the stated groups; $|\kappa|$ remains a modulus, and other finite data can introduce relative moduli.

The exact correction is to specify the lines **and** a coupling, together with an evaluation and normalization prescription. Moreover, $5Q(p\cdot x)$ fails to commute with the other coordinate compressions. It can be chosen as additional finite grading data after restricting to its even algebra.

That evades Result 11's inherited-grading hypothesis; it does not derive the grading from the even parent or construct the proposed point interface.

**10. Correct with amendment.** Equation (21) gives a constant matrix connection after a unitary change of domain, with relative endpoint rotation angle $\theta = ml$. On an interval, the open-path transport depends on endpoint frames; the physical quantity is the transport together with the specified boundary conditions. A closed Wilson loop or a dynamical gauge field needs further structure.

The gauge-Higgs comparison is an analogy at this stage: the interval calculation supplies a fixed matrix connection and no Higgs fluctuation or effective potential. The reference is Yutaka Hosotani, "Dynamical mass generation by compact extra dimensions," *Physics Letters B* **126** (1983), 309–313. For the dynamical Wilson-line and gauge-Higgs construction in an orbifold setting, see Hosotani, Noda and Takenaga, §1.

**Part 3. Counting 11. Correct with amendment.** In Proposition 21.1, $(q_L - q_R)/2 = (T_3/2)(1, -1)$, so the additional term in lemma 18's (33) is annihilated by the photon vector $(1, 1)$. The stated sums and removal entries reproduce:

| removed | $\cos^2\theta_{\mathrm{NC}}$ | $\kappa_{\mathrm{em}}/\kappa_s$ |
|---|---|---|
| $u$-type | 5/8 | 8/3 |
| $d$-type | 59/92 | 46/15 |
| charged lepton | 17/28 | 7/3 |

The table assumes the stated three-generation charges and that the other species have the mass assignments used in the removal calculation. At the endpoint where the neutral mass vanishes, the massive neutral direction is no longer distinguished: read its angle as the limit from finite mass, as lemma 18 states. Other weak-angle definitions need not have these values.

**12. Error.** The trace $84+12e^{-x}$ is correct for a top-only Dirac mass and no Majorana mass. With the positive heat cutoff and $m > 0$, its radial derivative is strictly negative. The text says there is "no finite stationary point"; at $m = 0$ the derivative is zero.

Replace that sentence by: **there is no finite nonzero stationary point on this free radial ray**. This con-

cerns the literal flat-space action with no added potential or constraint. Compact support can give a plateau; sign-changing cutoffs, additional terms, constraints, and paths with an additional fixed noncommuting mass lie outside the strict conclusion. The result points to analyzing those changed actions or prescribed backgrounds.

**13. Correct with amendment.** Complexifying the real 30 gives

$$(\mathbf{5},\mathbf{3})\oplus(\bar{\mathbf{5}},\bar{\mathbf{3}}),$$

and the displayed exterior-square branching has dimensions

$$192+24+8+1+60+45+60+45=435.$$

The charge tables and the charged scalar-partner counts follow. The charge operator includes the $U(1)$ commuting with $SU(5)$:

$$Q=\mathrm{diag}(3/5,3/5,-2/5,-2/5,-2/5)+\tfrac{1}{15}I_5.$$

Also,

$$\mathbf{496}\to(\mathbf{435},\mathbf{1})\oplus(\mathbf{1},\mathbf{1})\oplus(\mathbf{30},\mathbf{2})$$

under $SO(30)\times SO(2)$.

These are representation and charge counts. Identifying the entries with physical scalar partners requires a field construction, spin and chirality assignments, interactions and dynamics. Rivero's preprint, Tables 1–2, contains the stated diquark and meson counting; its proposed particle interpretation remains a proposal.

**14. Correct with amendment.** Both decompositions and the overlap of 72 follow by counting four internal states per Dirac species and colour. Without $\nu_R$, the first family has 78, and the charged family remains 84. A massless three-form in eleven dimensions and its six-form dual each have $\binom{9}{3}=\binom{9}{6}=84$ on-shell polarizations.

The 84 in $H_F$ is a complex finite-Hilbert-space dimension; the three-form count is a real bosonic polarization count. A physical identification needs a map of representations and degrees of freedom.

**15. Correct with amendment.** Seiberg, "Electric-Magnetic Duality in Supersymmetric Non-Abelian Gauge Theories," §1–3, *Nuclear Physics B* **435** (1995),

129–146, gives the four-dimensional $\mathcal{N}=1$ supersymmetric result. For $N_c=3$, $N_f=5$, $9/2<5<9$; the magnetic group is $SU(2)$, with five flavours and a $5\times5$ gauge-singlet meson chiral superfield coupled by $W=Mq\tilde{q}$.

To turn readings 1–3 into statements, one needs:

- An $\mathcal{N}=1$ supersymmetric theory with the required massless matter and dynamics, plus an operator and anomaly map. Five-flavour ordinary QCD alone does not supply Seiberg duality, and the magnetic $SU(2)$ needs a separate identification with any electroweak group.
- A representation and dynamics relating the finite fermionic states to the bosonic form or brane degrees of freedom; equality of the number 84 is insufficient.
- For the KO-label reading, the required involution on the stated finite sector, including its action on $D$, $J$, the algebra and its opposite. The vectorlike $SU(3)\times U(1)_{\mathrm{em}}$ charges alone do not establish it, and neutral states still require treatment.

**Part 4. Comparison with repository results 16. Correct with amendment.** Example 12.5 gives

$$\tan^2(kl)=|X|^2, \qquad k_{\mathrm{light}}=\frac{\arctan|X|}{l}.$$

For small $X$, $k_{\mathrm{light}}=|X|/l+O(|X|^3/l)$; the half-integer spectrum occurs as $|X|\to\infty$. At $X=0$, zero has multiplicity two and must be omitted from the positive spectrum.

With $X(l)=\tan(ml)$, the light mass is exactly $m$ and the remaining levels diverge as $l\to0$. Identifying $X$ with a physical Yukawa coupling, or $1/l$ with $v$, remains an extra hypothesis. The Wilson-line wording needs the qualification in item 10.

**17. Error.** CCM, (4.12) gives

$$\mu_0^2=2\frac{f_2}{f_0}\Lambda^2-\frac{e}{a}.$$

Top dominance in the Dirac Yukawa trace alone does not force $e/a=0$: the Majorana–neutrino term can remain

relevant. Under exact top-only data, or the additional assumption $e = 0$, the stated truncated-potential relations give

$$\frac{m_t^2}{\Lambda^2} = 2\frac{f_2}{f_0}.$$

For the heat profile $F(u) = e^{-u^2}$, $f_2/f_0 = 1/2$, giving $x = 1$. This is a stationary point of the **truncated** potential; lemma 18's literal positive heat action has no finite nonzero stationary point on the free top-only radial ray. Its exact quadratic-background calculation at $x = 1$ gives the stated departure from the small-mass ratio.

**18. Error.** As written, "finite low-energy electromagnetic coupling" includes a coupling tending to zero. The trajectory in the transcript provides that case: take $\Lambda = \mu_* = v$, keep $\Lambda_5 > 0$ fixed through one-loop matching, and choose

$$\frac{8\pi^2}{g_2^2} = \frac{23}{3}\log\frac{v}{\Lambda_5} - \frac{1}{3}\log\log\frac{v}{\Lambda_5} + O(1).$$

Then $g_2 \to 0$, $m_t/\Lambda \to 0$, the top-only Taylor ratio tends to $\sqrt{8/3}$, and the low-energy electromagnetic coupling tends to zero. The text itself lists that endpoint as an escape.

The intended no-go can be stated with **nonzero limiting** $\alpha_{\mathrm{em}}$ and positive finite limiting $\Lambda_5$, a matching scale above the relevant thresholds, top dominance at matching, the stated one-loop running, and the top-only heat-cutoff Taylor mass. In a step-threshold regime with $\mu_0 < m_t < \mu_*$ and nonincreasing $\alpha_{\mathrm{em}}^{-1}$ toward $\mu_*$,

$$\alpha_{\mathrm{em}}^{-1}(\mu_0) \geq \frac{28}{3\pi}\log\frac{\mu_*}{\Lambda_5}.$$

This follows from $b_6 = 7$, $b_5 = 23/3$, and $\alpha_{\mathrm{em}}^{-1}(\mu_*) = \frac{8}{3}\alpha_3^{-1}(\mu_*)$. It bounds $\mu_*/\Lambda_5$; matching then bounds $g_2$ away from zero, so $m_t \asymp g_2 v$ eventually conflicts with $m_t < \mu_*$.

The signs quoted in the note are correct in their respective active-field ranges: $\sum Q^2 = 20/3$ for five active quarks and three charged leptons, and $b_2 + b_Y = -11/3$ above electroweak thresholds. The thresholds are set by masses, rather than simply by $v$. The written route

$m_t/\Lambda \to \infty$ cannot simultaneously use six-flavour matching at $\mu_* = \Lambda$ above the top. Extra thresholds or gauge mixing, altered matching, a different action or mass definition, and an $e \to 0$ endpoint are directions outside the amended no-go.

**19. Error.** The one-loop threshold relation

$$\Lambda_5 = \Lambda_6^{21/23} m_t^{2/23}$$

is correct under the transcript's matching assumptions. Its $n_f = 5$ and Seiberg's $N_f = 5$ count different theories; the equality of the numbers establishes no duality.

The transcript says pions *do not have to be* elementary KK modes or finite-algebra fluctuations. The comparison changes that into a stronger exclusion. Proposition 17.10 establishes absence of a kinetic term for the specified coordinate change of a fixed operator; it leaves a new operator or quantum effective action available. Proposition 17.8 gives the stated Banks–Casher order of limits under its measure and regularity assumptions.

For $T^3 = \mathrm{diag}(1/2, -1/2)$ and $Q = \mathrm{diag}(2/3, -1/3)$, the algebraic coefficient is $2N_c \,\mathrm{tr}(T^3 Q^2) = 1$. The reduced action supplies a starting point; a regulated fermion measure and anomaly calculation remain to be supplied.

**Part 5. Paper 20. Error.** Proposition I omits $\kappa \neq 0$ in item 2. At $\kappa = 0$, the block has rank zero and $\mathcal{A}(T_\kappa) = \mathcal{A}^{\mathrm{ev}}$, of real dimension 40. Add $\kappa \neq 0$.

Item 1 must also assume that the sheet-diagonal $D_F$ already has the stated signs and order one. In item 3, specify a symmetric cutoff satisfying

$$m < \Lambda_0 < (\pi - \theta)/l.$$

The Majorana identification in the following paragraph needs the basis and flavour qualifications in item 8. The marked-pair paragraph needs the coupling modulus and grading qualification in item 9.

The abstract's wall-coupling sentence is supported for a **nonzero coupling on the chosen decomposable channel**, with the constant matrix observable algebra and the stated fibres and grading supplied as inputs.

**21. Error.** The table at lines 330–349 needs the following row-by-row corrections.

| Row | Assessment and correction |
|---|---|
| Odd bulk | Say **no net chiral spectrum** under the specified standard smooth closed reduction. A periodic massless Dirac field on a circle has left- and right-handed zero modes. Boundaries and defects change the hypotheses. Baptista, §1, studies chiral interactions of massive non-Killing gauge fields; that is a distinct escape from an exact-isometry assumption. |
| Odd factorization | State the full chiral Standard Model representation and the particular involution criterion of Theorem D. Reducing to invariant vectorlike sectors changes the algebra and representation. |
| Only KO-7 | State the scalar-character neutral sector and the odd multiplicity, including zero singular values, in Theorem D. Even multiplicity and appropriate Dirac pairing are the stated alternatives. |

| Row | Assessment and correction |
|---|---|
| Empty light sector | Specify the **massless type-(c)** interval and fixed spectral cutoff. A massless type-(a) interval has a zero mode for every $l$. Tuned bulk mass and scaled boundary coupling give the stated alternative families. |
| Fixed compression | Specify the local gauge transformations and fixed light projection of Proposition F. A projection invariant under a given transformation does not have this defect. The covariant projection changes with the fluctuated operator. |
| First order | Specify the two-point compression and inherited pointwise $J$. Fluctuations without order one relax that axiom; the doubled parent changes the data. |
| Order zero | Add $J_0$-invariance of the window and the inherited pointwise reality. A doubled or otherwise changed opposite action lies outside those hypotheses. |
| Bounded-fermion matching | Keep the leading spectral-action order, positive kinetic form, stated trace and common profile space. A tower-scale fermion removes the bound's obstruction; matching the full actions needs more. |

| Row | Assessment and correction |
|---|---|
| $M_W \to 0$ | Specify **massless complete doublet**, the stated heavy-doublet family, the Taylor definition, and the cutoff hypotheses including $f(0) > 0$. Keeping a doublet with a fixed nonzero mass can leave a positive residual mass even for the heat cutoff. Moreover, for the nonnegative $f(z) = ze^{-z}$, $f(0) = 0$, the top-only Taylor mass instead has $M_W^2/\Lambda^2 \sim x^2/3$. The heat-cutoff pole result has its own analytic-continuation hypotheses. A joint cutoff limit, altered light sector, or quantum effective action changes the problem. |
| Radial vacuum | Specify an allowed nonzero ray $V = cV_*$, $c > 0$, flat background, adequate integrability, and no added term or constraint. For nonnegative cutoff of unbounded essential support the derivative is negative. Compact support can give a plateau; other listed changes are possible directions. |

| Row | Assessment and correction |
|---|---|
| Grading | Replace “grading not inherited” by “the proposed $\Gamma_5$ grading is not inherited” under the four hypotheses of Result 11. The Dolbeault window itself has an inherited scalar grading. A compressed height function can be added as finite data with a restricted even algebra. |
| Scalar gluing | Add closed empty-interior gluing sets and independent functions supported away from them, with unchanged separate windows. Matrix-valued multipliers alone need not evade the same separation argument; a restricted multiplier algebra, changed representation, or coupled window can. |
| Finite limits | Specify finite limits of **connected scalar algebras** in the stated spectral-propinquity class. Truncation, matrix-valued parent algebras, or another convergence notion changes the question and needs a separate construction. |

The vector-mass row points to exact calculations with the desired light masses and cutoff, while the grading and gluing rows point to an explicitly constructed coupled parent.

**Amendments and errors**

- Amend the pullback scope in **5**, the flavour and grading conventions in **8**, and the interval transport terminology in **10**.
- Amend the endpoint and degree-of-freedom interpretations in **11, 13–16**.
- Correct the missing connector data in **9**, the stationary-origin statement in **12**, and the missing Majorana term in **17**.
- Replace **18** by the explicitly nonzero-coupling, threshold-matched no-go, with the $e \to 0$ trajectory retained as an escape.
- Correct the transcript attribution in **19**, the hypotheses and marked-point summary in **20**, and the individual no-go rows in **21**.

Overall: 6 Correct; 8 Correct with amendment; 0 Gap; 7 Error.

# Part R21 (second check of the Terra referee of the 11D line)

## Second check of the Terra referee of 11D files 01 to 09, by Claude Sonnet 5.5

The user asked, on 4 November 2026, for a second check of each evaluation in the Terra referee report, with Claude Sonnet 5.5, since no other model had quota left. The checks ran in five batches, one at a time, each redoing the step on which a verdict turns and checking the records. The subagents were refused file writes, so each returned its report as text; the reports are filed below verbatim, each followed by the assessment of Claude Opus 5.5, who rechecked the claims against the text and the source lemmas before applying them. Scripts are in the session scratchpad and were not kept in the repository.

### Batch 1: files 01 and 02

**Report** A. Verdicts

1. Prop 1.1 and joint family: PARTLY. Map, domains, grading, spectrum (k+1/2)pi/l, r window states, gap and N(E)=2 r l E/pi redone and correct. Missing: the joint family lives on the periodic circle (B1).

2. Prop 1.4: AGREE. F(1)=93 zeta(5)/64, J(0)=3 zeta(3)/2 and slope (54/5-n_f)J(0) reproduced; normalization matches lemmas 23, 24.
3. Prop 1.5: AGREE. Counterexample G=theta^4 and small-angle claim hold; true-determinant roots of theta G'-4G are isolated (1.0094 at n_f=48, 1.0309 at n_f=16, A_0 from 11D/10). Remark 10.7 and Terra amend different sentences: no conflict.
4. Prop 2.5: AGREE. Cohomology table enumerated for common and independent chirality, both colour conventions: no flux in span(6Y, 3(B-L)) works, with or without nu^c; c_L=-3 c_Q forbids |P_2|=2 on both doublets.
5. Prop 2.6: AGREE. Exactly the two stated solutions; on P(3bar) the first components reverse; none without chi or B-L, for either chirality hypothesis.
6. Thm 2.1, Props 2.2, 2.4: AGREE. Freeness, Euler class (3,2), Z_6, congruence, transverse Einstein constants 10, 10, factor 5/4. M^{111} identification: COULD NOT CHECK (no repository source); label "conditional" is honest.
7. Prop 2.3: AGREE. Smith forms for nine pairs; (3,2): H^2=Z, H^3=0, H^4=Z/9, spin. Kreck-Stolz: COULD NOT CHECK.

Ledger and README rows match all seven verdicts (status and checkers). Recorded as sonnet_5_5, this check gives d11_flux_nogo a third identifier, in its stated scope.

B. Corrections, by severity

1. 11D/01 line 36 ("the parent unfolds to -i d/dx + D_F on this circle (lemma 25)") and line 44 ("In the circle language of Proposition 1.1"): Proposition 25.1 unfolds the lemma-22 parent to the periodic circle. Write "unfolds to -i d/dx + D_F on the periodic circle of length 2l; theta interpolates between it (theta=0) and the antiperiodic circle of Proposition 1.1 (theta=pi/2); the joint-family window is the n=0 mode".
2. 11D/01 Prop 1.5: line 5 says "amended at its statement below", yet heading (line 88) and lines 92-97 are unchanged. Replace line 97 by "If the roots

of theta G'-4G are isolated, a stationary branch has l=theta_/m *(alpha=1; Remark 10.7 for small theta_)*. A root interval (G=theta^4 near 0) makes every alpha<1 stationary and selects none." The universal wording also stands in STRATEGY.md line 77, INDEX.md line 497, 11D/06 line 63, paper/connes_kk_dictionary_note.tex line 112.

3. 11D/02 lines 62-77: lemma table, Statement ("the same image", "2j+1 … linear") and first two proof sentences need "common ten-dimensional chirality"; add the chirality-free proof (|M_2(c)|=2 on Q and L, c_L=-3 c_Q). STRATEGY.md line 77 still reads "pending check".
4. Stale headers: 11D/01 line 3, 11D/02 line 3 ("Referee pending"; "pending an independent check"; "Theorem 2.1 … proved here" against ledger "conditional").
5. Prop 2.4(c): ledger d11_z6, 11D/02 lines 50, 52, 101 and STRATEGY.md line 77 say "no bosonic Kaluza-Klein mode" without the Terra scope; add "tensor modes on a background invariant under the displayed action; spinors open".
6. 11D/02 line 26 "Sym^{3k}(3bar) x Sym^{2k}(2): for k = 1 this is the (10, 3)": Sym^3(3bar)=10bar; write Sym^{3k}(3) for functions. Line 60: add "convention for base P(3); for P(3bar) use (c_0,c_1)=(9/4,3/4) or (3/2,3)".
7. 11D/01 line 50 "(on the circle of Proposition 1.1, fibre C^2, …)": write "the scalar circle gives 2 l E/pi; fibre H_F gives 2 r l E/pi". Prop 1.1 item 3: "even f(|x|), Lipschitz at the fixed points".
8. 11D/10 Prop 10.6 item 3 ("no joint family results") needs the Prop 1.5 scope (V=0 or tuned H leaves flat directions).

C. Derivations

B1: one channel, walls Q_R=0, U_L=0 at 0 and Q_R=X U_R, U_L=-X Q_L at l, X=tan(theta): the determinant vanishes at l*lambda = n pi +- theta, so n pi at X=0 (zero mode), (n+1/2) pi as X tends to infinity. B2: (theta G'-4G+4A_0)/theta^2 tends to -2c, c=(54/5-n_f)J(0). B3: left-handed fields come from left-handed modes (pairing +(alpha+3/2, beta+1)) or conjugated right-handed modes (pairing minus that); common chirality gives P_2=+(2j+1), independent chi-

rality +-(2j+1). Third charge z: coefficients (-1,3), d^c gives -5 against 1; no vector of 117,649 passes both components. Scripts (Python): s1_oneloop.py, s2_flux.py, s3_topology_spectra.py, s4_periodicity.py, s5_third_charge.py, s6_unfolding_map.py.

**Assessment** Checked against the sources by Claude Opus 5.5. Correction 1: lemma 25, Proposition 25.1, is stated on the circle with periodic conditions, while Proposition 1.1 concerns the chiral interval, which is antiperiodic; applied in 11D/01 and in the dictionary note. Correction 2: G = theta^4 gives V = m^4, constant in l, so every l is stationary; applied, with the heading changed and the INDEX link updated. Correction 3: in the span of 6Y and 3(B - L), Q has charges (1, 1) and L has (-3, -3), so the chirality-free argument holds; added as a scope note. Corrections 4, 5, 7 (first part) and 8: applied. Correction 6: the tensor and function spans are now distinguished; the colour-convention point for Proposition 2.6 was already in the referee note of 11D/02. The Lipschitz point is in the referee note of 11D/01. The ledger records sonnet_5_5 as a second checker on the ten rows of files 01 and 02.

**Batch 2: files 03 and 04**

**Report** Second check of Terra sections on 11D files 03 and 04 (Claude Sonnet 5.5; Python 3 scripts). Ledger and README rows otherwise match Terra.

A. Verdicts 1. Prop 3.2: AGREE. Odd Lambda^k(5bar) = SM Weyl fields, even = conjugates, X = 3, -1, -5. 2. Prop 4.1: AGREE. Lemma 21 sums rebuilt from SM charges; among 12-state whole-species blocks only u, c, t and the three neutrinos keep (5/8, 8/3). 3. Prop 3.8, Cor 3.9: AGREE. All 17 nine-dimensional SU(3) representations decomposed; self-conjugate ones are 9(1), 3+3bar+3(1), 8+1; centre trivial on Y_{3,2}. 4. Props 4.2, 4.3: AGREE with both amendments. None of the 17 Lambda^3 equals F1 or F2; only 9(1), 3+6(1), 3bar+6(1) give {1,3,3bar}; 4.2(b) needs a factor-preserving map. 5. Lemma 4.A, Thm 4.4: PARTLY. Table, 45+45+6 and 33+33+30 reproduced from explicit E8 roots in two models; the chirality amendment covers only the top (B1). 6. Prop 4.5: AGREE. At root level Q u^c H_u, Q

d^c H_d, L e^c H_d, L nu^c H_u all carry family tensor eps_ijk; Y = eps.h has singular values (|h|, |h|, 0). 7. Section 8: AGREE; 14 operator grades redone. 8. Prop 4.6: AGREE. Dimension and Schur arguments sound; the blocks are congruent in R^8, so the grading carries the obstruction; the orbit of the top block under <W(A2 x E6), -1> (810 sets) misses the neutrino block. 9. Prop 3.5: AGREE. Normalization R_MN = (1/12)(G_MPQR G_N^PQR - g_MN G^2/12) (action R - G^2/48, mostly plus), checked on AdS4 x S7 and AdS7 x S4; all components and |f1| < 2|f2| redone.

B. Corrections, by severity 1. 04 lines 74, 105, 111: the chirality amendment also applies to the neutrinos. For t = +1/2, nu_L is in C_6, nu^c and nu_R in gravity, conj(nu_L) in C_3. Complete blocks give (C_3, C_6, gravity) = (6,6,0) for the top, (3,3,6) for the neutrinos, both signs of t (conjugation-closed blocks always have equal C_3 and C_6 counts). Restrict "the top on the M2 side, the neutrinos on the M5 and gravity sides" and "The particle neutrinos lie outside the C_3 sector" to the a = 1 fields and add these counts. 2. 04 line 34: the false "exactly" clause stays in Proposition 4.3; write "exactly when C^9 is 9(1), 3+6(1) or 3bar+6(1)". Ledger d11_colour_families and README row 4.2/4.3 ("With nonzero triplets and no other nontrivial irreducibles, C^9 is …") apply the condition to C^9, which 3+3+3(1) satisfies; apply it to Lambda^3 C^9. Line 28: add "tensor-factor-preserving" to 4.2(b). 3. 03 line 105, ledger d11_kk_triality, README row 3.8/3.9: add "tensor", "on a background invariant under the displayed action", "spinors open" (as 8b92f59 for d11_z6); likewise 03 line 25, 04 line 109. 4. 04 line 105: "does not preserve the family grading" -> "neither preserves nor inverts it". The proved scope, phi(s_f) = s_f^{+-1}, contains every automorphism normalizing <s_f, s_M>. Add that Proposition 5.1 excludes any W(E8) element mapping F1 onto F2 for aligned families. STRATEGY.md T14 repeats the claim ("must come from a colour-family exchange"). 5. Stale: 03 and 04 line 3 ("Referee pending", "pending an independent check"); STRATEGY.md T14 ("referee pending", "(pending check)"); 11D/README.md lines 5, 92. 6. Section 8 (ledger d11_discrete_z3, plus a note in 04): selection rule of the single element s_M, a symmetry only if the E6 breaking commutes with it; (nu^c)^2,

LLH_uH_u and LH_u, allowed by s_M, are forbidden by the family Z_3 (a = 2, 1, 2). 7. Wording: "closes" (03 line 105, 04 line 38) -> "excludes, in that scope"; two negation-first contrasts in 04 (lines 32, 105); 04 section 6 heading states a proposal as a result (fix INDEX.md line 527 with it); ledger d11_ads5: add flux quantization to "not checked".

C. Scripts (Python): b2_ext5bar_blocks.py (A1, A2); b2_su3_cases.py (A3, A4, B2); b2_e8_gradings.py (A5, A7, B1, B6; 240 roots, grades from Bourbaki marks and from b = a + 2T_3); b2_yukawa_family.py (A6); b2_blocks_weyl.py and b2_z3_group.py (A8, B4); b2_ads5.py (A9). Recalled-fact labels are honest; no conflict with 11D/10 or 11D/11. After B2 and B4, recording sonnet_5_5 gives d11_colour84, d11_colour_families, d11_m2m5_is_j three identifiers.

**Assessment** Checked by Claude Opus 5.5. Correction 1: from Theorem 4.4 with b -> -b for conjugates, conj(nu_L) has b = -2 = 1 for t = +1/2, so it lies in C_3; the counts (6, 6, 0) and (3, 3, 6) follow per colour and per generation, for both signs of t. Correction 2: C^9 = 3 + 3 + 3 (1) has 3 x 3 = 6 + 3bar inside Lambda^2, so the condition belongs on Lambda^3 C^9; the trivial 9 (1) gives 84 singlets, as Terra noted. Corrections 3 to 7: applied as stated; the section 6 heading of 04 is now "The Chevalley involution and the real structure J", with the INDEX link updated. The ledger records sonnet_5_5 on the eleven rows of files 03 and 04, and the 11D README now says that the checked no-go results have three model identifiers, each in its scope.

**Batch 3: file 05**

**Report** A. Verdicts

1. 5.1 AGREE. d = 22 on all 96 roots; F2: d = 20 (36 roots), 19 (48); F1: 16 (4), 22 (8), 20 (24), 19 (48); pair counts 42/816/1812/816; Terra 4x4 table and 12 slot/top choices reproduced. "Isometry of R^8" is the right notion (distance-preserving bijections extend; W(E8) is a subgroup). "Canonical up to W(E8)" holds for the chain E8 > SO(10) x SU(4) only (C2).

2. 5.2 AGREE. M = -P has 8 sign changes, lies in W(D8), M^2 = 1 on all 240 roots; T(0) <-> N(1,2,3), D(0) <-> E(1,2,3); nu_L, e_L of family 0 fixed; u^c, d^c, nu^c, e^c go to conjugate roots.
3. 5.3 PARTLY. Unitarity amendment and all stated properties confirmed (F^2 = 1, [F,gamma] = [F,J] = 0, rank 12, kernel 116). Missing: F differs from M (C1).
4. 5.4 AGREE. All 192 rerun; Terra formulas hold root by root; S - N: {19,20} in 48, {13,17,19,20} in 144 configurations; S - T pattern (16,16,22) in 96. The 11D/10 sentence is right.
5. 5.5 AGREE. No proved statement is contradicted; Props 5.1 and 5.4 concern E8-torus weights, which an irreducible SU(4) bundle lacks (argument). For aligned split bundles with n_i >= 0, sum 3, the obstruction persists (C3). Labels "recalled" (BHOP, Spin(9)) honest; 128 = 44 + 84 confirmed.

B. Corrections (11D/05 unless stated)

1. Line 66, "This is the finite-geometry form of Proposition 5.2." -> "F equals M on left-handed states and M followed by particle-antiparticle exchange on right-handed ones; F exchanges T_top and T_nu but lies outside W(E8); M flips gamma on 64 states." Reading 2 (line 91) likewise.
2. Line 72: replace "In every case S - N has d = 19 and d = 20 only, ..." by "S - N has d in {13,17,19,20} ({19,20} iff the down-quark and charged-lepton slot sets coincide, 48 of 192); each neutrino root of S - T has d = 16 + 6(1_A(f) - 1_{0}(f))." Line 74 ("scratch script not kept"): cite tools/check_11d_10_11/s5_e8_prop54.py and the referee proof.
3. Lines 84, 86: "their number" -> "their net number"; "fall under Proposition 5.1" -> "only for n = (1,1,1,0) up to order; for n_i >= 0 the obstruction persists (C3)".
4. Line 46 item 3 -> "nu_L, e_L of family 0 fixed; nu^c, e^c to negatives; u^c, d^c to conjugates". Line 58: "Fix an isomorphism" -> "Fix a unitary isomorphism". Line 36: "canonical up to the Weyl group of E8" -> "unique up to W(E8) for the aligned (16,3)+(16bar,3bar)".
5. Line 3: "pending an independent check", "Referee pending." -> "refereed by GPT-6.1 Terra and

checked a second time by Claude Sonnet 5.5".

6. Line 78, Reading 1 (line 90), 11D/README.md line 99: "three families" results stated beyond the proved scope (single roots, three of four slots); cite 5.1 and 5.4.
7. 11D/ledger_11d.pl: add checked_by(_, sonnet_5_5, python(…)) to d11_no_isometry, d11_transposition, d11_flip, d11_misaligned (dry run: tests pass; both no-go rows reach three models); replace "not kept" in d11_misaligned by the kept script. d11_flip: conditional without a named hypothesis, README row 5.3 says independently checked. Referee notes and README rows 5.1-5.5 match Terra.

C. Derivations

C1 (b3_flip_vs_M.py): uR, dR are negatives of the u^c, d^c roots. F = M on the 64 left-handed states, F = -M on the 64 right-handed. F changes 3072 of 16384 ordered root inner products, M none. M sends gamma to -gamma on 64 states, so the T_top pair (uL, uR) goes to (nu_L, conjugate nu_R), violating gamma(s1) = -gamma(s2); F keeps it. C2 (b3_embedding.py, b3_classes.py): the 96 roots are the spinor-type roots among the 162 of (27,3)+(27bar,3bar); the 7560 A3 subsystems of E8 form one W(E8)-orbit with complement D5; the 192 placements give 3 invariant classes. C3 (b3_multiset.py): S has n_f copies of the 32 roots of slot f; degree multisets of S - T and S - N differ in all 40 (n, top slot) cases. C4 (b3_prop53.py, numpy): iota unitary, scaled unitary, diag(2,1,1,1) and a non-unitary GL(3) block; F unitary only in the first two; all other properties hold in all four. Others (Python 3): b3_prop51.py, b3_prop52.py, b3_prop54.py, b3_spin9.py.

**Assessment** Checked by Claude Fable 5.1 (the dialogue model from this batch on). Correction 1 is right and matters: M = -P sends t^c = (-c | ++ | 1) to the negative of a nu^c root, that is to an antiparticle state, while F of Proposition 5.3 acts inside H_f and keeps particles as particles; so F agrees with M only on the left-handed states, and "the finite-geometry form of Proposition 5.2" overstated it. Applied, with Reading 2 qualified. Correction 2: the values {13, 17, 19, 20} agree with the rerun recorded in 11D/10; applied, with the formula for the neu-

trino degrees. Corrections 3 to 6: applied. Correction 7: the four rows carry sonnet_5_5 with the method named, and the "not kept" entries now cite the kept script. The ledger row d11_flip stays conditional on the unitary identification, which the file now names.

**Batch 4: files 06, 07 and 08**

**Report** A. Verdicts

1. Prop 6.1: AGREE. Ad(W_a) on so(32): eigenvalue 1 on 436 = 435+1, e^{±2 pi i a} on 30 each; (n±a)/R = (n pi ± theta)/l; light mass a/R on [0,1/2].
2. Prop 7.1: AGREE. Wall determinant gives lambda l = n pi ± theta; nullity 2 at theta = 0 and, as X -> infinity, pi/2.
3. Prop 7.2: AGREE. Centralizer S(O(30) x O(2)), two components, Lie dimension 436; minimum at t = 1 (n_f = 1, 6, 16, 48). Scope gap: B2.
4. Prop 7.3: AGREE with Terra. h^0(L S_0) = h^1(L^-1 S_0) = p; C^2 = -1 gives (+,+,-); rho(1) has rank p of 2p. M_p + C is unital, order zero, and is lemma 16, Corollary 16.8 with (k_X, k_Y) = (p, 1). Berezin-Toeplitz step: true (Gaussian times Weyl word, exactly; p = 2..5, four spin structures, rank p^2), unproved in the file.
5. 07 §6 remark: AGREE (no Terra check). Commutant dimension 14 for stacks 1, 2, 3; spectrum 0, ±|x_a - x_b|.
6. Prop 8.1: AGREE (j-commutant dimensions n(n-1)/2, n(n+1)/2 for s = +1, -1).
7. Prop 8.2: AGREE (k, 2k; lowest |D|: 2k, 2k, 4k).
8. Remark 8.3: AGREE on arithmetic (15+1: z(0) = 7cL; 7+7: x_1 + x_2 = L; 7+6+triple: x_b = 2L/3). PARTLY on sources: B4.
9. Recalled dualities: AGREE, labels honest. PW and BGL confirm positions = Wilson lines (wilmap), heterotic KK = type I' winding, the slope rule, E_{n+1} (Table 1); others unchecked.

B. Corrections, by severity

1. 11D/10 is right on 10.1 to 10.3 (rechecked; 10.4 to 10.6 unchecked) and conflicts with 07 §4 Reading "distance proportional to a", 06 §3 last row, 06 §6 items 1 and 4, 08 §5 bullet 2. Only 07 §6b

and 08 §4 point to 10. Append "(proportional only at constant dilaton; see 10, Proposition 10.1)" and equivalents.

2. 07 §4 note, append: "in SO(32). In Spin(32)/Z_2 the det -1 component sends the lift of W_{1/2} to its negative, so the heterotic centralizer is connected; a -> a+1 holds for SO(32) matrices, the lift has period 2."
3. 08 line 43 ("is not available with colour-respecting moves; E8 x E7 is:"): unapplied Terra replacement: "With colour triples, the proposed E8 x E7 arrangement has seven and six branes at the walls and a bulk triple at 2L/3. Two E8 walls require relaxing the colour-triple rule."
4. 08 §4 item 1: "7 + 7 for E8 x E8" is in neither PW (E8 x E8 left open) nor BGL (E8 x E8 at N = 0, R_h -> 0). Write "extrapolated from the single-wall rule".
5. 07 §5: cite lemma 16, Corollary 16.8; state the Gaussian formula in step (3).
6. 08 §2 Reading: add the (j, d) refinement and the label tension (label 9 with O^+ against the far O8^- wall).
7. 07 §6: "8 n_a n_b per ordered block, 16 n_a n_b per pair".
8. Stale: line 3 of 06, 07, 08 ends "Referee pending." (02, 03 too); README row 8.3 and the 06 §2, 08 §4 notes still call PW and BGL unavailable.
9. Records: add sonnet_5_5 checkers to d11_wilson_top, d11_angle_interval, d11_far_wall, d11_torus_flux, d11_orientifold_sign, d11_wall_multiplicity, d11_e_walls, d11_duality_readings. Notes, ledger rows and README rows otherwise match Terra. Also add the unitality amendment at paper/connes_kk_dictionary_note.tex lines 54, 104.

C. Derivations

B1: PW: lambda = W(x)/(2W(L)), W = int z^{-1/3}. For 15+1, z(L) = 0, rho = (z(x_t)/z(0))^{2/3}: theta = 4 pi (1-rho)/(8-rho), R^2 = 14(8-rho^2)/(8-rho)^2 (as 10.2); x_t/L = 1/2 gives theta = 0.5859 against pi/4. B2: lifts W~ = G_{2n-1}G_{2n}, g~ = G_1 G_{2n-1} (2n = 8, 12): g~ W~ g~^-1 = -W~. Only these enter, so 2n = 32 agrees; -1 differs from 1 and

from the volume element. B5: rho(a, lambda) = a (+) lambda commutes with J rho(b, mu)^* J^-1 = mu (+) b^T; a unital M_3 on C^6 has commutant M_2, too small for M_3^op. B6: for s = -1, j swaps the ±mu eigenspaces of d, so the symmetries of (j, d) form U(n/2); s = +1 gives O(n_+) x O(n_-). Scripts (Python, not kept): b4_walls, b4_spin_lift, b4_so32_wilson, b4_unital, b4_torus, b4_commutant_j, b4_channel, b4_matrix_theory, b4_oneloop.

**Assessment** Checked by Claude Fable 5.1. Correction 1: the conflicts with 11D/10 are real, since Proposition 10.1 is the statement of the sources and the four readings predate it; pointers added at each place. Correction 2: the lift of diag(1_30, -1_2) to Spin(32) is Gamma_31 Gamma_32, which the lift of a reflection pair in the determinant -1 component conjugates to its negative, so the connectedness statement holds; added to the note of 07 §4. Correction 3: the Terra replacement had not been applied on the branch; applied. Correction 4: neither source read on master states the 7 + 7 configuration for E8 x E8; marked as extrapolated. Correction 5: Corollary 16.8 of lemma 16 gives the unital datum with (k_X, k_Y) = (p, 1), and the Gaussian form of the compressed characters is recorded as checked and unproved. Corrections 6 to 9: applied, with the dictionary note's table and text amended for the unital datum. Terra's error verdict on the unital triple of Proposition 7.3 stands, and the row keeps the status conditional.

**Batch 5: file 09 and the sections added by the referee**

**Report** A. Verdicts

1. Prop 9.1: PARTLY. Integration, concavity and far-wall formula are correct; the slope rule matches PW eq. (soln). Omitted: x_k = (2L/pi) theta_k holds only for constant z (11D/10 Prop 10.1 is right).
2. Prop 9.2: AGREE. Splits, dimensions and counts (6,6,3), (3,6,1) match lemma 21. PW confirms SO(2n) on O8^- and the (A_i +- A_j)/R strings; USp on O8^+ is outside the sources ("recalled" is honest).
3. Prop 9.3: PARTLY. Sign algebra is correct and

consistent with 11.1 to 11.4; "the signs permit only a subspace of S^2 V" is wrong.

4. Section 1: AGREE on both branchings; the EPJC comparison COULD NOT CHECK (no source).
5. Del Pezzo: AGREE, the error is real. BGL Table 1 has E_1 (self-dual radius) at the empty wall; the 5d dictionary is outside the sources.
6. Weyl action: AGREE.
7. Audit section 4: AGREE; asymptotics redone from lemmas 9.4, 9.5; table recomputed.
8. Prop 2.5 third charge: AGREE; the argument holds for any real z and any source chiralities.
9. Prop 3.4: AGREE (Z_3 on U = 0, Z_2 on V = 0).
10. Wording audit: AGREE with all nine reclassifications and the 08 replacement; no first-person singular in 01 to 11.

B. Corrections, by severity (remaining notes, statuses and rows match Terra)

1. ledger d11_dilaton_spectral, README row 9.1, 09 note of section 2: z(L) = z(0) - c(8L - sum x_k) is exact in the positions; x_k = (2L/pi) l m_k holds only for constant z; in general theta_k = (pi/2) W(x_k)/W(L), W = int z^(-1/3) (Prop 10.1). In the 15 + 1 profile the endpoint values 7cL, 8cL stand; theta_t = pi/4 gives 7.358 cL, not 7.5 cL. Add depends(d11_dilaton_spectral, d11_angle_map).
2. Replace "permits only a subspace of the symmetric representation" (09 section 4 note), "The signs permit only a subspace of S^2 V" (ledger d11_real_structure_diquarks) and, by editorial annotation, the same phrase in the Terra audit by "The signs permit all of S^2 V; the algebra, grading and order conditions of a triple select a subspace."
3. paper/connes_kk_dictionary_note.tex, "X with JX = XJ, in the labels 9 to 11": at label 9, JX = -XJ and J^2 = -1; write "JX = eps' XJ with eps eps' = +1". The gauge block is antisymmetric only for s = +1 (usp(30): symmetric).
4. d11_flux_third_general: change to no_go, proved; add checked_by sonnet_5_5; delete referee_pending; update the 02 note, README (intro, row 2.5, next-step 1), PLAN and STRATEGY T14. The open-directions lists name "a

changed common twist": that is z = F and is covered. The Prop 2.6 solution (fluxes F, chi, B - L, no Y) lies outside span(Y, B - L, z).

5. "Referee pending." ends line 3 of 09, 02, 03: replace by "Refereed by GPT-6.1 Terra and checked a second time by Claude Sonnet 5.5."
6. 09 section 3 Reading and section 6: add that 11D/11 Prop 11.4 and Theorem 11.3 qualify the O8^-/O8^+ selection (the window sees (3,6,1) also at s = +1; parent scalars need M_15(C) at both walls); 11D/11 is right.
7. Add checked_by(_, sonnet_5_5, by_hand) to a11_localized, d11_flux_third_bounded, d11_higgs_m2, d11_dilaton_spectral, d11_diquark_strings, d11_so32_algebra, d11_real_structure_diquarks, d11_dp_walls, d11_transposition_tduality; d11_so32_comparison stays pending. README and PLAN: the localized table is recomputed.

C. Derivations and runs

9.1: PW (mass), (wilmap): lambda_i = W(x_i)/(2W(L)). For 15 + 1 with z(L) = 0: z(0) = c(8L - x_t), theta_t = (pi/2) 8(1 - rho)/(8 - rho), rho = (z(x_t)/z(0))^(2/3) (Prop 10.2). 9.3: JX = eps' XJ on (v, w) is equivalent to B* = eps eps' K B K and D = eps' K A K^-1 with A Hermitian free, so B ranges over exactly S^2 V (eps eps' = +1) or Lambda^2 V (-1). A gauge generator H = ia has jH = -Hj, so (eps, eps') = (s, -1) (11.1). 2.5: phi(v) = 2v_Q + v_u + v_d kills y, b, p. Since w_L/w_Q = +-1, not -3, w lies outside span(y, b), so span(y, b, z) = span(y, b, w); phi(p) = 0 gives t(4s_Q + s_u + s_d) = 0, so t = 0, a = 0, b_0 = -5/2, p_L = 15/2, not +-3/2.

Runs (python3 -I, scripts not kept): exact fractions, rank [y, b, p, w] = 4 for 1024, 256, 64 sign patterns (six, five, four fields), the Prop 2.6 solution has rank 4, a rank-3 pattern would have refuted the extension; bisection, the four audit rows agree to the printed digits; mpmath quadrature, the PW formula on 15 + 1 equals the Prop 10.2 closed form (theta_t = 0.5859 at x_t/L = 0.5; linear 0.7854).

**Assessment** Checked by Claude Fable 5.1. Correction 1 restates Proposition 10.1 at the places of file 09 that still assumed the linear dictionary; applied, with the dependency row. Correction 2: for eps eps' = +1 the condition on the block is B^T = B with the diagonal block free, so every symmetric B occurs, and Terra's report itself says that the algebra, grading and order conditions select the subspace; the note and the ledger row now say so, and the report needed no annotation. Correction 3: the KO table gives (eps, eps') = (-1, -1) at label 9, so JX = -XJ there; the note now writes eps' and scopes the antisymmetric block to s = +1. Correction 4: the two-vector steps were redone by hand (phi kills y and b and not p; every vector of span(y, b) has v_L = -3 v_Q, which w fails), and the rank-4 enumeration over all sign patterns is Sonnet's; the row is now a no-go with two model identifiers (Terra as author, Sonnet as checker), below the three-identifier threshold, which nogo_below_consensus reports. Corrections 5 to 7: applied. The EPJC comparison keeps its pending flag, since no reader had the paper.